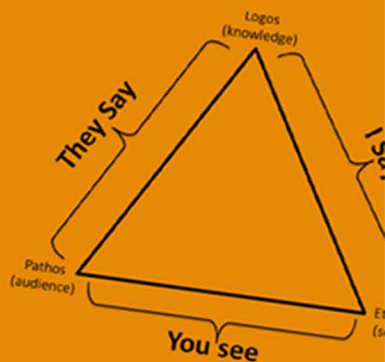
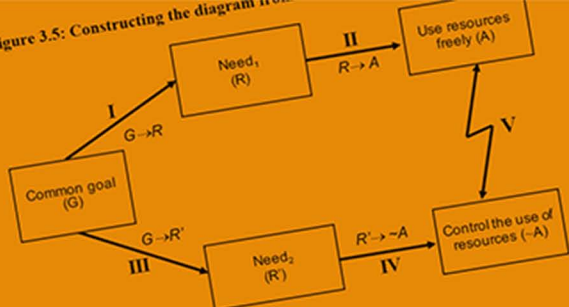


Figure 3.5: Constructing the diagram from

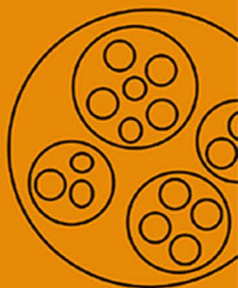


A GUIDE TO Argumentative Research Writing and Thinking

OVERCOMING CHALLENGES



ARNOLD WENTZEL



Arrogance can be good

How is that possible?

Meaning is misunderstood

It is not all the same

COMPANION
@
WEBSITE

ROUTLEDGE

A Guide to Argumentative Research Writing and Thinking

Research is difficult. Even though students are trained in the basic research methodology skills, when confronted with research writing, it feels to them as if they enter a bizarre world, with its own language and conventions, where it is hard to get things right. This book translates the apparent complexities of research writing into everyday ideas, language and skills, and will enable novice researchers to start overcoming the major stumbling blocks immediately.

This book focuses only on the greatest challenges in research writing, specifically those that supervisors find most difficult to explain to novice researchers. These challenges include both basic and more complex skills, such as: finding original research contributions; establishing one's voice while drawing on other authors; turning a vague idea into a feasible research question; generating literature reviews that are original in themselves; and avoiding list-like writing when discussing the research methodology.

Wentzel shows that it is easier to overcome these challenges, not with lists of prescriptions that are difficult to remember while writing, but rather by cultivating an argumentative mindset. Not only is such a mindset much easier to maintain, but it offers a central point around which one can organise any difficult writing task. The book shows how to use the argumentative mindset to approach every important writing challenge. It translates all the necessary skills into jargon-free language using a variety of visuals and simple step-by-step procedures that will enable any person to read the book quickly and start writing immediately.

The book is accompanied by a website containing an instructor's manual with guidance on the teaching and assessment of research writing, as well as lecture slides.

Arnold Wentzel taught research writing, education and economics for almost two decades in South Africa. He currently teaches innovation methodology and research writing at the Universidad Antonio Nariño in Colombia. His previous book, *Creative Research in Economics*, was also published by Routledge.

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A Guide to Argumentative Research Writing and Thinking

Overcoming Challenges

Arnold Wentzel

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Acknowledgements

I never really learned how to do research writing – I just somehow picked it up and produced a decent master’s dissertation in 1993. But then I left academia, only to return as a lecturer some years later.

In 2006, I started teaching a course on research writing and found it almost impossible to explain to struggling students how to overcome their main writing challenges. Instead, I resorted to the things that were easy to teach: the philosophy of research; lists of dos and don’ts; argumentative fallacies; details of different designs; and the rather superficial guidance found in most textbooks on how to do research writing. The result: I taught, but my students didn’t learn.

Out of frustration, I decided to take on the task of unpacking the tacit understanding of research writing that good researchers have. I changed my manner of teaching, and, eventually, by 2014, I had put most of the pieces together. The outcome was successful in that, today, the worst writing in my classes is better than the best writing in my class of 2006. The proposed book is derived from these ideas.

I could not have done it alone. Thank you to all my students who put up with my teaching and gave me the space to try out my ideas. I also want to thank my friend Jacqueline Jones, who helped me shape the initial version of this book into something much better. As a critical reader and friendly copy-editor her assistance was invaluable. She can be contacted at jacquelinejones94@gmail.com. Most importantly, I want to especially thank Dr Pia Lamberti from the University of Johannesburg who introduced me to the world of argument: many of the ideas in this book were inspired by working with her and learning from her.

Preface

About this book

Research writing is difficult. At first, it is overwhelming, complex and full of strange conventions that appear to be without logic. Paradoxically, once one accepts this, then like most things in life, it somehow becomes easier, as psychologist Scott Peck would have agreed. In fact, expecting research writing to be easy is the fastest path to frustration. Do not be fooled if this book makes research writing appear easy – it is, undeniably, difficult, but this book should help make the difficulties manageable.

The approach of this book

This book guides novice researchers through the major challenges of research writing. My own experience is that when learning a new skill, it is counterproductive to follow rules and prescriptions – there are just too many to keep in mind, let alone apply and knowing when and where to do so. I think that it is far better to identify a mindset that captures the spirit of all the prescriptions: a big idea into which everything fits. Once one grasps this overarching concept, it becomes far easier to master the details.

In research writing, this mindset about which I spoke is one of argumentative orientation. This is because research writing in the social sciences is essentially argumentative writing from start to end. It is the one differentiator that all research writing needs to have. Indeed, it is the ‘x-factor’ that readers expect, yet few people can tell you how to employ and develop it.

The approach of this book is to reduce the cognitive overload many novices, at first, experience when exposed to research writing and its apparently strange ways. This book helps by doing the following:

- *Keeping it short.* In any system of knowledge, only a fraction of the content enables most of the understanding. This book is therefore shorter than many books on research writing since it eliminates what would be *interesting* to the experienced researcher and includes only what would be *critical* knowledge to a novice. In fact, it is a basic pedagogical principle that one should (initially) abandon complexity in order to gain an

understanding of the critical ideas. This best prepares novices for building a richer understanding later.

- *Focusing on the bottlenecks.* Instead of explaining every single aspect of the research process, this book focuses only on the biggest challenges: especially those that professors find difficult to teach. These include: growing an interesting original contribution from within one's disciplinary knowledge; writing in one's own voice, while simultaneously drawing on the voices of others; knowing how to get started; writing a critical literature review that is not merely an exposition; and writing argumentatively about one's research design and methodology.
- *Tapping into the power of distributed cognition.* Cognitive scientists recognise that thinking is not something that is limited to what happens in the skull, but they acknowledge that we also think with the aid of that which lies outside it. Doing this makes our thinking more efficient since it shifts some of the cognitive load to other things or people. Some experts call this 'cognitive offloading' – for example, when we count on our fingers or use a dashboard. A good textbook should enable novices to similarly shift some of the initially excessive cognitive load onto the abstract objects in the book. The objects used for this purpose in this book include metaphors, diagrams, templates and step-wise processes, and they help to make the process of writing less overwhelming.

What this book is not

This book is not a book on the craft of writing as such or on producing edited writing; nor is it a book on research methods or a guide on how to start and finish a dissertation. Of course, a person who uses the proposed book, and applies it in his/her research, will write better, and write many parts of a dissertation in the process. This book will indeed complement the many existing excellent books in these areas, but it attempts to delve deeper. It will make it possible for novice researchers to gain more from these other books. This is because it focuses on the orientation and thinking approach that inevitably translates into better research writing.

This book puts the required skills within an argumentative context and shows how to apply these skills to overcome the major challenges while writing a thesis or paper. In the process, the reader will indeed complete many parts of a thesis, though not all parts, especially not the parts that are discipline-specific. In fact, the majority of the conventional books on research methodology tend to pay more attention to the discipline-specific parts of research, such as particular methodologies and referencing, and less to the overall approach of thinking and writing that forms the foundation of social science research.

The question of what quota of research writing is generic in relation to discipline-specific is debatable. My position is that both exist and should be

balanced. If there were no generic components, then different social science disciplines would find it almost impossible to communicate, and as such, generic components form the foundation from which the discipline-specific components are developed. Generic components often show the *form* – for example, that claims that *should be* argued with evidence – while discipline-specific components elaborate on the subject or *content*, that is, details like *what counts as* evidence in the discipline. Because they are so non-specific, generic components are often not taught. The reasons for this are: either because they are difficult to teach well due to their broad spectrum; or because it is hoped that novices will already have absorbed the skills through exposure to the discipline. Alas, the latter is rarely the case.

This dilemma regarding students not knowing, and not having been taught, *how* and *what* to write – and by extension, what should be generic and what specific – is evident across the board. To this day, across all social science disciplines, supervisors have remarkably similar complaints, which include: students producing list-like writing; their not knowing how to write in their own voice with the aid of authoritative voices; and the fact that they don't seem able to write critically about the literature or research designs.

Chapter overview and using the book

Since argumentation is all-pervading in research writing, it was difficult to separate the chapters completely. As a result, there is some overlap. All the techniques can be applied to the writing of both theses and papers, and to both qualitative and quantitative research. Novices are advised to read the chapters in sequence, whereas more experienced researchers could focus solely on the chapters relating to their challenges, while still bearing in mind the cross-references to other chapters. Using the textbook in teaching might require still a different sequence, as explained in the Instructor's Manual.

Chapter 1 Research as argument

Argumentation is the x-factor in research writing, so it is important that any novice researcher knows the language of argument. Drawing on the work of Stephen Toulmin, this chapter shows readers how argument permeates research writing; familiarises them with the terminology commonly used; and, with the aid of examples, classifies the different components of a single argument. It must be noted, however, that the Toulmin model is not a good guide to the *actual* practice of writing argumentatively. So, unlike many books, this book does not rely on this model as a template, but uses it only in the capacity of it being a way to create a shared language for recognising and evaluating arguments.

Readers who are already familiar with the basic terminology of argument can skip this chapter.

Chapter 2 Understanding the basic research skills in light of argument

This chapter explains the skills that novice researchers most likely have already learned, but with added insight and considerations. By looking at these skills through the lenses of argument, it helps the novice understand these skills more thoroughly and use them more effectively. These skills include: (1) strategies for searching for the sources that support argumentation; (2) how to read academic sources with your argument in mind; (3) summarising academic sources argumentatively; and (4) how to argue your point in your own words with the aid of academic sources.

Readers who are already familiar with these basic skills can skip this chapter; although, even for them, it may be useful, since this chapter reinterprets them as manifestations of argumentation.

Chapter 3 Finding an original contribution

This chapter explains what it means for a research contribution to be original, and offers systematic techniques for both ‘gap-spotting’ research – which extends previous studies – and for deriving interesting research contributions – which disrupts existing understanding. It simplifies and adapts, for the broader social sciences, notions and theories delineated in my book, *Creative Research in Economics*.

Readers who have already found an interesting contribution may skip this chapter.

Chapter 4 Writing argumentatively

Building on chapter 1, this chapter shows not only how to write arguments; but also how to present arguments that, themselves, contain smaller arguments. It develops the metaphor of argument as conversation and shows how to put it in practice. It does this through using: the ‘imaginary conversation technique’ for generating the smaller arguments of a complex argument; and the ‘issue-based talk show’ for developing the skill of writing in your own voice, especially while drawing on the ideas of others. In the process, it also demonstrates how to logically connect arguments and explains how to use templates, such as those of Graff and Birkenstein, to ensure a writing style that follows the common forms of argumentative writing.

This chapter is the foundation of all the chapters that follow, and thus should not be skipped.

Chapter 5 How to get started and write introductions

This chapter takes a researcher through a ten-step iterative procedure for turning a vague idea into a feasible research question and research design. This process gives very explicit guidance on how to start the research and

writing process. Most novices don't realise how much argumentation goes into this process, and, I believe, that this oversight lies at the root of the problems when writing proposals or introductory chapters.

This chapter is essentially aimed at readers who struggle to get started on their research projects. Those who feel stuck may also find it useful as I have found that researchers often can't move forward because they didn't start their research well.

Chapter 6 Writing an original and argumentative literature review

Literature reviews are some of the worst written parts of theses and papers. This chapter explains step-by-step how one can produce high quality literature reviews once one understands argumentative writing. It redefines the literature review as 'a series of connected arguments supporting the research question' and shows how to apply the techniques of chapter 4 to write literature reviews that contribute to the originality of the research.

This chapter is especially useful for readers who feel that their literature review is not well integrated with their thesis or feel that their review is artificial or list-like.

Chapter 7 Principles of writing argumentatively about your research design and methodology

While this is not a book about research methodology, this chapter focuses on the common arguments across all research designs. Novice researchers rarely recognise that they are making arguments here, and one finds much of the list-like, expository writing in this area. This chapter sensitises readers to the argumentative richness of writing about research design and methodology in general, though it should be supplemented by discipline-specific textbooks about how to execute particular research designs.

Similar to the previous chapter, readers who feel that their research design chapter is not well integrated, or is list-like, would benefit from this chapter.

Chapter 8 is a short conclusion.

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1 Research as argument

A student visited me once complaining that she was not getting anywhere in her research. “I read 100 articles and I am stuck”, she told me. A few weeks later, she came to me again saying: “I have now read 200 articles and I am even more stuck.” I only had an answer for her on the day when she told me she had read 400 articles, and, you guessed it, she was more ‘stuck’ than ever. It was only after that last visit that I discovered the solution. What advice would you have given her?

The first thing to understand is that ‘research writing’ is not ‘writing about research’: it wrongly creates the impression that you just read research and then write about it. On the contrary, you can only write if, based on the research, you have something to say *personally*, something to argue. The problem is that simply reading more will never get you there. On the contrary, you have to do it in reverse: *first* you find something to say and *then* you read about it. Once you have read some literature on the said topic, you may change what you want to say. Whatever the case may be, organising everything you do around your arguments makes reading and writing much easier.

It reminds me of a classic school physics experiment. You take a flask of cold water and keep on dissolving sugar in it until it will dilute no more. Then you heat up the solution and continue to dissolve sugar in it until it will dissolve no further. Finally, you let this super-saturated solution cool down and dip a thin wooden stick into it. Slowly, the solution will start to crystallise around it.

I draw the comparison to reading in that reading is like adding more and more sugar to the solution, but nothing happens except that it becomes completely saturated after a while. Things only start to happen when there is something, like the stick, around which the solution can crystallise. That thing (the proverbial stick), as you will see in every chapter of this book, is your argument. Without it, research will be overwhelming and your writing lifeless.

This is what happened to my student: there was nothing around which her research could crystallise. So what was my advice to her? ‘Find something to say first.’ Now, this is easier said than done, so I decided to write this book.

2 Research as argument

It explains how thinking about research in terms of argument helps you to overcome all the major research writing challenges.

Unlike the rest of this book, which aims to be practical, this chapter is more theoretical. It aims to make you familiar with the role of argument in research and the language of argument. Just reading this chapter will be enough. The details of how to do everything explained in this chapter will be covered in subsequent chapters, so don't be concerned if you don't fully understand everything – just try to get the big picture.

1 Academic research is not so strange

Research is difficult, but you already know more about it than you realise. It is a basic human survival skill. Research is nothing more than 'finding things out'. We do it every day: when watching news; finding out about people and their idiosyncrasies and complexities; checking out good movies to watch based on reviews and recommendations; determining the best way to do something, and so on.

There is just one big difference between everyday research and academic research: in everyday research, you try to find out stuff that other people already know; but, in academic research, you try and find out things that nobody knows yet. In academic research, you are the first one to find out something: you are a knowledge creator.

It is for this reason that to do academic research you will need to forget and unlearn many of the intellectual habits you learned in school and undergraduate studies up to now. What you had to do to pass in school and undergraduate studies is sometimes the exact opposite of what you need to do in order to excel in academic research.

For now, I will highlight just two of the habits you need to 'unlearn':

- In your studies thus far, you may have passed by finding out what is already known and then simply repeating it in your own words. This is not research. In fact, you will fail in research if you do this, because research is about finding out things that *nobody* knows at that moment in time.
- And since you will no longer simply *reproduce* knowledge, but rather *create* it, you will have something to say to those in your discipline. Instead of mainly repeating what others have said, you now have to find something you can say to them. Whereas, in the past, you studied other authors as an end in itself, other authors now only become a means to the end. The end is what you want to say, and you read other authors only so that they can help you say what *you* want to say. You will no longer simply repeat – you will argue.

Research therefore centres on finding out something that has not been said before – this is knowledge creation – and then saying it – that is, your argument. Very simply put, the process of research involves the following:

- First, you must have a reason to say something. To do this, you have to find a **problem** that others in your discipline either care about or *should* care about.
- Second, you need to find something to say about this problem: something that has not been said before and suggests that what we know about the problem is inadequate. When you find this, you have your **research question**.
- Then you need to get the help of other authors to improve your understanding of the problem, and also show how other authors have not been able to fully answer the question. This involves reading and reviewing **the literature**.
- Once you can see that the literature is insufficient, you need to tell us what you will do to find the answer to the question. This means you have to lay out your research strategy and how you implement it – this becomes your **research design and methodology**.
- Finally, once you have executed your research, you will have found the answer to your question: you will have found the one big thing you wanted to say. This you will do when you present your **results and conclusions**.

In this book, we will discuss all of this, and how maintaining an argumentative mindset makes it easier.

2 Argument in research

Argumentation is the x-factor of academic research. You cannot be a good researcher without making arguments. However, what is problematic is that few people can tell you how to write argumentatively. In fact, argument expert Richard Andrews calls it the ‘hidden criterion’ of research because, although it infuses all research, examiners rarely make it explicit that they are looking for argumentative writing or what exactly it is that they are looking for.

This is because, like anything indefinable (x-factor), it is very difficult to pin down exactly what ‘argument’ is. Argument is not just giving your opinion or criticising others: it is much more.

You really learn what it is by *doing* it, but, for now, I will attempt a working definition to get us started: argument is the act of making claims. Take the following two statements: ‘one plus one equals two’ and ‘Colombia is the best country in the world’. Only one of these statements is a claim. The first statement is not a claim, but a fact, and a fact is something that everyone agrees on as it has become an established ‘truth’. The second statement is a claim because it is a statement about which there are different perspectives.

If all authors had the same view about everything, there would be no reason to write anything, because one person could say everything there is to say. But in the social sciences, it does not work this way. There is a great deal to talk about. The talking takes place around points where there are different perspectives and where there is room for difference – and this is what the

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concept of argument is all about. Unless there is an interaction between different ideas, there is no argument.

In school and your undergraduate studies, you were probably taught not to argue: to pass, you simply had to repeat facts. This is probably a mindset that has become engrained in you and become a habit, which needs to be ‘unlearned’ and it won’t be easy. You cannot be a good researcher if you simply write down facts.

Argument is inevitable in the social sciences. In the social sciences, we study a reality that is not purely objective and thus not based only on ‘facts’. Instead, it is often created by people and their beliefs, and may even be influenced by *your* thoughts as well. Many elements of this reality, such as money, markets, organisations, poverty etc., would all disappear if humans were no longer on this planet, or if humans were differently conditioned. So, human beliefs and the concomitant systems influence most of the things we study. These beliefs are usually inconsistent and change constantly, which means that differences in perspectives will always be around, and almost everything we study will be open to argument.

Before getting into argumentative writing in detail, I first want to help you see what goes into one single argument and to become familiar with the language of argument. This will enable you to understand the jargon of argument and help you later when you have to put arguments together in your writing.

3 The language of argument

An argument is when you have a message – that is, you have something to say or a point to make – and because of your unique idea, it is then possible for there to be an interaction between different views around this message.

3.1 The Toulmin model

Let’s look at a single argument in more detail. Stephen Toulmin (1958) identified the following elements that should appear in any good argument (see Figure 1.1):

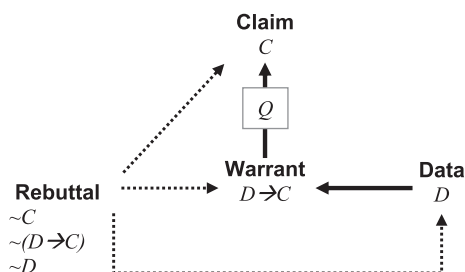


Figure 1.1 The Toulmin model

Source: Adapted from Mitroff & Mason (1980:337)

- There should be an explicit statement of the claim (C). A claim is any statement that allows for differences. The differences can be opposing views, but can also be different ways of interpreting the statement.
- The facts or evidence that support the claim are referred to as ‘data’ (D) by Toulmin. Data comes from the Latin word *datum*, which means ‘what is given’, so data is anything (quantitative or qualitative) that the discipline accepts as being correct (usually facts). Data may include statistics from credible sources, accepted theories, references to peer-reviewed studies, and more.
- Toulmin calls an explanation of why the data is relevant in support of the claim a ‘warrant’ (W). A warrant explains why the data leads us to the claim. So, if someone says your claim is ‘unwarranted’, they don’t mean you don’t have data – they mean that the data does not seem relevant enough to support your claim.
- All claims are open to different views, so no claim is certain. Therefore, claims must have qualifications (Q), meaning we deliberately weaken the claim so as to appear ‘confidently uncertain’.
- The conditions under which the argument might not be valid are the ‘counter argument’ or ‘rebuttal’ (R). The rebuttal can focus on challenging the claim, the data or the warrant.

Let’s take an example and transform it from a poor argument into a good argument. With each attempt, see if you can identify which element/s of argument it contains, and which elements are still missing. Here is the first attempt:

Universities do not prepare graduates to become employable.

This is a poor argument, but it has at least one element of an argument – the claim. This statement is a claim because it is possible for there to be more than one view about it: some people may say that universities do prepare graduates, while other may say that it depends on various factors (like the direction of study etc.). But a claim needs to be supported by evidence; so, to improve it, the next attempt should provide some evidence.

The youth unemployment rate is 55% (SSA, 2012). Universities do not prepare graduates to become employable.

Now evidence has been provided (in italics) with a credible reference from a governmental statistical agency. However, the data seems to be disconnected from the claim in that it is not clear how the evidence leads us to the claim. A ‘warrant’ is needed to explain how the data is connected to the claim.

The youth unemployment rate is 55% (SSA, 2012). *A quarter of these young people have studied towards a qualification at a university.* Universities do not prepare graduates to become employable.

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The phrase in *italics* links the data to the claim, because it refers to the data (a quarter of the 55%) and to the claim (they studied at a university). So, now it is clearer how the data leads to the claim. The argument becomes stronger.

But, at the moment, the argument is not credible enough yet, because the claim is too strong. The ‘claim’ assumes that all universities are the same, and therefore generalises, suggesting that they are not doing their job at all. What is missing is a qualification that makes the claim weaker, but the overall argument stronger. There are several ways to qualify. Let’s look at them, using the claim that, ‘Universities do not prepare their graduates’ as an example. What follows is adapted from the excellent book by Swales and Feak (2004).

i) Reduce the possibility of the claim being true

Here you create the impression that the truth of the claim is not absolute. By using certain words, the degree of certainty can change: for example: ‘Universities *might* not be preparing their graduates’ or ‘It *could be* the case that universities are not preparing their graduates’; or you could be more explicit, as seen below where some adjectives are placed in sentences, creating sentences that range from most certain to least certain:

It is certain thatuniversities do not prepare their graduates

It is probable that ... for employment.

It is possible that ...

It very unlikely that ...

The first allows no equivocation or argument or room for differences. This is not conducive to the practice of argumentative writing. The others allow room for a counter-argument, and therefore allow for argument.

ii) Create distance between you and the claim

When applying this, you acknowledge that the veracity of the claim depends not so much on you; but rather on other people and their research, or even your own point of view. For example:

It seems that ...

According to recent studiesuniversities do not prepare their

The limited data suggests that... graduates for employment.

iii) Make the claim more specific

By making parts of the claim more specific, you create the impression that the claim is not universally true. For example:

- Qualify the verb: in this case ‘prepare’. Universities do not *adequately* prepare graduates *for positions in the medical industry*.
- Qualify the noun: in this case, focus on either ‘graduates’ or ‘universities’. *Most universities in European countries* do not prepare their graduates.
- Add exceptions such as: ‘Except for Engineering qualifications...’, ‘Besides the top 100 universities...’ or ‘Under certain conditions...’

Now let’s return to the example to see how the author qualifies the claim.

The youth unemployment rate is 55% (SSA, 2012). A quarter of these young people have studied towards a qualification at a university. So *it appears that* universities *play a role* in graduates not being employable.

Note the two phrases in italics that qualify the claim. Even though they seem to make the claim weaker, they, on the contrary, actually make the claim more difficult to attack because the author is saying it ‘appears’ that universities are failing, not that they *are* necessarily failing. And it is also more accurate because universities are not the only cause of lack of employability, although they could ‘play a role’.

At this point, the argument is much stronger, but it still presents only one view. What is missing is a rebuttal or counterargument. But you have to be careful how you employ a counterargument. The next attempt just adds a counterargument without doing anything with it. Notice how this harms the argument here:

The youth unemployment rate is 55% (SSA, 2012). A quarter of these young people have studied towards a qualification at a university. So it appears that universities play a role in graduates not being employable. *However, Smith (2011) argues that individual characteristics are more important when it comes to determining employability.*

The last sentence ‘throws in’ a counterargument, and the author expects the readers to decide for themselves whether they agree or not. That does not work because the point of making an argument is to *convince* the reader of your point. So, if you introduce a counterargument, you have to show that what was stated was flawed and thus use the counterargument to refine what you are saying. The author tries to use the counterargument in the next attempt. Notice how it improves the argument.

The youth unemployment rate is 55% (SSA, 2012). A quarter of these young people have studied towards a qualification at a university. So it appears that universities play a role in graduates not being employable. However, Smith (2011) argues that individual characteristics are more

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important when it comes to determining employability. *While this is true, Jones (2010) showed that universities can change individual characteristics. This suggests that universities are indeed failing their graduates because instead of developing them as individuals, they focus mainly on conveying generic subject knowledge.*

In the italicised text, the author shows that even though individuals' own abilities influence their employability, it does not mean that only individual abilities matter. This is because universities can, in fact, improve individuals' abilities, which suggests that universities focus on the wrong thing. What the author did is agree with the counterargument, but then turned it around to make it support the original claim. In a way, you do to the counterargument what you did to the data: you connect it to your claim, that is, warrant it. And by refuting a counterargument, you indirectly strengthen your own argument.

3.2 The importance of counterarguments

Some people think that counterarguments are unnecessary, but they actually make your claims stronger and clearer if you use them correctly. To make your argument clearer and stronger, you should introduce the arguments of those who have different views than you, and then critique them.

A counterargument can focus on any of the following:

- The claim, by showing it is flawed;
- The qualifier, asserting that the claim is too strong or too weak;
- The warrant, showing that the author draws the wrong conclusions from the evidence;
- The data, explaining that the author uses erroneous or inappropriate data;
- The method, suggesting that the method by which the author generated the data is wrong or inappropriate.

You can criticise those with whom you disagree in your own voice, but this may not come across as credible. It is more credible if you use other authors to criticise those you disagree with in order to substantiate your argument.

As an example, I am going to use an extract from a controversial thesis by one of my students, Anthea (Paelo, 2015), for the purpose of illustrating how one handles counterarguments to strengthen one's own view. She wanted to argue that ethnicity plays an important role in influencing career choices. Notice below that she raises a counterargument – which is focused on criticising the 'method' – and then in turn shows that the 'claim' of the counterargument is wrong. Also, notice how she uses other authors to refute the counterargument, rather than doing so directly.

Hofstede (2000) found that different ethnicities have particular cultural traits ... (Here she elaborated by providing specific evidence – not shown).

However, his theory was not without controversy and a number of studies questioned it (McSweeney, 2002; Baskerville, 2003; Fang, 2003). McSweeney (2002) appears to have provided the strongest criticism ... McSweeney's criticism of the method of study arose out of the fact that in some countries Hofstede made use of sample populations that were too small ... While McSweeney's criticism that the sample size Hofstede used was too small is valid, it does not negate the final results concerning the dimensions found. Van Oudenhoven (2001) replicated Hofstede's results using a different and larger sample ... The fact that the dimensions remained true 20 years later among varied samples offers strong evidence that people from different nationalities fall into particular dimensions showing cultural traits.

The criticism of Hofstede's model has failed to discredit his theory; in fact, it has gained increasing support. His book has been cited in more than 26,000 publications since its publication. The general acceptance of Hofstede's model and its consistent replication (see Vaara, Sarala & Stahl, 2012 for a recent example) offers strong support for the claim that different ethnicities and races possess different cultural traits.

Anthea could simply have argued the one side: maybe just writing the first paragraph and leaving out the rest. While this would be fine in some cases, it is very dangerous not to engage the counterarguments if the issue is critical to your overall argument. The issue of whether ethnic groups have cultural traits was critical to her research, but also very controversial. Pretending that there are no alternative views would have left her claim open to severe criticism and made her look as if she was trying to hide something. By anticipating the criticism and addressing it at the outset, her argument became stronger and also more interesting.

In some ways, counterarguments can do more to strengthen your claim than data can. The only way to show that you are the strongest is to defeat your most formidable prospective opponent. Similarly, if you can refute the strongest counterargument to your claim, your claim immediately appears much more robust. Counterarguments also clarify your argument since it allows your readers to see the contrast between what you see as true and false. We only recognise what is false if we know what is true. Knowing what you see as false makes it easier for the reader to understand what is true.

4 Using the Toulmin model

The Toulmin model is not a good guide to actually writing argumentatively, but it does help you to understand the language we often use when talking about and evaluating arguments. We will return to it in more detail in

chapter 4 where it will be one of the tools I use to explain argumentative writing. It does have some uses, but it is fallible in terms of its efficacy in presenting arguments.

4.1 Awareness of argumentation

Earlier, I identified the parts of the research process: problem, question, literature, design and results. Later, I will discuss them in more detail. For now, I just want you to notice how the Toulmin model illuminates how each of these parts is an argument made up of a claim, data, warrant and sometimes counterarguments.

Problem: Argue that you have a reason to say something:

- Claim: There is a problem in the real world.
- Data: Discover and outline facts about the real world from credible sources.
- Warrant: Show how these facts lead to a problem for some group (explained in chapter 5).
- Counterargument: Refute those who might say that this problem does not exist in the real world or is not that urgent to resolve.

Question: Argue that what you have to say about this problem has not been said before:

- Claim: There is something about the real-world problem that we do not know as researchers.
- Data: Give references to previous studies about the real-world problem.
- Warrant: A systematic review of the studies that is organised so as to show that they are inadequate in some way (as explained in chapters 3 and 5).

Literature: Use this to argue that previous research supports the question:

- Claim: The literature shows that the question is valid and that well-designed research is likely to produce an answer to the question.
- Data: The collection of theories and empirical studies that is relevant to the question.
- Warrant: Shows how the literature speaks to different parts of the question (as explained in chapter 6).

Design and methodology: Argue that your research is likely to lead to an answer to the question:

- Claim: I have a good idea what is the best way to find the correct answer to the question.

- Data: All the possible designs, especially designs used or considered in similar studies.
- Warrant: Show that the design is appropriate given the question and available data (to be explained in chapter 7).
- Counterarguments: Refute those who would argue that other designs are more effective in terms of answering the research question.

Results and conclusions: Argue that the results you generated have answered the research question and have made a contribution to the literature compared to previous studies.

- Claim: The results answer the question.
- Data: The presentation of your results in various forms (e.g. text, diagrams, tables, and so on).
- Warrant: Show how the results, combined with previous literature, answer the question.
- Counterarguments: Acknowledge alternative interpretations of your results and explain why your interpretations are superior.

4.2 *Noticing your own voice*

Understanding the Toulmin model also helps you to see that, even though you are using other people's ideas, there are many places where your own voice, as writer, comes out. Again, I will show you how to do it in more detail in chapter 4. For now, the Toulmin model will help you to see that it is impossible to make arguments without your own voice coming out. It emerges in the following places:

- The claim: *You* are the one who chooses which claim to make, even though you have to support that claim with evidence drawn from other people's research.
- The data: *You* decide which data is best and which authors to use in support of *your* claim.
- The qualifier: Based on *your* understanding of the evidence, *you* decide how far to go in qualifying your claim.
- The warrant: A warrant interprets the data, and, as the author of your research, *you* are the one who explains what we should see in the data.
- The rebuttal: *You* pick the counterarguments that *you* think are the best, and *you* decide how to handle these counterarguments.

4.3 *Evaluation of arguments*

Finally, the model helps you to evaluate arguments – those you read and those you write.

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After writing, the model guides you in checking and evaluating your own arguments: Is there a clear claim? Is it supported by credible data? Is the data connected to the claim? Are other views acknowledged and are they treated in such a way that they clarify or strengthen the claim?

Critical reading requires that you know what to look for in other people's arguments. Toulmin's model helps you to identify the argument and its components and to see if anything is missing. Does the author make a claim? Does the author use credible data and interpret the data appropriately in support of the claim? Is there recognition that there may be differences and does the author engage with them?

4.4 *Writing single arguments*

While the Toulmin model is not effective in formulating the complex arguments in academic research, it may be useful to you when you have just one argument to make.

Follow these steps when you write a single argument:

- 1 Identify an issue where there are different views.
- 2 Summarise the different views fairly.
- 3 Decide who you agree with and how you will position yourself.
- 4 Determine what evidence you will call on to support your position and how the evidence warrants your position.
- 5 Use the arguments of the view you agree with to critique the arguments of the view you don't agree with (based on the claim, qualification, warrant, data and/or method).
- 6 With all this at hand, write up your argument by letting the authors interact with other and using appropriate argumentative language.

Steps 1–5 are the thinking steps, but may involve some rough written work as well. Step 6 is, however, the actual writing step.

When you write up the argument, there is no 'correct' order in which to place the claim (C), data (D), warrant (W), qualification (Q) and rebuttal (R); but here are some of the more common approaches:

- $C \rightarrow D \rightarrow W \rightarrow Q \rightarrow R \rightarrow Q$
- $D \rightarrow W \rightarrow C \rightarrow Q \rightarrow R \rightarrow C$
- $R \rightarrow C \rightarrow D \rightarrow W \rightarrow Q$

There are other approaches too. The only real requirement here is that, whatever approach you follow, the argument needs to be logical.

No essay, proposal or thesis consists of just one single argument. So, once you understand how to assemble a single argument, you need to generate smaller arguments and link them together to lead to a larger argument. This will be discussed in chapter 4. Before we get there, though, I need to show

you how argument helps you understand what you are really doing when preparing for research (chapter 2) and how to find something interesting to write about (chapter 3).

5 In summary

Herewith a summary of the elements of an argument:

Table 1.1 Summary

Claim <i>What you want to say</i>	When you write, you need to make a point and convince others of it (claim). It is only a claim if there can be different perspectives.
Data <i>What is taken for granted</i>	Evidence that is accepted by the discipline (quantitative or qualitative) that supports the claim.
Warrant <i>So what?</i>	Link data to the claim – show how and why the data should lead us to believe the claim.
Qualifier <i>Admit the possibility of different views</i>	All knowledge is problematic and open to questioning, and we express the degree of certainty with qualifiers.
Rebuttal (the counterargument) <i>Different views</i>	There may be counterarguments to the claim, qualifier, warrant or data used. Rebutting strong counterarguments makes your argument stronger, or conceding to rebuttals helps you to make your position more flexible.

6 References

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2 Understanding the basic research skills in the light of argument

This chapter explains four of the basic skills that novice researchers have most likely already learned, specifically those skills that are applied in preparation for the actual writing of the thesis. Instead of explaining them in detail, this chapter interprets these skills through the lenses of argument, specifically: (1) searching for sources; (2) reading sources; (3) summarising sources; and (4) writing about them in your own words. Keeping argumentation in mind at this early stage ensures that you will have something to say and that your research will already start to crystallise around clear arguments.

We start with the skill of searching for sources. If you don't know what you want to argue, you will quickly become overwhelmed by the vast quantity of research and most likely miss many important sources. Once you have found sources, you will have to read them. Again, there are probably too many sources to read, so if you don't know what you want to argue, you will not know what sources to look for and what to look for specifically in these sources. Once you have read your sources, it is quite useful to summarise them briefly; and, if you summarise them argumentatively, you will find that your writing will take far less time and become easier. While summarising, you will get practice in a skill that you will use extensively while writing the thesis: the skill of writing in your own words. As stated previously, you cannot write in your own words unless you actually have something to say.

1 How to search for relevant academic sources

To do academic research, you need to search academic databases such as Ebscohost, JSTOR, ScienceDirect or Google Scholar for sources that can help you. What you will find in these databases is academic research, which means you will find sources that are written by experts who draw on other expert sources, all of whom show their sources and follow systematic methods in generating new knowledge.

This research may be peer-reviewed or non-peer-reviewed. If it is peer-reviewed, it has been checked by other experts for quality and you will find

most of them published in academic journals or edited academic books. Academic research that is non-peer-reviewed will usually appear as working papers and research papers of universities, government institutions and reputable research institutions. When searching for academic sources, it is preferable to draw on peer-reviewed sources; but, because it can sometimes take years for research to appear in academic journals or edited books, it is acceptable to consult non-peer-reviewed academic sources for the most recent research. What you should avoid, if possible, are non-academic sources. Such sources are usually not the product of academic research and not peer-reviewed. Such sources come in the form of: websites of non-research institutions, blogs, magazines, newspapers, and most popular books.

Before searching, you need to know the kinds of things you should be looking for. Knowing the arguments you have to make at this point will help you know what to search for; especially arguments about your problem, your question and possible designs (as mentioned in chapter 1). So, at the start of a research project, you need to look for academic sources that help you with the following:

- **Problem:** Look for facts that show there is a problem and studies that explain the nature and extent of the problem. You will need this, because, later on, you have to convince your readers that you are researching a problem that exists, and thus warrants the attention of researchers.
- **Question:** Look for studies that have been done about the problem, so that you can establish if there are any areas where the studies are inadequate: specifically aspects on which researchers disagree and gaps in the current research. If you can find places where the studies are wrong, conflicting or lacking in something; you will have found a subject that still needs to be researched, and where you can make a valued contribution.
- **Design:** Look at how previous researchers have approached similar research. This will help you to see what is involved in the research of your problem – for instance, the methods used, the data needed etc. Your observation of past studies, and the way the researchers went about doing their research, is critical in order for you to establish, firstly, whether it is feasible for you to conduct your research at all; and, secondly, it should give you some indication of which research methods have been effective and may therefore lend themselves favourably to yours. In other words, you will get ideas on what you will need to do in your own research in order for it to be empirically sound.

To search academic databases, you will need to use keywords. Academic databases look for sources that contain the keywords you enter. But here is a problem: these databases cannot ‘think’ for themselves; so, if you use the wrong keywords, you could miss many important sources.

You therefore need to spend time investigating, trying and finding the best keywords.

The main sources of ‘good’ keywords are the words (usually nouns) found in the description of your problem or in the statement of your question. The problem is, due to individuality being an inherent human trait, you will not always use the same words as other authors when you describe your problem or question, so you should also look for alternative keywords. The easiest way to do this is to find common synonyms for the keywords you have entered at this point. One can find synonyms by consulting a thesaurus or a dictionary. An even more effective way of doing this is to take an article or book pertaining to your area of research and then to go to its list of references. Look at the titles of the articles that appear in the list of references, and there you will find many words commonly used in research on your topic. It is important to spend some time finding these keywords, because, if you use the ‘wrong’ words, you may miss many important sources in your searches.

Let’s take an example. Suppose my problem is that many companies spend lots of money on their brands in the hope that this will increase the value of their brands, and, in this way, improve their sales. However, it is not clear from the research whether a higher value of a company’s brands always improves its profitability. So the question that could follow from this is: What is the relationship between brand valuation and company performance?

From the question, the keywords that stand out are ‘brand valuation’ and ‘company performance’. Looking deeper into the description of the problem, alternative keywords may be ‘sales’ and ‘profitability’ as indicators of the degree to which ‘company performance’ is measured. Looking at marketing textbooks or the titles of related articles, one may find that ‘brand valuation’ could be assessed by ‘brand equity’ or ‘brand recognition’.

Once you have a list of keywords, you enter them into the search box of the database you are using, but you need to know how to combine the keywords, using the following main operators:

Table 2.1 Search operators

<i>Operator</i>	<i>Main purpose</i>	<i>Effect on number of results</i>	<i>Example</i>
AND	Look for combined ideas	Reduces	brand AND profitability
“...”	Look for whole phrases	Reduces	“brand valuation”
OR	Look for synonyms	Increases	(sales OR profitability)
*	Look for word variations	Increases	profit*
NOT	Exclude words	Reduces	brand NOT cows

You will use the AND operator when you have different ideas that all need to appear in the sources you are looking for, as shown in the table above. With this operator, all the words will appear in the same source, but not usually together as a phrase.

However sometimes you need a phrase. Double quotation marks are necessary if you need to look for a phrase. If for example, you wrote *brand valuation* just like that, the database will treat it as if you entered: *brand AND valuation*. Thus, you will then get results where the word *brand* and the word *valuation* appeared anywhere in the same document, but not necessarily together as a unit. So, if you want to search for the phrase, you need to signal this by putting the phrase in double quotation marks like this: “*brand valuation*”.

If there are synonyms for the same word or phrase, you can search for all of them in one go by combining them with OR. But whenever you use OR, remember to enclose all the keywords combined in this way with brackets, as shown in the example for the operator OR in the table above.

Sometimes the same word has many variations with the same root, and, to capture all of those variations, you can use an asterisk (*). For example, with *profit** you will find documents containing the root word and suffixes thereof, for example: *profit*, *profits*, *profitable*, *profitably* and *profitability*.

And finally, sometimes you get many results that you don’t want. For example, the keyword *brand* may bring up results pertaining to the branding of cows, which is irrelevant to you, and so they can be excluded by adding *NOT cows* in the search box. In the box below it shows that you can exclude more than one idea at a time by using the OR operator. Notice how it excludes the research about a particular country by using the phrase “*United States*” and its acronym *USA*.

The real power of these operators lie in the way you combine them, or, as you will type it in the search box, for example:

(“brand valuation” OR “brand equity” OR “brand recognition”)
 AND
 (“company performance” OR profit* OR sales)
 NOT
 (cows OR USA OR “United States”)

You should not see keyword-searching as something you do once. Throughout your reading you will discover new keywords, or think of new combinations of keywords. Keyword searching is an iterative process and you will go through several rounds of searches. This is because it is unlikely that you will find all the important academic sources after just a couple of searches.

Also, what is likely to happen is that your arguments will change as a result of your reading. As you read, you may realise that: there are more

serious real-world problems that need to be solved; your question has already been answered; or that there are better designs to use. If you have really been rigorous in your searches and read multiple sources, this is to be expected, and actually desirable. If it happens to you, take it as a positive sign rather than making the false assumption that your research has failed. It means that you have the open-mindedness needed to succeed in research. Indeed, it means you will probably go through several rounds of keyword searches, but as long as you keep focused, knowing why you are doing it and what you are looking for; the process will be manageable, and promises a fruitful outcome.

2 How to read

Once you have found enough sources, it is time to read them in order to find out how they can help *you* say what *you* want to say. Some sources will help you argue the nature of the problem; some help you make the case that the problem is serious; some help you to show where the current research is inadequate; and others help you understand the best way to conduct the research. Whatever sources you decide to read, they need to help you in some way to say what you want to say. This implies that you need to know *why* you are reading a source before you start reading it. It also suggests that you do not need to read every source from beginning to end: you simply need to find those parts in the sources that serve your purpose. Once you find the parts that pertain to your research, you need not read more.

What I present here is a reading technique, called the ‘organic study method’, developed by Tony Buzan, the inventor of mind maps, which will help you read academic sources much faster. It involves reading in layers: almost like peeling an onion. But first you need to get rid of the idea that you need to read everything. In undergraduate studies, this may have been the key to passing courses, but, in research, there is always too much to read and you need to find what you are looking for as quickly as possible.

The steps to follow that will enable you to read more efficiently are:

- 1 **Purpose:** Spend a minute to figure out *why* you want to read the source, and *what* you hope to find in it.
- 2 **Overview:** Page through the source, looking at anything that catches your attention, be it: headings, diagrams, tables, the table of contents, illustrations, capitalised words, the glossary, the index, photographs, italics, graphs, statistics, etc. Get a sense of what the source is all about. If you find what you are looking for, make notes and move on to the next source – don’t read any further. If not, move on to the next step.
- 3 **Review:** Read the beginning and end of the source, that is, the first and last chapters (in the case of a book); and the abstract, introduction and conclusion (in the instance of an article). At this point, you should have a good sense of where you might find useful information: mark these

areas and make notes. If you are still uncertain of what is relevant, read the beginning and end of each chapter (in the case of books) or the beginning and end of each section (with regard to articles).

Don't spend time on the difficult parts. This is not the correct time to do so – leave them for the next step. This is because you need to keep your focus without informational overload. Often, you may find that they are unrelated to your study. Sometimes they may be relevant, but this is best determined after you understand the concepts better. If, by the end of this step, you find what you are looking for, make notes and move on to the next source: I reiterate, don't read any further.

- 4 **Inview:** If you still have not found what you are looking for by this stage, you will at least have a much better idea of which chapters or sections you may find it in. Focus on these areas, especially those areas you marked in the previous step, by reading them in detail and going back to the difficult sections that still seem relevant to your research. Make notes on how the source is relevant to your research and how exactly it will help you.

On the surface, this process might *appear* to take longer, but it actually goes much faster than normal reading. This is for two reasons: (1) since you focus on first getting a sense of the bigger message of each source, you will remember the details that you read; and this saves you a great deal of time later; and (2) you will rarely need to read the whole source, because you are only looking for what is relevant to your research.

There is much out there about speed reading that I consider erroneous and find that literature on this is often aimed at taking advantage of desperate students. One of the more rational voices about effective reading is that of Scott Young, a prolific writer and autodidact who completed the four-year MIT Computer Science degree in less than one year. I recommend reading his blog post: *I was wrong about speed reading: Here are the facts*. His four tips on reading are very similar to the organic study method explained here. One of them is: "Know what you want before you read it", in other words: if you know what you want to argue, you'll read faster.

3 How to summarise

Above, I mentioned taking notes while reading, and this involves a specific kind of summarising. It is different from the kind of summarising you did at school. When you studied at school, you only needed to repeat what others said. You summarised to produce a shorter version of what you needed to reproduce in an exam. There was good reason for this: contracting a lengthy piece made your studying more efficient and less time-consuming. I call this 'descriptive summarising'.

A good descriptive summary of an academic source would state, in your own words, its main message in one sentence. This is followed by a few

sentences that support the main message. These should be based on the ideas in the article and also be in your own words. The summary should be unbiased: in other words, after reading the summary, it should not be possible for me to determine whether you agree or disagree with the author. You should not introduce any other ideas outside the article into a descriptive summary.

It is exactly this lack of criticality or connection to other ideas that makes a descriptive summary less useful for research. It also prevents you from making the essential mind shift that is necessary for good writing. Descriptive summarising may wrongly lead you to believe that research is merely about writing other authors' ideas in your own words.

In fact, most students think that, in research, they are supposed to say what others are saying, with the only difference being that this should be relayed in their own words. This is not only counter-productive, but also very difficult, because published authors usually write their *own* ideas so much better than anyone else can. In contrast, in research, you are supposed to say what *you* want to say in *your* own words, but with the support of what others have written. Think like this, and writing in your own words will become much easier.

To help you make the mind shift from a student who simply reproduces knowledge to a researcher who creates knowledge instead, you need to summarise academic sources differently from now on. I call this new way of summarising 'interpretive summarising' because it involves *interpreting* each source through the lens of your own argument, rather than simply providing a shorter version of it. It is the kind of summarising that you need to do when you are trying to say something through your writing.

Before reading anything, you should always ask yourself the following: Why am I reading this? What is my purpose? What am I hoping to get out of it? When you read, you will therefore look for useful ideas that you can use to strengthen or improve what you are trying to say. Obviously, not everything in a source will be useful to you. Some may contain only one useful sentence, while others may contain several useful paragraphs. But no one person will ever have tried to say exactly what you are trying to say, so you will not find a single source that is 100% useful to you.

This is why you cannot rely on the descriptive summary of a source alone. The descriptive summary of an article, usually the abstract, captures what the author thought important. But, because you are trying to say something *different* from the author, you will also find other ideas useful and will want to highlight those ideas in your own summary. The ideas you will highlight are those that are relevant to your specific purpose. The ideas you find useful in an article may therefore be completely different from those that the author of that article highlights in his/her abstract.

An interpretive summary is a paragraph or two which summarises why a particular source is useful to your *own* research. And, because different

people have different questions they are trying to answer, each one will therefore produce a different interpretive summary of the same source.

Here is a basic procedure for producing a good interpretive summary:

- 1 Read the text with your purpose in mind, and note the part or parts that may be relevant to your research.
- 2 Write down, in a single sentence, how the source – book, chapter or article – is useful to your research.
- 3 Expand, in your own words, why you believe the source is useful by referring to relevant parts of the source. You may also indicate how this source relates to other sources you have read.
- 4 File the summary for later because it may be some time before you need the source again, and having the summary available then will save you from having to read the whole thing again.

As an exercise, and an example, read the following passage:

Crime is probably the greatest concern in South Africa, and has been the subject of many political controversies. For politicians to address the issue, we need a better understanding of the causes of crime. Some commentators have suggested that the country's high unemployment is a major cause of crime. Figure 2.1 (see page 22) depicts the unemployment rate in each province with its corresponding contact crime rate.

It reveals a negative relationship at a given time between unemployment and contact crime. This finding is surprising, but exploring the meaning of the terms might shed some light on it. Unemployment is a measure of economic activity, and it is evident that provinces with the lowest unemployment rates are also those that are known to experience the highest economic activity. So, it might be that crime is not 'pushed' by unemployment, but rather 'pulled' by opportunity. One could therefore argue that policing should be concentrated where the opportunities for crime are the greatest. The finding should be treated with caution since inconsistent reporting rates between provinces reduce their reliability. Such an analysis can be improved by tracking the variables over time and taking factors such inequality and race relations into account.

Suppose there are three different people, with three different research questions. Now read the passage again, and see if you can provide an interpretive summary for each one based on their different questions:

- Researcher 1: Does poverty cause crime?
- Researcher 2: Will the severity of punishment for crimes reduce the crime rate?
- Researcher 3: Should spending on police services be increased?

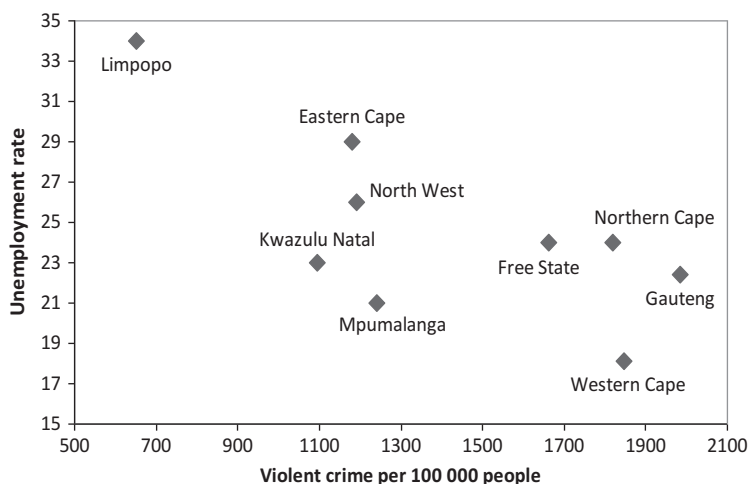


Figure 2.1 Crime and unemployment

The passage above does not directly answer any of these questions, but contains information that may be used to contribute to the answer. The three different researchers, with their different questions, will find different ideas useful, which will, in turn, lead to different interpretive summaries. For example:

- 1 The article states that there is a negative relationship between violent crime and unemployment. I know from other research that unemployment leads to poverty. Hence, based on this connection, this article offers evidence that there may also be a negative relationship between poverty and crime. In other words, in areas with high poverty rates, the crime rates may be lower.
- 2 From the article, it appears that crime is more likely to occur where there are opportunities for criminal activity. This suggests that criminals do not act out of desperation, but rather take action based on the probable benefits of the crime. Making punishment more severe could act as a deterrent, thereby helping to reduce the desirability of crime, and lowering the crime rate in this way.
- 3 The article provides evidence that crime predominantly takes place where there are opportunities, which are the areas with more businesses and therefore less unemployment. Other articles I have read would disagree, but this article therefore suggests that higher spending on police services is more necessary in wealthier areas.

If you do interpretive summarising, you will put yourself into a researcher mindset from the beginning, even while you read. You will always keep your own problem and question in mind and continually look for those ideas that: serve the purpose of clarifying your problem; help to answer your question; or get you to think about how to approach your research. The

Table 2.2 Types of summarising

<i>Student mindset</i>	<i>Researcher mindset</i>
Say what others have said, but in your own words	Say what you want to say (purpose) with the help of others (connection)
Descriptive summarising: creating a short version of others' messages	Interpretive summarising: summarising with your purpose in mind

next table tabulates the differences between descriptive summarising – that is, having the student mindset of simply reproducing ideas – and interpretative summarising, where one adopts the position of researcher and delves into more depth and individuality of thought.

Once you get into the habit of thinking about what you want to say, it becomes easier to write in your own words. This is because, as you may realise by now, writing in your own words is not just rewriting other people's ideas in different words; it is writing about how other people's ideas help you make your *own* ideas stronger and clearer.

4 How to write in your own words

We are usually told to write in our own words in order to avoid plagiarism. You would plagiarise if you pretended that someone else's ideas are your own. By now, you should realise that there is more to writing in your own words than simply trying to avoid plagiarism, but let us spend a little while on discussing how to avoid it.

Figure 2.2 is a simple typology that helps us understand what plagiarism is and what it is not. Whenever you get ideas from someone else, but don't acknowledge the source of the idea by means of a reference, then it is plagiarism because you give the false impression that those ideas are yours. This is shown in the first column of Figure 2.2. Copying and pasting, or making minor changes, without a reference, is plagiarism. But

	No reference	Reference
Copy or small changes		
Own words		

Figure 2.2 Plagiarism

even if you write completely in your own words about someone else's ideas, and you don't reference them, it is still plagiarism. Referencing involves providing an in-text reference when you write about the idea, and ensuring that the full details of the source appear in a list of references at the end.

You can make in-text references in the following different ways, that is when you refer to the author(s) within your text. This always involves providing the author (or authors') surname(s) and year of the publication. From now on, for ease of relaying the concept, I will refer to the author as singular, although it should be noted that if there are more, the same format applies when referencing several. There are two possible scenarios:

- If the author surname is not read as part of the sentence, put brackets around both the surname and year of publication. In such a case, you can either: (a) add the reference at the end of a sentence, for example: "Coaching is an important skill to learn to become an effective manager (Smith, 2011)"; or (b) insert the reference inside the sentence, for example: "Some authors (Smith, 2011) have argued that coaching is an important skill to learn to become an effective manager."
- If the author surname should be read as part of the sentence for the sentence to make sense, then you insert the reference inside the sentence, but now the brackets are only placed around the year of publication. For example: "According to Smith (2011), coaching is an important skill to learn to become an effective manager"; or "Smith (2011) stated that coaching is an important skill to learn to become an effective manager."

Note above that when the author's surname is embedded in the sentence, *only* the year is placed in brackets. When the author's surname is not read as part of the sentence, the whole reference is enclosed in parenthesis.

There should be a list of references at the end, in alphabetical order by author surname. Following from our example above, where we had an author named Smith with a publication dating from 2011, we should find the same surname and year in the list of references, for example:

Smith, A.F. (2011). The manager as coach. *Journal of Management Coaching*, 21(3):56–81.

Only sources referred to in the text should appear in the list of references. There are several referencing formats, for example, the Harvard and APA systems. These are used for several source forms. I'm going to simplify things and focus only on the three most common sources and the information you need to provide for each and in which order. Bear in mind that this differs from one referencing technique to another:

- Journal article:
Author's surname, Initials. Year. Article title. *Name of Journal*. Volume, issue, pages.
- Book:
Author's surname, Initials. Year. *Book Title*. City of publisher: Publisher.
- Web source:
Author's surname, Initials. Year. Document title. Web address. Date accessed.

Returning to Figure 2.2, there is one case in which it is acceptable to copy and paste someone's words, and that is when you treat it as a quotation. That means you need to enclose the words you have reproduced verbatim in quotation marks, and add an in-text reference that includes the page number. Again, there are two scenarios, for example:

- When the author's surname is read as part of the sentence: Jones (2015:54) argued that, "policing should be concentrated where the opportunities for crime are the greatest."
- When the author's surname is not read as part of the sentence: However, it can also be argued that, "policing should be concentrated where the opportunities for crime are the greatest" (Jones, 2015:54).

Despite this, as Figure 2.2 indicates, it may still be plagiarism when you quote. This happens when you quote excessively. When you read academic sources, you will note that authors use direct quotations very rarely. So, unless it is absolutely necessary that you use the exact words of another author, rather use your own words. If you quote too much, you are no longer providing your own work.

It is not always easy to know when to reference, but here are some guidelines:

- If you are using or writing about an idea that is not common knowledge, identify its source. A simple rule of thumb is to regard everything that is taught in your discipline up to first-year university level as common knowledge. Thereafter, it is not always clear what common knowledge is or not; so, when in doubt, reference.
- Whenever you quote directly, reference the quotation accordingly, as already explained above.
- When you are presenting arguments that other people may call into question, use references to show that there are others with the same ideas because this gives credibility to your argument. The same applies when you explain the research methods you propose to employ.
- Unlike verbal conversations, in writing, you cannot see or hear the authors. Thus, referencing can help in these instances since it shows who is talking. A reference like 'Smith (2011)' shows that the ideas are derived from Smith. In the same way, 'Jones (2016)' signals that Jones is

talking. Where there is no reference, this usually means you are writing about topics that are common knowledge or that you are providing your own ideas.

Referencing is an important part of writing in your own words. As suggested above, it is not only about avoiding plagiarism; it also helps to support and clarify what it is you are arguing.

Writing in your own words does not involve rewriting other people's ideas sentence-by-sentence in different words. That would simply be reproducing their ideas, rather than writing about your own ideas. Instead, writing in your own words means that you use other authors to talk on your behalf about your *own* ideas – just like we do with interpretive summarising. If you think like that, you will automatically write in your own words when using other authors' ideas.

To write about someone else's ideas in your own words, you: firstly need to know what you want to say; then identify the relevant idea in another author's work that will help you make your point; and, finally, write about how this author's idea clarifies or supports what you are trying to say. In other words, it is not their argument just re-phrased, but yours, and they substantiate what you are saying.

Suppose your research question is: 'Does higher education reduce unemployment?' Based on your research, you are convinced that universities can help to reduce unemployment, so this will be the argument you will try to make. Let's say you found the following source, which we will call 'Kelly (2014)':

The youth unemployment rate is 55% (SSA, 2012). A quarter of these young people have studied towards a qualification at a higher education institution (HEI). So it appears that HEIs play a role in graduates not being employable. However, Smith (2011) argues that individual characteristics are more important when it comes to determining employability. While this is true, Jones (2010) showed that universities can change individual characteristics.

Looking at Kelly, you will establish that she provides an idea: that universities can influence the characteristics of students that make them more employable; which supports your argument. You can then take this idea of Kelly's (2014) and show how it helps you to make the argument that universities can reduce unemployment. You can do so by writing something like this:

Education policy is based on the assumption that education can help to reduce unemployment by improving the employability of the youth. Kelly (2014) questions this, but still argues that universities are able to influence the most important determinant of employability, which is individual characteristics.

Notice that you don't repeat Kelly's (2014) ideas in different words, but rather integrate Kelly's (2014) assertions with your own ideas, and this can only be done in your own words.

5 Conclusion

This was a very brief tour of the most basic skills you need to master in order to conduct academic research. While these skills are applied in preparation for the actual writing, you should return to them regularly in every part of the research writing process. For example, if you don't yet have a clear idea of what research contribution you want to make, you should have at least one potential interesting contribution after using the tools of chapter 3. You will then need to return to searching for, reading and summarising sources. Similarly, you might return to using these skills after chapters 5, 6 and 7. Chapter 5 will force you to clarify your ideas about your problem and question which will lead you to search for new sources or require you to read and summarise your existing sources with new eyes. The same will happen after chapter 6 helps you to discover new connections in the literature or after chapter 7 gets you to think more deeply about your research design.

3 Finding an original contribution

One of the defining features of a thesis or article is that it seeks to make an original contribution to a field of study. It is strange then that most research courses prefer to teach the details of executing research designs, as opposed to how to find the original contribution that inspires the actual research. This chapter challenges this formula, suggesting that a person's original contribution is the most important factor and should therefore be a central topic in research courses.

This chapter is a simplified version of my previous book *Creative Research in Economics* (Wentzel, 2017) and uses examples from various social science disciplines, not just economics. Consult it if you want to know more about the theory behind the techniques discussed here, or want to discover more techniques not discussed here, such as the 'cross of creative questioning'.

To make an original contribution, you need two things: a *reason* to be original and something to *be* original about. Finding a reason for originality is the easier and obvious starting point. It is best captured for me in the idea of 'calling' which is, as Frederick Buechner described it, the place where your passion and the world's needs come together. The Japanese call it *ikigai* or 'reason for being'.

1 Finding a reason to be original

One of the primary rules of research writing for me is that you must only write about things you care about. Most people think they will find this passion in a topic, but this is a mistake. Topics don't excite. It is easier to care and write about that which stimulates you. So, start by looking for a problem or concern that stirs your emotions – that makes you angry or curious or sparks your interest. Problems are more likely to do this than topics, which tend to be staid and neglect active engagement, for the most part.

In addition to being somewhat unyielding, topics are mostly very broad, so it is incredibly difficult to see what research they call for. By contrast, problems are more specific and require actions, like research, to solve them. Problems also naturally lead to questions. Unanswered questions, like unsolved riddles, will get other people interested. Since all research is about

finding answers to questions, identifying a problem makes it easier to find questions to which the world needs answers. It is therefore in problems that you are likely to find something that you care about and that the world also needs.

Finally, if you write about something about which you're passionate and that which excites you, your zeal is bound to be infectious. This means that you will get others to care as well, and thereby make a difference in the real world. Many people think research is an emotionless exercise, when, in fact, people do their best research when it arouses strong emotions inside them. When you find that which you are passionate about within, you have found the reason for your research, and this is what will get you through the difficult times.

You can find such problems by asking yourself the following: Are there any problems within your discipline that have always interested you? Any issues that make you angry or energise you? Which issues did you find most interesting in lectures? Which issues do you find yourself talking about often? Which ones do you love reading about?

Even though having a reason is what makes a research contribution worthwhile, it is not enough by itself. You need to give your problem or concern more substance and structure so that your research about it becomes publishable material, as explained in chapter 5. Also, you need to find something to say about this problem that is truly original, which is easier if you have found an area that is ripe for an original contribution, as explained in the rest of this chapter.

2 What does it mean to be original in social science research?

Sometimes, it is easy to feel that the experts in the field have made most of the contributions worth making in a discipline. That would be correct if there was nothing more to originality than finding something new. But originality is so much more than that. If you know what you are looking for and how to find it; then you realise that with each new article and thesis, your discipline creates more opportunities for really interesting research.

The naïve view of originality holds that we simply discover new ideas and add them to the literature. What this implies is that the current literature is essentially 'correct', and merely incomplete. This perspective is limiting in that it implies that researchers are simply expected to fill the remaining gaps. If you hold this view, it will be difficult to see what you can contribute, because over time, the gaps get smaller and smaller and less interesting.

Fortunately, this is definitely not the case in the social sciences. Social reality does not exist independently from humans, for example like fossils waiting to be discovered, because humans *create* this very reality, and thus are pro-actively involved and don't simply 'discover' or 'stumble across' a pre-existing truth. Because humans change faster than the research about social reality, there will always be areas where our understanding is no

longer adequate, and the existing literature insufficient, offering further opportunities for research. To complicate matters, social science researchers not only describe this reality, but through their work, participate in, question, and often transform this reality. As a result, research always plays catch-up to the existing reality, and thus, new contributions are always needed.

Collectively, we as researchers therefore don't merely discover new contributions, but we *create* them through the arguments in our research. However, this reality does not simply change by the actions of a single researcher. Social reality is shaped collectively, so it exists independently from any single researcher's actions. For research to have an impact it has to be accepted by other researchers, and hopefully, also by non-researchers.

There is a lot of persuasion needed when making an original contribution: a researcher cannot simply present facts, but also has to make a contribution possible by appealing to the discipline's values, methods and shared understanding of reality. A researcher may come across an opportunity for a contribution, but its originality will only be recognised if the researcher can convince others in the discipline that it is relevant to the discipline and it challenges the discipline's prior understanding.

Argumentation enters here in that you will need to: first, convince others that you have noticed an opportunity for a new contribution that is relevant or useful to the discipline; and, second, make the case that this contribution promises to alter or supplement current understanding and therefore offers progression. From this, one can construct a typology of new contributions that is inspired by Davis (1971), as seen in the next figure. In that typology, only one of the four kinds of new contributions is truly original: those that are labelled 'interesting'.

A new contribution has to be relevant, that is, it should be clear to other researchers that it has value to the discipline. But it is possible for

Not relevant	Irrelevant	Absurd
	Obvious	Interesting
	Not significant	Significant

Figure 3.1 Kinds of original contributions
Source: Adapted from Davis (1971)

contributions to be useful without significantly changing our understanding. This happens when a contribution fills a gap. An example of this is when one simply extends a previous study to another country or applies an existing method to more recent data. Such studies are insignificant and fall into the 'obvious' quadrant.

Contributions that are not useful will be seen as 'irrelevant', and, in some cases, 'absurd'. An absurd one is one that has no value because it goes too far in terms of radically subverting the understanding of the discipline. This might seem strange, but a new contribution that raises too many new questions or challenges the most basic assumptions of a discipline is one that pushes the limits too far, so that the fundamental tenets of a discipline are questioned. An example of this would be a study in biology that challenges evolution or denies the role of DNA.

For a contribution to be significant, it has to disrupt something in a discipline's understanding; but be careful not to push the boundaries so much that it brings into question the veracity of the entire discipline. This disruption is essential, because as already explained, research is always chasing reality, and without some disruption, our understanding would be stagnant.

But it should not be disruption for its own sake. It has to be one that brings about a useful change in our understanding, but is not so severe it is utterly disconnected from what we know. Karl Weick (1989:524) summarised it for me when he wrote: "The contribution of social science does not lie in validated knowledge, but rather in the suggestion of relationships and connections that had not previously been suspected, relationships that change actions and perspectives." Science Nobelist, Peter Medawar (1979: 94), went even further when he said that: "If research does not hold out the possibility of causing one to revise one's views, it is hard to see why it should be done at all."

In light of these revelations, researchers in search of an original contribution therefore have to strike a balance between relevance and significance. Those who do that will find themselves with 'interesting' contributions.

In a letter Einstein wrote to his friend, Solovine (Holton, 1979), he explained something similar. I adapted Einstein's diagram in the next figure. It shows that an original contribution has to start with a good grasp of a discipline's understanding and then break away from it (E-J-A). This is attained by: challenging an assumption; raising a new question, reformulating an old problem; or suggesting an idea that answers a question or solves a problem. However, that contribution will not be seen as original until it is connected to the previous understanding (A-S-E). This can be achieved by developing the idea using the methods of the discipline and then writing up the ideas.

One makes a contribution relevant by linking it to the shared understanding of the discipline (as contained in the existing literature). This requires argumentation, and will be explained in chapter 4. The more a contribution breaks away from the existing literature, the more significant

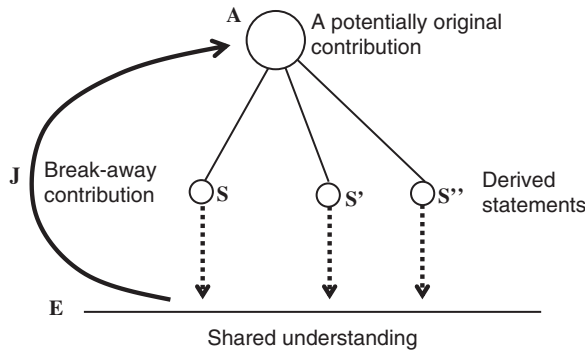


Figure 3.2 The process of making an original contribution
Source: Adapted from Holton (1979)

it is. More significance requires more argumentative effort when making it relevant.

Novice researchers at undergraduate, or possibly even master's level, should take care not to deviate too much from the body of existing literature. Initially, it may be more prudent not to break away too much and rather make contributions that are closer to being 'obvious'. However, experienced researchers, at doctoral level and beyond, should aim to break away more and thus make 'interesting' contributions. In the next section I present an easy way to quickly find more obvious contributions. In the sections thereafter I will show you techniques that make you think more deeply and so find the more interesting contributions.

3 A fast way to find some original contributions

One way to make an original contribution is to search for and find places where previous research is inadequate. In general, we find four kinds of inadequacies:

- **New:** Something that has never been researched, theoretically or empirically, before. This is rare in the social sciences, and such opportunities are usually not around for long.
- **Gap:** Something that has been researched before, but there are some areas that have been neglected. This is the most common and easiest approach: what Sandberg and Alvesson (2011:23) call 'gap-spotting'. Often, this involves extending previous research or existing methods to neglected areas – such as other regions, different groups or more recent time periods. Many papers and theses will point out such gaps when they discuss areas for future research.
- **Conflict:** Research about something that contains competing explanations or inconsistent findings. If there are different schools of thought in your discipline, these are easy to find.

- **Wrong:** Parts of the research (theoretical or empirical) may be incorrect or make wrong assumptions. Researchers with opposing views will often point out each others' mistakes or faults.

In any real-world problem, we may find five areas where a research contribution is possible:

- **Theory** is a generalised explanation of things we can observe in the real world and is not specific to a specific time or place. It could be that there is no theory to explain something or that some theory has not been tested before. Often, things that are confounding, and for which there are no known explanations, call for the creation of a new theory. An example of this is germ theory, which in the 1800s solved the puzzle of the transmission of disease.
- **Application** is the use of a theory in practice. It could be that a particular theory: has not been applied to certain places; has not been used to explain observations; or has a new application that has not been investigated before.
- **Method** is the technique by which results are generated. One may develop a new method, adapt or improve old methods, or apply methods to theories or data in ways not done before.
- **Prediction** is when you make an informed guess based on the theory or data about something that is not known. It could be a forecast or positing of the existence of unknown phenomena. The classic example of this is that of Mendelev, who predicted the existence of new chemical elements based on his periodic table. It could be that a theory makes predictions that nobody thought of before or can enable the construction of a model that can better anticipate future events.
- **Generalisation** is when you extend a theory or finding to a wider range of groups or phenomena than that which was originally intended. For example, geographically speaking, it might be that findings made in one South American country could be generalised to all countries on the continent; but that nobody ever tested this; or that a theory about families can be generalised to the whole economy, but that nobody ever considered this.

When we combine the kinds of inadequacy in the research with the possible areas of contribution, we get the 4×5 matrix of 20 kinds of research contributions, shown in Table 3.1 (see page 34).

This is an imperfect matrix, with much overlap between the cells, but it does give you a sense of how much room there is to contribute to the research. Some contributions may fit into one cell, but is quite common for studies to combine contributions from more than one. Each cell may trigger ideas for several contributions; for example, let's take four of the cells and see how they can do so:

Table 3.1 A matrix of contribution possibilities

	New	Gap	Conflict	Wrong
Theory				
Application				
Method				
Prediction				
Generalisation				

- **Application–Gap:** Are there any theories in your discipline that have not yet been applied to certain phenomena? Are there theories from other disciplines that could be applied to your discipline?
- **Method–Conflict:** Can any of the methods in your discipline be improved to help settle a disagreement amongst researchers? Are there any new methods available that could do the same?
- **Theory–Wrong:** Are there any theories in your discipline that are wrong or are becoming wrong due to changes in social reality? What are they and where do they go wrong?
- **Prediction–New:** Are there any theories or models, in your discipline or in others, that could make predictions about certain phenomena that are better or have not been made before?

Alvesson and Sandberg (2011) show that the vast majority of research tends to focus on filling gaps. The result is research that contributes less and less with each study as the gaps increase in number, but decrease in size and significance. The number of publications grows exponentially, but they become boring and predictable, to the extent that we become people who know more and more about less and less until we know everything about nothing.

In contrast, the techniques that follow will get you to delve more deeply into the problems and questions of your discipline and become aware of its assumptions. It aims to develop the mindset of *shoshin*: a Japanese word that describes the attitude of openness and awareness of preconceptions that one finds in true experts. The really interesting original contributions can only be made with this mindset – which enables you to recognise the often invisible constraints of a discipline’s assumptions, problems and questions, and then disrupt them.

4 Originality through the intellectual problems of the discipline

Without problems, there would be no way of knowing whether science is making progress. Creative scientists are motivated by the need to solve unsolved problems and this drives research as we move from old problems to new ones. Without unsolved problems, science would become sterile

since unsolved problems motivate scientists to generate and pursue innovative ideas.

The kinds of problems that innovative researchers are interested in are problems with our ideas. Such problems are not mere ‘puzzles’, as Thomas Kuhn (1970) called problems that keep our understanding unchanged. Rather, such problems are what Hattiangadi (1978) called ‘intellectual’ problems. These cannot be solved simply by gathering more data or filling more gaps. Intellectual problems call for new understanding, so finding them is a good place to search for interesting new contributions.

Intellectual problems don’t appear out of nowhere. They are the result of countless arguments that accumulated and connected to other arguments in the regulated chaos that characterises the development of scientific knowledge. Arguments get patched together over time to deal with a changing reality and to respond to new arguments that arise. Every argument is based on assumptions and when it becomes part of the conversation, it smuggles in those assumptions, and this is where the trouble starts. Many of those assumptions are inconsistent with each other, and, because researchers are not aware of all the assumptions, it takes time for the inconsistencies to become clear. They become clear only when they manifest as intellectual problems, and then it takes researchers with *shoshin*, who can see the assumptions, to make the kinds of interesting contributions that improve our understanding.

There is a more detailed discussion of this in Wentzel (2017), but suffice to say that we now realise that inconsistencies are an inevitable part of the development of knowledge, and they will never disappear. This is a good thing for us, as researchers, because it means a continuous flow of opportunities for originality: if we know how to capture these inconsistencies.

The secret lies in representing problems as logical conflicts. This is because we know from the study of classical logic that if there is even one small inconsistency in a knowledge system, then it is possible to prove anything – even things that are false or the exact opposite. If a discipline produces wrong theories or opposing views, like all disciplines do, it is usually a sign of some undetected inconsistency in its assumptions.

Representing problems as logical conflicts, besides offering a more fruitful perspective on problems, has the added advantage of making it easier to see the assumptions that give rise to the problem. Just being able to represent a problem differently can sometimes be an original contribution. In fact, some of the greatest breakthroughs in science came from reformulating an old problem. This is because new formulations change the theory employed, the questions asked, the assumptions made and the kind of answers that are considered.

4.1 Understanding intellectual problems as logical conflicts

Before I show you how you create a logical conflict for a problem, let me first help you to read it. The next figure shows an example of a simplified

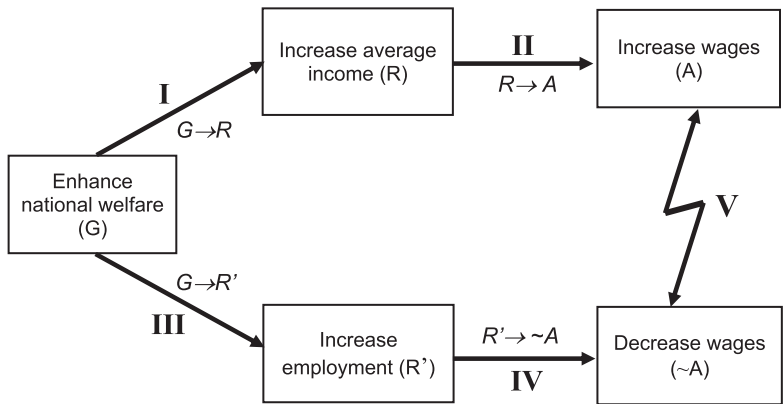


Figure 3.3 Example of a problem as a logical conflict

problem in economics, with respect to wages. This particular representation is taken from Goldratt (1994), except that the direction of the arrows is reversed, to suggest a logic of deliberate intention by the people involved in the problem.

From Figure 3.3, we can see two opposing views that end up in actions (A and $\sim A$) that conflict with each other. Both sides do, however, have a common goal (G), which is to enhance the welfare of the country. To achieve this common goal, we must meet two different needs or requirements (see R and R' above).

On the one hand, it is postulated that, if we want to enhance welfare, we must increase the average income level of the country's citizens. This gives us the arrow labelled 'I', which connects our common goal to one requirement ($G \rightarrow R$). On the other hand, if we want to enhance welfare; we must increase the employment of the country's citizens. This gives us arrow III, which connects our common goal to the other requirement ($G \rightarrow R'$).

The conflict arises because it is argued that, to increase people's income, we must pay them higher wages; but, to increase employment, we need to reduce wages. To achieve the first requirement (R), we need to take one action (A), and the two are connected by arrow II ($R \rightarrow A$). To achieve the second requirement (R') we need to take an action ($\sim A$) that conflicts with the first action; and the two are connected by arrow IV ($R' \rightarrow \sim A$). The ' $\sim$ ' signifies 'not', so $\sim A$ means 'not A' or 'the opposite of A'.

The argument to increase wages (A) is in conflict with that of decreasing wages ($\sim A$) and this is shown by the kinked arrow labelled 'V'. This tells us that, given the accepted knowledge of the discipline, it is not possible for A and $\sim A$ to occur together. But, since reasoning based on the current understanding of the discipline leads us to two opposing actions, it means that there is some inconsistency in the assumptions on which this reasoning is based.

Knowledge is derived from assumptions, thus, the collection of all the assumptions is a discipline forms its shared knowledge. If there is an inconsistency, we have an intellectual problem that brings the existing knowledge into doubt, because it should not happen that our knowledge leads us to a contradiction.

To find the cause(s) of the inconsistency, and the source of ideas for original contributions, we need to locate the assumptions and bring them to the surface. If you found yourself disagreeing with one side of the logical conflict, you have probably already noticed some of the assumptions. Look at the figure again: is there any place where you disagree with, or at least think, the reasoning is questionable? Can you pinpoint it?

This is what makes this way of representing problems so useful: it makes the reasoning underlying the problem visible. If you disagreed, you probably pointed at one of the arrows, and it is exactly here where you find the assumptions.

Every arrow is an argument and it is based on assumptions. For example, in arrow IV, what do we assume when we say that to increase employment, wages must fall? One assumption is that companies will only employ more people if they can pay them less. Under arrow II, what do we assume when we say the income levels will rise if we increase wages? The other assumption is that when companies pay their workers more, they will still employ the same number of workers than before.

Through this diagram, not only do we become aware of each side's assumptions, but the cause of the conflict also becomes apparent. In short: one assumption is that that companies pay attention to the wage, and therefore employ more people when the wage falls; while the other maintains that that they don't care about the wage, and employ the same number of people even when the wage rises. There is an obvious inconsistency.

Taking arrow III as an example, the simplest way to find the assumptions under each arrow is to ask ourselves: "What do we assume when we argue that to enhance national welfare we must increase employment?" Do this with every arrow, including arrow V, where you ask, "What do we assume when we argue that it is not possible to increase and decrease the wage at the same time?"

You will probably find it easier to find the assumptions underlying the arrows you disagree with. If you struggle, take an arrow and ask yourself: "Is it always true?" and, if not, "Why not?" You can probe further, asking "Is it true under all conditions? For all time? In all places?" For example, this questioning could lead you to realise that a company does not *always* employ more people when wages fall because it assumes at least three things: that the new workers will be as productive as the current ones; that there is a need for the company's products; and that the company is sufficiently profitable.

With some practice and sufficient knowledge of your discipline, it is usually easy to find 10 assumptions. With some effort, it is possible to find 20 to 30 that constitute the shared understanding of the discipline about a

problem. Among these, you find that there will be a few that are inconsistent with each other. You will find that this will usually be the case between the assumptions in arrows I and III and those under arrows II and IV. Each one of these assumptions and inconsistencies is a source of potentially interesting original contributions, as I will explain later.

With this example in mind, it will now be easier to understand the conflict diagram in its generic form, as given in the next figure. I show this generic form because I want to point out some common mistakes that people make when constructing their own conflict diagrams. We read this diagram from left to right as follows:

- To achieve G, we must have R, and to achieve R, we must do A.
- To achieve G, we must have R', and to achieve R', we must do $\sim A$.
- We find that A is in conflict with $\sim A$.

The common goal (G) has to be a goal that both sides want to achieve, so it usually has to be very broad in order to capture the needs of both sides. Some people make this goal too specific, but it is totally acceptable if expressed in vague terms, such as: 'welfare', 'happiness', 'success' or 'profitability'.

Many people mix up the requirements and the actions. A conflict diagram is constructed with a system (e.g. community, government, firm) in mind and the requirement boxes (R and R') have to be expressed as a need of this system or group. These needs are not actions or decisions, but the state that the system or group has to be in to make it possible to realise the goal. To meet these needs specific actions or decisions have to be taken by some group in the real-world, which are found in the last boxes (A and $\sim A$). It is not always necessary that A and $\sim A$ be polar opposites, but it is

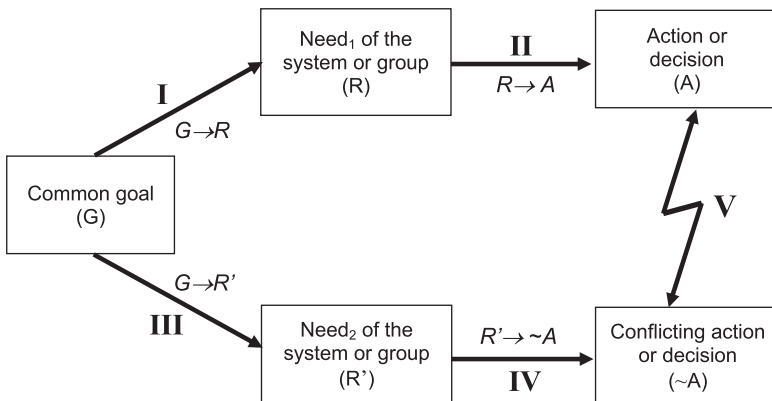


Figure 3.4 Generic logical conflict
Source: Adapted from Goldratt (1994)

critical that they are perceived by the discipline as being in conflict, or at least inconsistent.

Another common mistake is that people feel they have to be unbiased and agree with both sides of their conflict diagram. This is neither necessary nor desirable. You will most likely find yourself disagreeing with one side, and that is fine. At this point, you are simply trying to set up a conflict so that you can derive the assumptions and questions that will lead to an original contribution. If you put down only what you agree with you might never find a conflict, and by extension, never formulate an argument that will lead to an original contribution.

4.2 Constructing a conflict diagram from actions or decisions

To find original contributions you should know how to construct a conflict diagram for your chosen problems. There are several different ways to do this, and this sub-section and the next one will demonstrate two possible methods, as explained in Wentzel (2017:50–55)

One limitation to be aware of is that conflict diagrams are best suited to real world problems, meaning problems affecting people in the physical world. For purely theoretical problems, it is better to use the techniques for questions (as will be explained in the next section) or the ‘cross of creative questioning’ developed by Wentzel (2017).

Sometimes, we can see problems in policy arenas; especially where stakeholders are caught between two actions or decisions. Where groups encounter trade-offs, dilemmas or problems, you’ll encounter phrases like: ‘On the one hand we should ... but on the other hand, we should...’ Thus, when we see these deliberations, it is easy to see these conflicting actions or decisions. In such a case, we start from the action boxes and work our way backwards.

Let’s take a simple real-world problem from disciplines dealing with sustainable development. Often, when opportunities arise for communities to lift themselves out of poverty (such as in mining or tourism), there is a conflict between those who want to allow people to use the resources of the community as they see fit; and those who argue for some regulation of how the resources are used. So, let’s construct the diagram step-by-step.

Step 1: Since we have identified the conflicting actions/decisions, we start by inserting them into the appropriate boxes, as shown in Figure 3.5 (see page 40). It does not matter which one goes in which box, as long as they are perceived as being in conflict. The result so far is shown in the next figure.

Step 2: Derive the needs of the system (in this case the communities struggling with this issue) from the actions. These should *not* be expressed as actions or decisions. To find these needs ask: “What needs of the system will be satisfied by this action or decision?” You can also just ask: “Why does the system need this action or decision?”

So, in this case, we ask: “What do some people believe will be achieved if the community or firms outside the community are allowed to use its

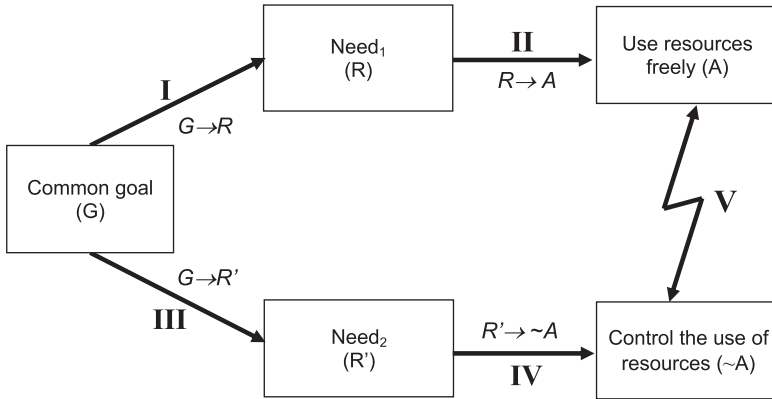


Figure 3.5 Constructing the diagram from actions: step 1

resources freely.” One could argue, for instance, that this will enable the community to create the maximum number of jobs in the shortest time. Insert this deduction into the R-box. Then, in accordance with the contrary viewpoint, we could ask: “What is the reason for controlling and regulating how the community’s resources are used?” One may argue that this will ensure that the community’s resources are used sustainably and not depleted quickly. Insert this inference into the R’-box. You can see this in the following figure.

Step 3: Ask, “What do both sides hope to achieve by meeting their respective needs?” They may be in conflict, but both are trying to achieve the same broad objective: to sustain natural resources, while uplifting the community. In this case, one may argue that both employment and sustainability will lead to the development of the community. Insert this into the G-box. The result is also seen in the next figure.

Step 4: Finally, we test the logical flow. This is an important step, so you should spend a significant amount of time on this. It is extremely rare that you will get the logical flow right the first time round. If the logical flow does not feel right, continue revising it until it does.

To test the logical flow, it is a good idea to read the diagram from left to right and in reverse. When reading from left to right, read as follows: ‘If we want [G], then we must [R], and if we want [R] we have to [A].’ This uses the logic of obligation, which I prefer because it recognises that the problem arises from human action. To double check, I recommend also reading it as Goldratt intended, from right to left. When reading from right to left, read as follows: ‘If we [A] it will lead to [R], and if [R] it will lead to [A].’ Here, we use the classical logic of causality. If, in both cases, the reading is grammatical and reasonable, according to some view in the discipline (even if you don’t agree with it) and the actions conflict in some way; then the logical flow is acceptable.

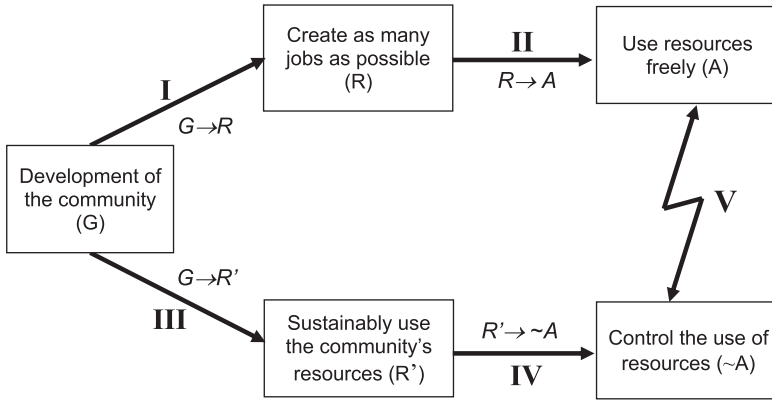


Figure 3.6 Constructing the diagram from actions: steps 2 and 3

Figure 3.6 achieves this. Reading it from left to right makes sense: If we want development of the community, then we must create as many jobs as possible; and, if we want to create as many jobs as possible, we must use resources freely. If we want development of the community, we must sustainably use the community's resources; and, to sustainably use its resources, we must control the use of resources. Using resources freely is in conflict with controlling their usage. Similarly, reading from right to left also makes sense: If we use resources freely, then we will create as many jobs as possible, and, if we create as many jobs as possible, then we will have development of the community. If we control the use of resources, then we will use the community's resources sustainably; and if we sustainably use the resources, then we will have development of the community.

As a side note, there is another approach – on which I won't expound here, but those who wish to explore this in detail can consult Wentzel (2017). This method involves starting the construction process from either the common goal or the requirements, but is more challenging.

4.3 Constructing a conflict diagram from an undesirable state

Sometimes the actions are not so clear, but you have the sense that there is a situation in which the current state of some system is not ideal – let's call it an 'undesirable state'. This undesirable effect can be derived from a question or a narrative of the problem which suggests that what is actually happening in the system is different what was hoped would happen. In this sub-section, I will demonstrate how to derive a conflict diagram from such an undesirable state.

Let's take an issue that most people are familiar with, namely, migration, especially the way countries manage the flow of migrants. Let's construct a conflict diagram using the steps in Figure 3.7. The final product is shown as Figure 3.8.

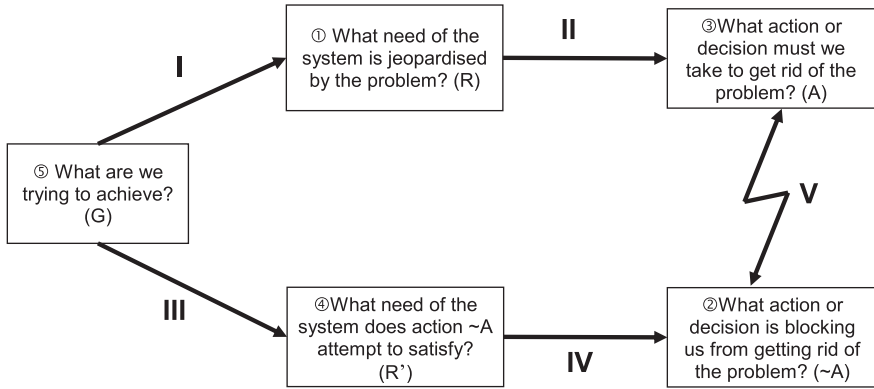


Figure 3.7 Stepwise method for finding a conflict from an undesirable effect

Source: Avraham Y. Goldratt Institute (2000)

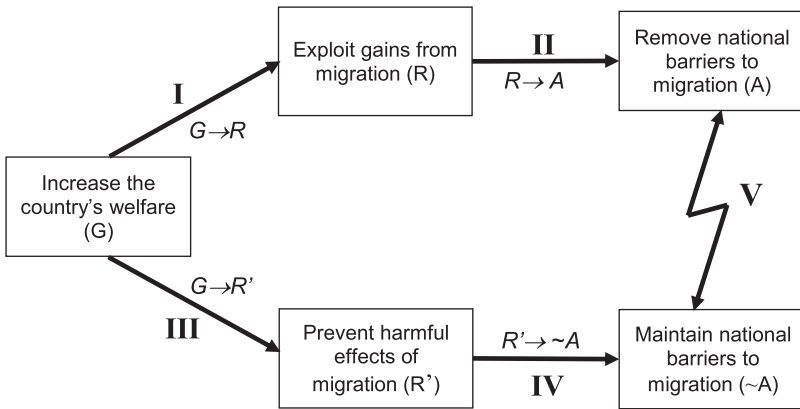


Figure 3.8 A conflict diagram for the problem of migration

Source: Wentzel (2017:158)

Step 1: We need to answer the question: What main need of the system is being jeopardised by the way countries are handling migration flows? In order to do this, we first need to know what we mean by ‘system’. In this case, it would be governments of countries (especially those of developed economies) that receive migrants from less developed countries, and then try to limit the rights or inflows of new migrants.

Step 2: Referring to the problem (the way they manage the inflow of migrants), we now try and find R by asking: “What is the need in the system (developed countries) that is being jeopardised by the problem?” It has to be expressed as a need, so consider why it is important to solve the problem. What need is not being satisfied due to the problem? Maybe it is that the way they manage the influx prevents them from

obtaining the benefits from migration; for example, turning away migrants who can make positive contributions, such as skilled workers and entrepreneurs. So, they neglect to 'exploit the gains from migration', and thus jeopardise their ability to gain from this phenomenon, and so we put that in the R-box.

Step 3: The next question goes to the opposing side of the conflict as we ask: "What action or decision is blocking us from getting rid of this problem?" This has to be expressed as an action or a decision. What action or decision stops us from solving the problem, or obtaining the benefits of migration? Maybe it is because these countries are maintaining, or even raising, their barriers that impede migration. So 'maintain national barriers to migration' is filled in inside the ~A-box.

Step 4: To find the opposing action, the next question is: "What action or decision must you take to get rid of the problem?" Once again, note that it has to be an action or decision. In this case, it is simply the opposite approach, that is, 'to remove national barriers to migration' and this should be filled in inside the A-box.

Step 5: The next question tries to find the reason for the opposing action or decision by asking: "What need of the system is being satisfied by this action?" This has to be a need. So, why do countries maintain barriers to migration? What need did this satisfy? This is usually to prevent the perceived harmful effects of migration into a country, so inside the R'-box, we have: 'prevent harmful effects of migration'.

Step 6: The last question acknowledges that both sides try to achieve the same thing, so we should ask: "What are we trying to achieve overall?" This has to be expressed as a goal. What is the common goal between the two needs? What do both of them want? Box G might then be filled in with some very broad common goal such as 'a labour force that contributes to economic growth' or 'economic development' or 'increase the country's welfare'.

Step 7: Test the logical flow by reading the boxes from left to right and from right to left (as explained earlier) and check that the actions/decisions are in conflict. Irrespective of whether you agree with the view or not; in either case, the reading should be both grammatically sound and reasonable, and in accordance with some theoretical or empirical view of the discipline. If the actions also conflict in some way, then the logical flow is acceptable.

There are other ways to approach the construction of a logical conflict. For example, you may start with a question like: "What action or decision do you complain about?" (which becomes box A), followed by asking: "What action would you prefer?" (which becomes box ~A). From the action, the requirements can be obtained by asking: "What need are you trying to satisfy with A?" (which becomes R); and, "What need are you trying to satisfy with ~A?" (which becomes R'). Then, finally, one can ask: "What is the common objective achieved by R and R'?" (which becomes G).

Once the conflict is constructed, it can be used to elicit the assumptions as explained earlier. In so doing, we identify the significant assumptions and

these will become the raw material that goes into the process of generating original contributions.

4.4 Using the diagram to identify original contributions

Once you know how to construct the diagram, you can start ‘mining’ it for opportunities to make original contributions, starting with the diagram itself. This is pertinent, as by simply formulating a problem or by noticing assumptions, a researcher may at this point already have conceived of an original contribution.

a) Problem formulation and reformulation

For me, the predominant strength of this technique is that it allows for more than one way to formulate a problem. Someone else may have gone through the same steps for the same problem and created one that makes sense, but has different content in the boxes. Take, for example, the case of community development in Figures 3.5 and 3.6. It could have been constructed as a conflict between, for instance: efficiency and equity, or individual choice and collective action; and a myriad of others. This is an advantage because every new formulation reveals new ways of seeing the same problem, and new perspectives on how to resolve it, which can in itself, be an original contribution. Problem formulation is therefore often seen as an important achievement. Indeed, one of the salient skills of a creative researcher is the ability to reformulate the problems in their field.

b) Identifying assumptions

Sometimes the simple act of making the assumptions of a discipline apparent is, in itself, an original contribution. As a discipline grows, assumptions may creep in without researchers noticing.

Assumptions are statements that we take for granted. We are forced to make them because our knowledge has to be based on something. While assumptions make our thinking more efficient, they can also put constraints on our thinking. This is because, in a logical system, we will avoid ideas that are not derived from these assumptions. But assumptions are most dangerous when they are implicit, i.e. when we are not aware of them. To understand the power of assumptions, I suggest you search for the ‘nine dot problem’ on the internet, before reading any further.

People who are not aware of their assumptions will be unable to come up with ideas other than those pre-determined by their assumptions. In all disciplines, the majority of researchers produce boring research – the kind of research that leaves the understanding of the discipline unchanged – because they cannot go beyond the assumptions of the discipline. As mentioned,

true experts, those with *shoshin*, are aware of their assumptions. They still base their thinking on assumptions, but because they are aware, assumptions don't control or determine their thinking.

As with the 'nine-dot problem' in many disciplines problems remain unsolved because researchers are not aware of their assumptions. In some cases, implicit assumptions are, ironically, themselves, the cause of problems. One way to make an original contribution then is to show the existence of an assumption that was unknown to the discipline up to that time. If you can show how that assumption causes problems for the discipline, you have further original contributions.

As explained earlier, the conflict diagram enables you to pinpoint assumptions quite easily. Only some of the many derived from Figure 3.8 are shown in Table 3.2.

If all the assumptions are known to the discipline, original contributions may appear when you question one or more of the assumptions. Assumptions can only be questioned if they are explicit, which is why it is so important to bring them to the surface.

c) Challenging assumptions

Since assumptions are unproven statements, which we take for granted as being 'truths', they are open to challenge. For a problem to exist as a logical conflict, at least one assumption has to be inconsistent with one other. Hence, not only are assumptions open to challenge, they should be challenged.

Every assumption is a potential source of original contributions, because if the truth of any assumption is in doubt, it has repercussions in terms of the understanding of the entire discipline. Take each assumption and ask: "Is it always true? For all time? In all groups? Under what conditions would it be invalid?" If you find that any assumption is not true, or not universally true, and you can demonstrate this, you have made an original contribution.

Table 3.2 Some assumptions

Connective I	Gains exist and are not exploited. Migration raises profits in rich (receiving) countries and raises wages in poor (sending) countries.
Connective II	Barriers are based on ignorance or narrow interests. Restrictions on migration limit gains from migration.
Connective III	Gains from migration don't translate into national welfare. Migration harms infrastructure in rich regions and does not raise labour productivity in poor regions.
Connective IV	Countries design policies to serve national interest. Barriers to migration can counter the harm from migration.
Connective V	Restrictions have to be relaxed in the same place/s. There can be no differentiation between restrictions.

Source: Adapted from Wentzel (2017:159)

Since the knowledge of a discipline is based on assumptions, you can extend this contribution by working out the repercussions that a false assumption has on the understanding of the discipline. You could go further and suggest a replacement assumption and its implications. For example, Copernicus was not the first person to say that the assumption that the sun moves around the earth was incorrect, nor was he the first to propose the heliocentric assumption as a replacement. His real contribution was working out the implications that this change had for how we understand movements in the solar system, and how this new assumption solved some of the problems in astronomy.

Look again at the scope offered by the four questions: “Is it *always* true? For *all* time? In *all* groups? Under what conditions would it be invalid?” These queries help guide this process of challenging assumptions in a plethora of ways. If an assumption is not universally true, you can work out the repercussions of the exceptions as well, and propose new ones for different conditions.

Related to this, is when assumptions are true, but not well defined. In such cases, you can ask if some of them can be improved or refined, and work out the repercussions. See how endless the possibilities are for original thought in the social sciences?

The possibilities are endless in the social sciences and that is what makes it exciting and open to ‘excavation’; however, don’t be over-zealous: be careful in choosing which assumptions to challenge because the discipline is not always open to such confrontation. As mentioned before, some assumptions are so basic to the discipline that challenging them would undermine its foundations, and such challenges will be resisted, and possibly even be counter-productive. To challenge such fundamental tenets, even strong evidence may not be enough – you’ll need to be a recognised authority to even be taken seriously.

In the conflict diagram, you are most likely to find some of these basic assumptions under arrows I and III. So, if you are a novice researcher, you may find it easier to get published if you challenge assumptions that are not as critical to your discipline to begin with. Challenges to assumptions in arrows II, IV and V will still be resisted, but with strong arguments and evidence, such challenges may be more acceptable to the discipline, so this is a safer area to operate in at the outset.

Again, in terms of being over-zealous, be cautious: it is key to be aware of the fact that not all assumptions need to be challenged. It could be that the discipline knows about an assumption, but has never bothered to (or doesn’t want to) test it. Nevertheless, if you can show that an important assumption has never been tested and then test it adequately, with methods acceptable to your discipline, you would have made an original contribution.

d) Addressing inconsistencies

If a discipline tries to solve its intellectual problems, it has to identify its assumptions in order to test or challenge them. Of course, the obvious place

to introduce tests and challenges is in investigating those assumptions that are inconsistent.

Considering Table 3.2, it is possible to see several inconsistencies: for example, ‘restrictions on migration limit gains from migration’ and ‘barriers to migration can counter the harm from migration’ are clearly inconsistent. An obvious contribution would be to design research that is able to decide which assumption is correct under which conditions, or whether both are wrong.

An original contribution is already made in simply highlighting an inconsistency that the discipline was not aware of. One might think that it is not possible for a discipline to be unaware of the inconsistencies in its understanding, but that is not true.

Cherniak (1984) estimated that the strongest computer imaginable would need the equivalent of the time that our universe has existed to determine whether there are inconsistencies between only 138 unambiguous statements. Researchers are not super-computers, so they are likely to struggle with fewer than 138 statements. Even four statements present some difficulty: for instance, it took the economics profession several decades to even see the logical inconsistency between just four known propositions from which Arrow’s Theorem was derived.

The body of knowledge in any discipline consists of more than 138 statements, and many of them are quite ambiguous. It is inevitable, therefore, that numerous logical inconsistencies exist in the discipline, and that many exist undetected, that is, until experts with *shoshin* apply their mind.

e) Finding provocative questions

The simplest approach to finding provocative questions is to simply question assumptions. As already mentioned, questions can be generated from every assumption under each arrow, by asking:

- Is [the assumption] necessarily so? Do we know this?
- Is [the assumption] true under all conditions? When or where is it not true?
- Would anything make [the assumption] invalid?

If one of these questions leads to an assumption being invalidated, then the arrow that relies on that assumption is also invalidated. By implication, this resolves the problem because it removes one side of the logical conflict. The problem with invalidating one side of the conflict is that it destroys the information on which their arguments are based. Sometimes this may be preferable, though it is very rare that an opposing view contains no useful knowledge.

An approach that leads to more provocative questions, one that may maintain information and can lead to new knowledge, is to synthesise the

two sides of the conflict. From the conflict diagram, it is easy to find questions that can direct research toward synthesis. These synthesising questions do not challenge assumptions as much as they provoke the creation of new ones. Wentzel (2017) explains the process of finding new assumptions in more detail, but again, in this chapter this will not be explored extensively – I will only show how to find the questions.

As illustrated in the next figure, there are three main groups of questions designed to bring the two sides together. Each one creates new arrows with new arguments and assumptions:

- $R \rightarrow \sim A$: How can we increase average income by reducing wages?
- $R' \rightarrow A$: How can we increase employment by increasing wages?
- $A + \sim A$: How can we both increase and decrease wages?

If any of the three questions lead to new insights, the conflict collapses by leading both sides of the conflict into the same action (e.g. when $R' \rightarrow A$ leads to the same action as $R \rightarrow A$ in the original conflict) or by combining the two actions into one ($A + \sim A$). In any of these cases, the question, by itself, may be a new contribution if it has never been considered before, or may direct research down unexplored paths.

f) More insightful literature reviews and cutting-edge empirical research

Assumptions need not be only theoretical, but can also lead to questions that guide empirical research. If an assumption appears to be accepted, but without strong empirical support, questioning it can lead to research that is ground-breaking should the assumption turn out to be wrong. This is especially likely to occur after dramatic changes in reality, or after a period of

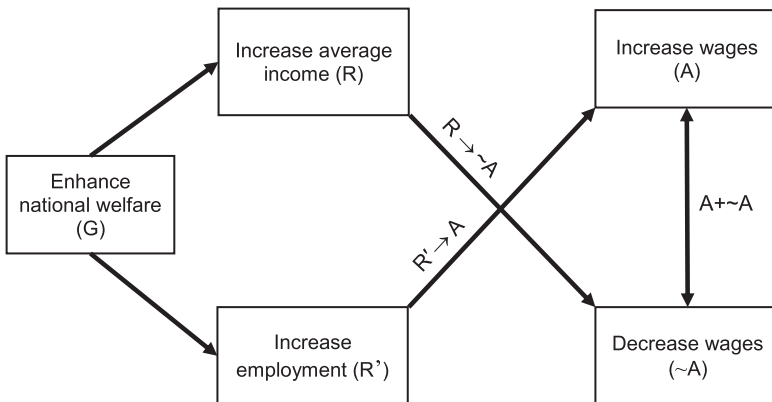


Figure 3.9 Three questions to provoke new understanding
Source: Wentzel (2017:160)

theoretical disruption in the discipline. Under such conditions, every assumption that structures a problem may be open to question, and turning each of these into a question can generate a new research programme for original research.

Cutting-edge research is, by definition, research that explores the implications of new ways of thinking – that is, exposing assumptions, or even better, the creation of new assumptions. Assumptions determine the edge of a discipline, so it is here where you will find the likely future research. Current developments in the theory or methods of the discipline may indicate which assumptions are most vulnerable and hence the places where cutting-edge research will occur.

This also has implications for literature reviews (even though it will be discussed in chapter 6 in detail). Literature reviews, especially those that are published, are not simply summaries of the literature. Instead they explain why the literature developed in certain ways, where the key challenges remain and the most likely directions that future research will or should explore.

Everything in this chapter so far will help you to write such literature reviews. Seeing the discipline from the perspective of solving intellectual problems helps to explain its evolution. After constructing several logical conflicts in a discipline, certain patterns will emerge, and you will either start to see: that the discipline has been struggling with the same fundamental problem without much progress; or how its problem changed as it challenged a succession of assumptions. Once you represent a problem as a logical conflict: you can also see what assumptions it is making, which ones are the greatest stumbling blocks to progress and which ones are currently the most likely to be the focus of cutting-edge research in the future.

5 Originality through the questions of the discipline

Science may progress from problem to problem, but research into these problems only happens if they give rise to questions, because research pursues answers to questions. While questions may give the impression of being open to many answers, in reality, every question is an argument that limits us to only a small range of answers.

In this section, I will focus on questions; specifically how questions can become problematic and a technique for finding interesting new questions. Even, in and of itself, a new question is an original contribution if it opens up people's minds to the possibility of new ideas and methods. Thus, the key to original contributions is to challenge questions and find new ones. Because questions can be deceptively limiting, when aiming to produce original research, one has to question the questions. If one knows how to interrogate even a conventional or common question, original contributions can arise.

When asking even the simplest question, we make many assumptions. If I ask, "When did you stop smoking?" I not only presume that you stopped

smoking, I assume that you did smoke at some time in your life. With a question such as, “How do we alleviate poverty?” I assume that poverty exists, that it is harmful and that it is in our power to control it. This is why questions constrain thinking. When asking a question, you automatically make the assumptions on which that question is based, without being aware of it. And it is these implicit assumptions that are dangerous because they limit us to only deriving answers that follow from these. For example, given the poverty question above, I will avoid arguments based on alternative assumptions; such as, the assumption that poverty is something outside our control.

However, there is something even more dangerous when it comes to questions: that is, the possibility that we are asking problematic questions: ones that don’t make sense. Such questions are unanswerable because their assumptions make an answer impossible. For example, the question “Why are you late?” assumes that you are late. If that assumption is false, the question does not make sense. So, if you were on time, it is not possible to answer the question, because the question is problematic.

As with problems, problematic questions can only be answered by adding, changing or rejecting the assumptions imposed by such questions. A discipline that does not question its own questions, will most likely waste valuable time and energy on problematic questions that don’t have answers. Disciplines need researchers who can see the assumptions of its questions, challenge them and propose new questions.

To question a question is quite difficult in the absence of a systematic approach. In this section, I will offer one approach to finding new questions, which will help direct you. Later, in chapter 6, I outline another approach.

5.1 *Van Fraassen analysis*

There is a group of philosophers who specialise in the logic of questions. Bart van Fraassen (1980) is one of them. He created a theory of why-questions. While he applied his theory only to why-questions, other philosophers showed it can be applied to any question that calls for an explanation; exactly the kind of questions that we encounter in research.

In this section, I show how to analyse questions using Van Fraassen’s method of analysis and how it helps us to question our questions and find new ones. Unlike the conflict diagram approach that is most appropriate to real-world problems, a Van Fraassen analysis can be used with purely theoretical issues as well, provided you have a question.

Van Fraassen identifies the main elements of a question as being:

- 1 The topic (what you try to explain, which is also the most important assumption of the question);
- 2 The contrast class (alternative ways to answer the question);
- 3 The relevance relation (what counts as a relevant answer).

Van Fraassen uses the ancient question from the Garden of Eden as an example: “Why did Adam eat the apple?” If I ask that, my most basic assumption is that ‘Adam ate the apple’ so this is the topic of the question. One can easily find the topic of a why/how-question by removing the ‘why’ or ‘how’ and then adjusting the remaining words until it forms a grammatical statement rather than a question.

There are actually many answers to that question. Each answer comes from a different contrast class, since every contrast class defines an alternative way to answer the same question. Here are some possible answers: Adam was hungry; Eve was on a diet; there were no bananas; he preferred meat; Eve tempted him. Of these possible answers, the only one that is taken as a valid answer (based on the assumption as derived from the narrative written in the Bible and corpus of literature reinforcing it) is the last one.

To find the contrast classes, simply identify the verbs and nouns in the question, and state the question each time, putting emphasis on a different verb or noun. This can be seen in Table 3.3 in the ‘question’ column. By stressing the importance of one word over others, notice how the meaning of the question changes.

When I put emphasis on ‘Adam’, I am asking why it was *Adam* who ate the apple as opposed to someone else. This gives us one contrast class of that question. A different contrast class calls for a different answer (and also a different relevance relation). For example, when I put emphasis on the word ‘apple’, I am interested in answers that explain why an *apple* was chosen over other fruits, and not why Adam ate it.

When asking a question like this one, I make many assumptions, and, if any of them are wrong, my question becomes problematic:

- I assume that the topic – ‘Adam ate the apple’ – is true. If Adam did not eat the apple or if Adam never existed, this question is problematic, that is, it does not make sense.
- I assume that one contrast class is the appropriate one and that the others are incorrect. The conventional answer ‘Adam ate the apple because Eve tempted him’ assumes that the second contrast class (in

Table 3.3 Analysis of an ancient question

	Question	As opposed to: (contrast class)	Answers will focus on: (relevance relation)
1	Why did <u>Adam</u> eat the apple?	Other people (Eve)	Why Adam was the one who did it
2	Why did Adam <u>eat</u> the apple?	Not eating it or doing something else with it	Why the eating happened
3	Why did Adam eat the <u>apple</u> ?	Other fruits or types of food	Why the apple was so appealing

Table 3.3) is the correct one. This is because it is the second contrast class that focuses our attention on why Adam ate the apple instead of ignoring it. There is usually nothing that helps us decide which one is correct except the decision of the group of people who want to find the answer. In other words, the chosen contrast class is assumed by the discipline (in this case the theological literature).

- What counts as an answer depends on the contrast class chosen, so I assume that there is at least one true and relevant answer within the chosen contrast class.

Once we understand this way of analysing a question, we can see that there are, in fact, many questions that are hidden in a question, and therefore many ways to answer the same question. Some ways of answering are ‘correct’ answers, but we don’t regard them as valid because they fit into other contrast classes. The answer ‘Adam ate the apple because there were no bananas’ sounds more like you are trying to make a joke. It is the kind of answer that you might give if you assumed the third contrast class. It is not necessarily wrong, but it is not taken seriously because it doesn’t fit the apparently correct contrast class.

Sometimes, there is more than one answer within a contrast class. The implicit assumptions within a discipline leads it not only to favour one contrast class, but also to prefer one answer within a contrast class over others. In the second contrast class, for example, we may find two answers: ‘He ate it because he was tempted’ and ‘He ate it because he was hungry’, but only the first is regarded as being correct. Again, this is not because the second one is necessarily incorrect, but because the discipline rules it out by assumption.

As Table 3.3 suggests, there may be a great deal of interesting questions that remain to be asked, and thus answers to be considered; and this can be achieved by simply changing the contrast class. Rethinking a question’s contrast class leads one to ask a different kind of question. So, looking for contrast classes that have not yet been seriously considered by the discipline can generate new questions that can then also generate original contributions. Also, considering alternative answers that fall within the same relevance relation may reveal answers not considered before.

5.2 *Doing a Van Fraassen analysis*

In this section, I’ll take you, step-by-step, through a Van Fraassen analysis. Suppose someone has a question like ‘Why do countries manage migration in the way they do?’ This question is much too vague, so the first step will make the question more specific and meaningful.

Step 1: Refine the question until it is sufficiently meaningful to be researched. This is explained in chapter 5, so I won’t go into detail here. The more meaningful the question, the easier it is to find contrast classes. Suppose

that, after this step, the question changes to “Why do developed countries resist the gains they can obtain from unrestricted migration?”

Step 2: Find the topic and check it. Remove the ‘why’ and turn the remaining words into a grammatical statement. The topic in this case is: ‘Developed countries resist the gains they can obtain from unrestricted migration.’ This is also the fundamental assumption of the question, and if this were not true, the question would be problematic, and the analysis could not continue. So, ensure that the evidence is strong enough to warrant the topic that developed economies are, indeed, resisting unrestricted migration.

Step 3: Identify the different nouns and verbs. These are: ‘developed countries’, ‘resist’, ‘gains’, ‘can obtain’, ‘unrestricted migration’. (I put ‘developed countries’ and ‘unrestricted migration’ together as they form one unit of meaning.) Construct a table in which you emphasise a different one each time (as in the question column of Table 3.4).

Step 4: Define the contrast classes from the changes in the meaning of the question. This is done in the contrast column of Table 3.4.

Step 5: Identify the possible relevance relation/s in each contrast class, that is, consider what the answers will focus on if it assumes that contrast class is valid. If possible at this stage, form some hypotheses as to what these answers may be. Be open to the possibility that a relevance relation can produce more than one answer.

Step 6: Ask yourself the following to find new questions that can lead to original contributions and more interesting research: Which contrast class does the discipline favour? Does it miss the significance of other contrast classes? What assumptions does it make that blinds it to other contrast classes? Does it even realise that there are other contrast classes? Do any of the other contrast classes suggest better or more interesting questions? Do any of the relevance relations suggest answers or approaches not previously considered?

This simplified adaptation of Van Fraassen’s theory demonstrates how even the simplest of questions can be questioned and potentially problematised. As with the conflict diagram, a Van Fraassen analysis suggests interesting empirical research and opens up new directions existing research can take. In fact, the two techniques interact quite well, because once a question is problematised, the question (through its contrast classes) can be represented as a logical conflict, and this opens up further possibilities. For example, the last contrast class in Table 3.4 can be framed as a conflict between encouraging migration versus other kinds of influx. For the sake of brevity, this will not be explained here, but the reader who understands the construction of a logical conflict should be able to create one.

5.3 Relational analysis

There is one technique called ‘relational analysis’, as developed by Wentzel (2017), which helps you to identify the assumptions on which a question is based. Since this technique is useful when writing literature reviews, it is

54 Finding an original contribution

Table 3.4 Why do developed countries resist the net gains they can obtain from unrestricted migration?

	Question	As opposed to... (contrast class)	Answers will focus on ... (relevance relation)
1	Why do <u>developed</u> countries resist ...?	...developing countries.	What causes developed economies to manage migration inflow differently to developing economies.
2a	Why do developed <u>countries</u> resist the ...?	...provinces, states, cities.	Why national migration cannot/ should not be restricted in the same way as international migration.
2b	Why do <u>countries</u> resist the net gains...?	...other institutions.	Reasons why national governments restrict migration.
3	Why do developed countries <u>resist</u> the ...?	...ignore, allow, encourage.	Explanation of why countries actively resist possible gains from migration.
4	Why ... resist the <u>gains</u> they can obtain...?	... losses.	Identifying the gains that exist and how they came to be.
5	Why ... resist the net gains <u>they can obtain</u> ...?	...others obtain.	The mechanisms by which countries can appropriate the gains from migration.
6	Why ... obtain from <u>unrestricted</u> migration?	...restricted, prohibited.	Arguments that gains mainly follow when there are no restrictions.
7	Why ... obtain from unrestricted <u>migration</u> ?	...trade, capital flows, idea flows.	Why gains from migration are more important than those from other forms of economic integration.

Source: Wentzel (2017:162)

explained in detail in chapter 6. Using this technique produces assumptions, which can then be challenged; so becoming the seeds for original contributions, as already explained in this chapter.

6 Conclusion

To make an original contribution you need two things, encapsulated succinctly by the Japanese words: *ikigai* ('reason for being') and *shoshin*

(‘beginner’s mind’). Cultivating *ikigai* is what makes an original contribution *worthwhile*, while it is the mindset of *shoshin* that is pivotal in that it makes this interesting original contribution *possible*. Most books will focus on *ikigai*, though in this chapter, I put emphasis on *shoshin*, which is more difficult to cultivate; but without which original thought would not be possible at all.

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4 Writing argumentatively

With hindsight, I can't help but feel sorry for my students of years ago. I would read their writing and know that it wasn't quite right, but couldn't put my finger on it. So I would send them away with vague directives like 'write more argumentatively' or 'be more critical', which, of course, they didn't understand. Next time, I saw them, their writing would still not meet my expectations – they would either just pile on more facts or references, become opinionated or criticise everything in sight. And we would repeat this until one of us gave up in frustration.

That x-factor I was looking for and that most professors want to see, but which is very hard to define, is that argumentative quality to the writing. Fortunately for my later students, eventually I found some ways to explain more clearly what I meant by the broad instruction of 'be more argumentative' and show them how to do it. In this chapter, I share this process with you by: using metaphors, like talk shows; employing techniques that involve imaginary friends and puppets; and having recourse to templates developed by other authors.

This chapter follows from chapter 1. In chapter 1, I explained the inner workings and intricacies of a single argument using the Toulmin model, but mentioned that it is not a good model to guide writing. Then, in chapters 2 and 3, I covered basic skills and originality; all of which featured the concept of argument, but not yet the actual writing found in a thesis or paper. As we now get into specific writing tasks in chapters 5 to 7, the techniques and metaphors of this chapter will be essential. To start, let's review some of the important ideas from previous chapters that you need to keep in mind in this chapter.

Argument is the act of making claims. Facts are not claims because everyone agrees with facts. A claim is a statement about which there can be *different* perspectives, so argumentative writing involves playing with these differences (Andrews, 2010).

If all authors had the same view about everything, argumentative writing would not exist as there would be no reason to write anything, because one person could say everything there is to say. But in the social sciences, it does not work this way. There is so much to talk about and discuss. The talking takes place around points where there are different perspectives and room for difference: and this is what argument is all about. Unless there is an interaction between different ideas, there is no argument.

In school and your undergraduate studies, you were probably taught not to argue; in order to pass, you simply had to repeat the facts you'd been taught, like a parrot. This is a bad habit you will have to unlearn quickly – you cannot be a good researcher if you simply write down facts and just repeat what other have said.

Argument is inevitable in the social sciences. In the social sciences, we study a reality that is subjective. This reality comprises social constructs like: money, poverty, marriage, jobs, organisations, traditions, laws, etc. that would all disappear if humans were no longer on this planet, and so human beliefs create most of the things we study. These beliefs are usually inconsistent and change constantly, which means that differences in perspectives will always be around, so that almost everything we study will be open to argument.

No paper or thesis consists of a single argument alone. In this chapter, I go beyond chapter 1 and show you how to create and write the kind of complex arguments that are expected in a paper or thesis.

1 Arguments within arguments

Any complex piece of writing will try to say (or claim) one big thing – we call this overall claim the 'thesis'. But to argue such a thesis, smaller arguments need to be made, connected to each other and then placed in a logical order, so that one leads to the other, until the last argument finally concludes with the thesis. This applies to chapters, sections and paragraphs.

A thesis is broken up into chapters with each one making an argument until the last chapter concludes with the overall claim. Then the chapters are broken down into sections, each one making an argument, until the last section concludes with the claim of the chapter. And the same applies to sections then being divided into paragraphs. This is represented in Figure 4.1 (see page 58), where: the big circle is the thesis (the overall claim), made up of the smaller claims of the sections, which comprise the even smaller claims of the paragraphs.

These claims cannot, however, simply be placed in any order. They have to be developed into arguments – with data, warrants and counterarguments – and then each argument has to feed into the next argument.

We can use the Toulmin model to help us see this. Data does not just have to be what other people have found. Your own argument, in one section or paragraph, becomes the foundation on which you base the argument of the next section or paragraph. Figure 4.2 (see page 58) shows this: that your own arguments *must* become data for your next argument. In fact, if the argument you just made doesn't feed into your following argument, then the value of that argument becomes debatable, questioning whether it was even necessary.

Arguments, of others and your own; therefore, feed into the next argument, and this creates the complex argument of a thesis, a chapter and a section. To write any complex argument made up of many smaller arguments, especially those in the chapters that follow, you need to master two skills: (1) discovering the smaller arguments inside a larger argument; and (2) connecting those smaller arguments logically.

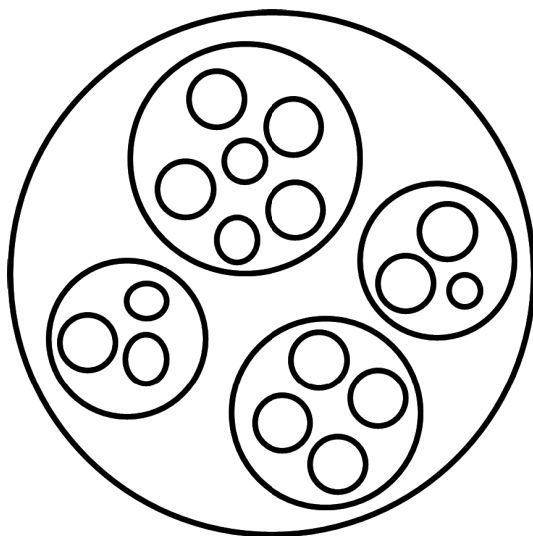


Figure 4.1 Complex argument

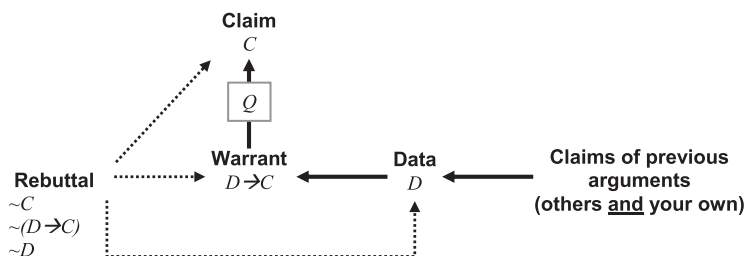


Figure 4.2 Your own arguments are also data that feeds into the next argument

2 Finding smaller arguments

To find smaller arguments, you must first know what your overall message, your thesis or overall claim, is. This becomes the ‘seed’ from which you will grow the smaller arguments.

Before writing anything, you must be able to summarise your thesis in a single sentence. The sentence must be clear and concise, and you must be able to clearly define every term in your thesis. In Figure 4.3, this is the stem of the tree (A) from which everything grows. Once you have a thesis (A), you will use it to create the smaller arguments that make up your chapters (B). You will use the arguments of your chapters to create the smaller arguments of the sections (C) and so on. Every chapter, section, subsection and paragraph must convey a single argument.

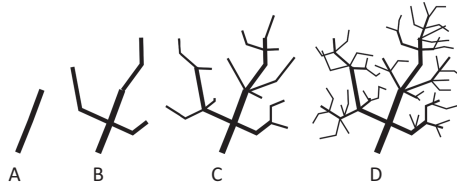


Figure 4.3 Growing an argument

2.1 How to do it

To discover the smaller arguments, I'm going to teach you a technique that might sound strange, but is used by many good writers. You start with the thesis, and then have an 'imaginary conversation' with an imaginary friend (or yourself). When this technique is used effectively, it gives writing that pleasing flowing quality where ideas appear in the writing just as the reader thinks about them.

The process works like this:

- 1 Make a statement to your imaginary friend.
- 2 Imagine the friend responds with a question. The friend is only allowed to ask a question, and the question should be an open-ended question that encourages you to explain more. Such questions are usually 'why' or 'how' questions. The imaginary friend is not allowed to ask closed-ended questions; that is, those that can be answered with a single word or phrase. A closed-ended question with a simple answer stops the conversation, and this is not what you want at the very beginning of your argument.
- 3 You then respond to this question, and your response becomes the next statement that starts the process again. Return to step 1 until you have enough smaller arguments to give you clear guidance in your writing.

The aim of this process is to produce a hierarchical structure with all the important arguments that can be used as an outline for writing. To illustrate the process, read the article in Appendix A, which will be our example.

You always start with the overall claim, which, in this case, is that 'arrogance can be good'. It is a claim because it is possible to take different perspectives or have different views on this. Following the process, I would start by stating this claim and let the conversation develop:

ME: Arrogance can be good.

FRIEND: How is that possible?

ME: For two reasons: we don't really understand the meaning of the word 'arrogance', which is why we think it is 'bad'; and secondly, there are actually different kinds of arrogance, and, therefore, we should not put them under one umbrella, thinking about each in the same way.

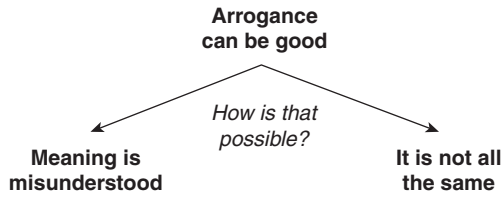


Figure 4.4 First level of the imaginary conversation

This creates the first level of arguments derived from the thesis, with two branches, as shown in Figure 4.4.

Now let's take each branch at a time, and using the same process, derive further arguments.

ME: We don't really understand the meaning of the word 'arrogance', which is why we think it is 'bad'.

FRIEND: How can we know what arrogance really means?

ME: We can understand its meaning by looking at the origin of the word and by considering what the opposite of arrogance is.

ME: There are actually different kinds of arrogance and we should not think about them in the same way.

FRIEND: What are these kinds of arrogance? (Note that, although this is a 'what' question, it is one that encourages me to expand, rather than responding with a phrase that stops the conversation.)

ME: There is destructive arrogance, which we should discourage, but also constructive arrogance, which we should encourage.

This produces an expanded hierarchy, which is shown below. The first level produced the main sections; and the second level, the important groups of paragraphs.

What remains, then, is to ask questions for each one of the second-level claims to direct the writing about each second-level argument, as shown

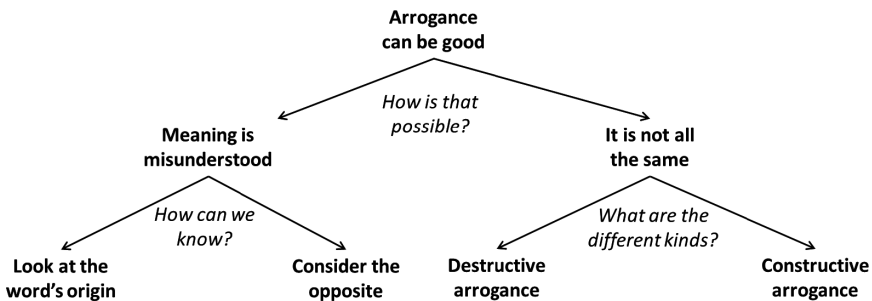


Figure 4.5 The second level of the imaginary conversation

below. If this is the last level of the conversation, it is acceptable to ask more closed-ended questions that bring the conversation to a close. Starting with ‘look at the word’s origin’ in Figure 4.5, we take each of the claims and apply the technique.

ME: We can understand the term ‘arrogance’ by looking at the origin of the word.

FRIEND: What does the word’s origin tell us about its meaning?

ME: We can also understand the meaning of arrogance by considering its opposite.

FRIEND: How does the opposite of arrogance reveal the meaning better?

ME: There is destructive arrogance, which we should discourage, but also constructive arrogance, which we should encourage.

FRIEND: What is the difference between the two? Why should we discourage destructive arrogance, and encourage constructive arrogance?

Guided by these questions, and using the Toulmin model, you can now expand on each second-level argument, looking for support to substantiate your claims. This is done through: analysing data, finding warrants and studying counterarguments and the merit of each. The result of this process is a gradually expanding structure as shown in Figure 4.6. The longer and more complex the piece of writing, the more levels you will create. Also, unlike the example above, at some levels you will often find more than two branches.

Writing cannot start yet, because the arguments are not yet connected, but we have now broken down the thesis into smaller arguments. Before explaining how to connect arguments, let me first explain why it is so important to spend time dividing the thesis into sections and paragraphs.

2.2 Why we do it

The human mind needs ideas to be dissected and hierarchically grouped into categories, because it helps us to remember those ideas. And if a reader can

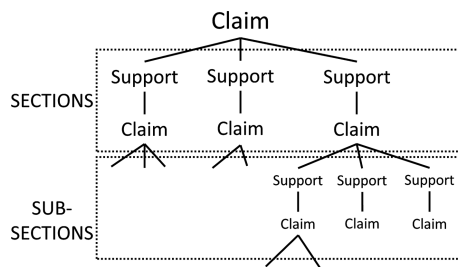


Figure 4.6 The hierarchical expanding structure of a complex argument

remember your previous arguments, they will be able to better follow your next argument.

To see what I mean, try the following exercise: read the following shopping list only once and try to remember all the items: eggs, cucumber, grapes, milk, carrots, apples, cream, bananas and lettuce. Now close the page and do something else for two minutes.

Next, write down the items on the list without looking at it. You will probably find that you forgot a few items. To deal with this, humans naturally categorise. Categorisation helps us cope with the complexity around us because it reduces the load on our short-term memory.

Similarly, most pieces of writing contain too much information to remember, and the more your reader forgets of what you wrote, the more difficult it becomes for them to follow your argument. However, if you provide categories – in the form of chapters, sections and paragraphs – you make it easier for readers to remember and understand your arguments. Hence, your argument has a far greater chance of persuading them.

Going back to the above example of the shopping list, let's categorise the items in order to illustrate how this helps us remember more. You probably notice that there are three categories, as in Figure 4.7.

Now, close the page, do something else for a while and try to remember the list again. I'm confident that you will notice a substantial improvement in your memory. The kind of categorisation you would have done could be represented as a pyramid (or an inverted tree). This pyramid structure is the fundamental (and conventional) configuration in the presentation of knowledge in academic writing.

Readers expect to see this structure in academic writing. If your writing does not follow this structure, many of them will be lost and struggle to remember what you wrote. The 'imaginary conversation' technique helps you create such a structure.

[Many of these ideas are derived from Barbara Minto's (2002) many insights into business writing. I highly recommend her book *The Pyramid Principle*. Also useful is the 'issue tree' methodology developed by Linda Flower (1998).]

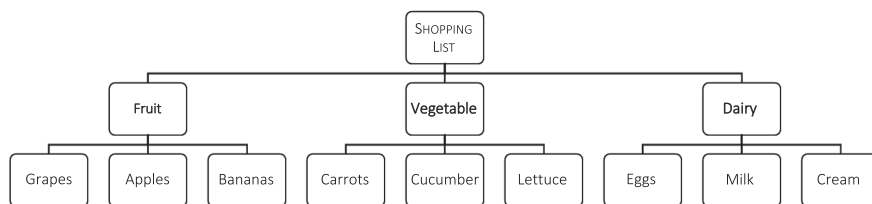


Figure 4.7 Categorising the items on the list

3 Connecting arguments

Once you have all the argument units – chapters, sections, sub-sections and paragraphs – you have to connect them. You need to create connections between sections, between paragraphs, even between sentences. Every sentence, paragraph or section needs to come from somewhere, and lead to some point. In other words, any new unit has to be based on its previous unit, and also be the seed from which the next unit will grow.

If you don't do this, your reader will not follow your logic. Thinking of it in concrete terms: every unit must draw something from the previous unit, and point towards something in the next unit, as represented in Figure 4.8.

Consider the following passage:

The double-entry accounting system was formalised by Luca Pacioli in 1494. Single-entry bookkeeping is appropriate to simple business ventures. Every transaction enters only once as an inflow or outflow of cash. It provides a quick method for determining accumulated income at a certain date. Single-entry bookkeeping provides no information on changes in wealth.

While the individual sentences are understandable, the passage as a whole is not clear. The reason is that the sentences all seem to be disconnected. In this passage, the flow between sentences is unclear and this can lead to a misunderstanding of the argument.

This problem can, however, be fixed by making the connections clear. There are three methods to create connections: (1) refer to previous ideas; (2) repeat key words and phrases; and (3) signal connections.

One simple technique that uses all three methods is called the 'old-new information' technique. It is easier to understand a sentence/paragraph/section if it starts with something you already know (old information) from the previous sentence, paragraph or section. The old information is drawn from the previous unit, and then new information is added to it.

Consider the improved version of the previous passage to see how this works between sentences. Every sentence takes something from the previous sentence, often repeating it. This is highlighted by underlining the old information that each new sentence takes from the previous sentence.

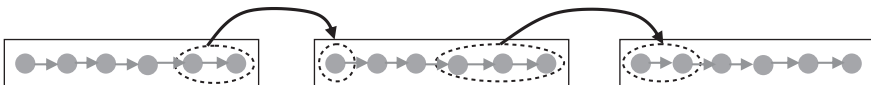


Figure 4.8 Connecting units of an argument

The double-entry accounting system was formalised by Luca Pacioli in 1494. It was a significant improvement over single-entry bookkeeping. Single-entry bookkeeping is appropriate to simpler business ventures. It is simpler because every transaction enters only once as an inflow or outflow of cash. Knowing cash flows provides a quick method for determining accumulated income at a certain date. However, single-entry bookkeeping provides no information on changes in wealth.

The word ‘because’ is used in the middle sentence and ‘however’ in the last. They are also underlined because they signal that the next sentence is connected to the previous sentence.

Words like these make a big difference to your writing because they indicate connections between sentences and ideas. We call them ‘transition words’ as they either continue with the idea from the previous sentence or break away from it. So, there is a ‘transition’ between ideas, hence the name. In short, transition words send messages to the reader about how you see the relationship between ideas.

Using the correct transition words and phrases can help improve the reader’s understanding of your arguments. There are many good web sources that provide such transition words, and the kind of connection signals they send. Table 4.1 provides a summary of these signals.

You can also employ ‘pointing words’ that refer to something in a previous sentence. These are words like: this, that, these, those, their, such, it, he or she. However, you should use these words with care.

Only use such words if it is very clear what you are referring to. Just because it is clear to you as writer, it does not mean it will be clear to the

Table 4.1 Transition words as signals to the reader

<i>Signals of continuation</i>	<i>Signals of breaking away</i>
Now I am going give you a list that offers more detail about the previous idea (<i>first, second, to begin, next, finally</i>)	Now I am going to contrast this against another idea (<i>by contrast, on the other hand, whereas, yet</i>)
Now I am going to add something to the previous idea (<i>also, in addition, furthermore</i>)	
Now I am going to elaborate on the previous idea (<i>in other words, actually, after all</i>)	Now I am going to contradict the previous idea (<i>however, nevertheless, regardless, even though</i>)
Now I am going to give examples of the previous idea (<i>for example, in fact</i>)	
Now I am going to show you how this idea is very much like the previous idea (<i>likewise, similarly</i>)	Now I am going to concede a point (<i>admittedly, granted, of course</i>)
Now I am going to explain how this is a result of the previous idea (<i>because, as a result</i>)	

reader. Take, for example, the sentence: “Poverty and unemployment rise together, and economic policy should respond to this during recessions.” What does ‘this’ refer to: poverty or unemployment or the relationship between the two? Sometimes it is better to specify what you are referring to, rather than using pointing words, even if it means repeating some words.

You can use the same connection techniques between paragraphs too. One way to apply these techniques is to use ‘hooks’: you ‘hook’ onto something from the previous paragraph in the next paragraph. Let’s return to a previous example:

The youth unemployment rate is 55% (SSA, 2012). A quarter of these young people have studied towards a qualification at a university. So, it appears that universities play a role in graduates not being employable.

The paragraph that follows this one can continue in many different ways. Let’s consider four different ways below. Notice that each time I use a different type of hook, which I emphasise by putting it in *italics*.

- A **shallow hook** is when one hooks onto one of the last words from the last sentence by repeating it; in this case, this is done by starting the next paragraph with the word ‘employable’, as follows:
Employability is determined by various factors such as...
- A **deeper hook** is when one hooks onto one of the words a little deeper into the paragraph by repeating it; in this case, the phrase ‘youth unemployment’:
A formal education is only one of many factors that influence *youth unemployment*.
- **Multiple hooks** hook onto several words in the previous paragraph. For example:
A formal *university qualification* should make a person more *employable*.
- An **idea hook** is that which hooks onto the meaning of the previous paragraph without necessarily repeating any of the words; in this case, the phrase, ‘this state of affairs’, refers to what was explained in the previous paragraph. Also, note the use of the transition word:
However, *this state of affairs* cannot be blamed on universities alone.

The basic structure of a paragraph makes it easy to connect it to other paragraphs. A paragraph communicates only one idea and consists of two or more sentences. Usually, the first sentence is the main sentence: it tells the reader what idea the paragraph will discuss and so creates an expectation in the reader’s mind. The sentences that follow elaborate on the main sentence and these elaborating sentences set up the next paragraph. The first sentence of the next paragraph will then draw on something that was said in the previous paragraph, and so on.

The length of a paragraph also allows for some repetition of key words and phrases mentioned in a preceding paragraph. This repetition establishes a further connection between paragraphs. Don't be afraid of being repetitive: it is better to be repetitive than to be misunderstood. Repetition binds ideas.

In order to make connections between sections, you can use all of the techniques mentioned above for sentences and paragraphs, but I also recommend another technique: that of creating connecting sentences.

The technique works like this: you express the argument of each section as a *single full sentence*, and then write a linking sentence that connects the arguments. The secret is to find the correct order of arguments so that a linking sentence makes the one argument flow into the next one. There has to be a good reason that explains why you put the ideas in a particular order, and the linking sentence makes that clear.

Figure 4.9 shows a possible order and link at the first level of the example from Appendix A. Notice that when we read the arguments and the linking sentence, it reads like one long flowing sentence that makes sense: "The meaning of arrogance is misunderstood, but this does not mean that arrogance is always good, because there are different kinds of arrogance."

There are often many different ways to connect the claims. The only real test is whether the claims and their connections read like one long flowing sentence that makes sense. This means that connections between claims are rarely simple words like 'and' or 'furthermore'. Linking sentences usually have to be longer phrases that make the one claim flow into the next one without it being contrived.

If you can't find a connection between arguments or the connection feels forced (indicated by the broken arrows in Figure 4.10) there can be only one of three diagnoses: (1a) the claims are in the wrong order; (2a) there is a missing claim; or (3a) one of the claims is redundant. The diagnosis leads to the solution: (1b) try a different sequence of claims; (2b) determine what other claim needs to be made to connect the two claims you are trying to connect; or (3b) see if the connection can be made if one claim is removed.

Let's return to the example of the article about arrogance, showing the possible sequence and linking sentences at the second level. Figure 4.11 puts the claims in the order used in the article.

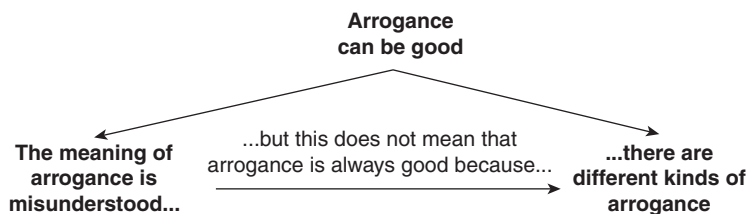


Figure 4.9 Connecting sections with a linking sentence

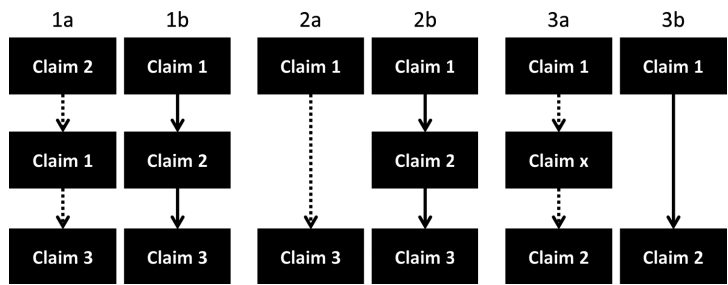


Figure 4.10 Diagnosing and fixing a lack of flow between arguments

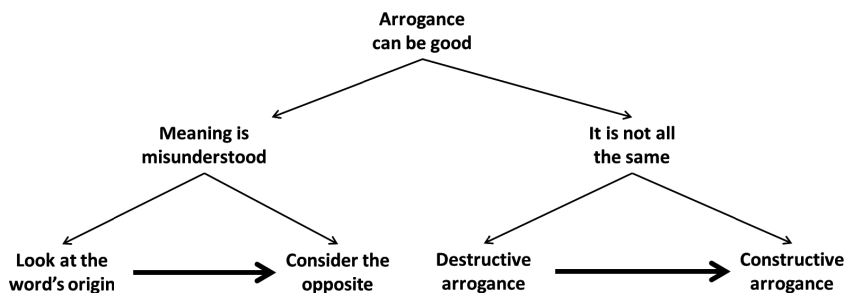


Figure 4.11 Linking the claims at the second level

Due to lack of space, I cannot show the linking sentences in Figure 4.11, but I do illustrate them in Table 4.2. Remember, there is no single correct order or linking sentences – as long as it makes sense to the reader and logically flows towards the overall claim.

If these were sections, where would one place the linking sentences? They have to be positioned between the end of the previous argument and the start of the next one. So, you could either place the linking sentences at the end of the first section or at the beginning of the next section.

Table 4.2 Linking sentences used in Figure 4.11

First argument...	...and the reason it leads to...	...next argument
The origin of the word tells us a lot about its meaning...	...and to see that this meaning is still applicable today we should...	...consider the opposite of arrogance today.
Destructive arrogance is bad and should be discouraged...	...but people want to discourage all kinds of arrogance because they don't know that...	...arrogance can also be constructive.

4 Writing complex arguments argumentatively

At this point, you will be able to produce a hierarchical outline structure for a piece of writing. So far, I have shown you: how to construct a single argument; how to derive smaller claims from a bigger claim; how to connect them; and what to put in them. Once the time comes to expand on each claim, you will want to ensure that your writing has an intellectual and argumentative quality. To do so you have to know how to write in your own voice while using other authors' ideas, and what kind of language to use when doing so. The next sub-sections show you how.

4.1 *Writing with your own voice using other authors' ideas*

To learn argumentative writing, it is best to compare it to something you already know how to do: having a conversation. Argumentative writing is like having a really good conversation. A good conversation happens when there is interaction between two or more people with different views all trying to move toward some kind of conclusion. Compare that to bad conversations, which are usually those where everyone just agrees or just disagrees with each other. Poor conversations are also those that don't go anywhere, like your typical boring small talk conversation or heated argument.

Most people don't realise it, but, when you write, you are actually having a conversation: you are talking and other authors are talking through you as you refer to them to substantiate your argument. Not only are you and these authors talking to each other, all of you are also talking to the readers. If it is a good conversation, you will all be talking about an issue about which there can be different perspectives, and the conversation will lead to a point. The hierarchical diagram we constructed in the previous section helps you to create such a conversation.

To better help you understand how to use the idea of conversation in your writing, you need to think of a particular kind of conversation: the issue-centred panel talk show. This is a kind of talk show where a talk show host gets guests with different perspectives together to have a discussion around a specific issue. In such a talk show, the talk show host integrates different voices by means of a guided conversation with an end to reaching some conclusion. Can you think of examples of such talk shows?

Imagine you are such a talk show host. You choose an issue to talk about in front of an audience. Obviously, you will select an issue about which you and your audience care, or which you can convince them to care about. But what kind of issue do you select? Probably one about which there are different views. What kind of guests do you get to talk? Ideally, those who would bring different perspectives, so that you can get a lively conversation going.

Table 4.3 Argumentative writing as an issue-based talk show

<i>Issue-based talk show</i>	<i>Argumentative research writing</i>
Talk show host	You as the author
Guests	Other authors you reference
Audience	Your readers
Issue	Overall claim or thesis
Conversations	Written arguments
Talking points	Smaller claims

You do the same in argumentative writing. You are the talk show host of your thesis or paper. You present your claims; that is, statements which allow for differences. Your talk show happens on paper and the guests are the authors that bring different contributions that help you argue your claim. You make the guests talk to each other and to your readers through your arguments.

How do you get the conversation going in an actual talk show? In the talk show, you would identify ‘talking points’; that is, particular aspects of the issue where differences exist. If you select good talking points, there will no boring patches or uncomfortable silences in your talk show.

Similarly, in argumentative writing, you find talking points around which you can make arguments. We did this in the previous section when we derived smaller arguments from the overall thesis. Every branch in that hierarchical structure is a ‘talking point’ and you will draw in different authors to discuss each point and come to a conclusion.

Table 4.3 summarises the metaphor so far. Now let’s take it a step further.

Do you, as the talk show host, form an opinion before the show? Most likely, you do, because you would have needed to do some research before the show to help you identify the possible perspectives.

Similarly, in argumentative writing, you need to know what you want to say; that is, you need to know your thesis. You need to know this before you start writing. Otherwise your writing will have no direction, and you won’t be able to locate and outline your talking points. This means you must have done a lot of reading *before* you start to write.

What do you do with your opinion as the talk show host? Since you care about the issue, you would steer the conversation in favour of your argument, in such an inconspicuous way that you, without coming across as biased, make it look more convincing. For example, instead of directly disagreeing with the guests, it looks better if you have the guests criticise each other and then use these critiques to steer the audience’s perception in favour of your view.

You do the same in argumentative writing. Let’s return to one of the previous examples:

The youth unemployment rate is 55% (SSA, 2012). A quarter of these young people have studied towards a qualification at a university. So it appears that universities play a role in graduates not being employable. However, Smith (2011:15) argues that individual characteristics are more important when it comes to determining employability. While this is true, Jones (2010) showed that universities can change individual characteristics. This suggests that universities are indeed failing their graduates because instead of developing them as individuals, they focus mainly on conveying generic subject knowledge.

Notice that the paragraph makes a clear argument: universities are not doing enough to make students employable. You can also see that there are four people involved in making the argument: the writer; the statistical agency (SSA); Smith; and Jones. The writer leads them to this conclusion, but when Smith mentions a counterargument, the writer does not disagree directly, but uses Jones to refute Smith's argument.

Now, we come to the most important part of the metaphor that helps you understand writing in your own voice. Who do you think dominates the conversation during a talk show? It is the talk show host, of course. And who talks the least during the talk show? The talk show host again!

But how is it possible that the person who talks the least can dominate the conversation? The talk show host dominates the conversation because it is the host who: decides the topic of the conversation; decides who will be the guests who will participate in the conversation; decides the talking points; guides these guests by the questions he/she asks; and, when required, switches from one guest to another in order to get the argument across.

Similarly, as the 'talk show host' of your own piece of writing, you will dominate the conversation without ever directly having to say what you think. This is because *you*: decide what the thesis is; decide how it will be broken into smaller arguments; decide which authors will participate; decide what they will say; use the authors to support or criticise certain points; and decide when to switch from one author to another and how to connect their ideas. In this way, you control the direction of the conversation and make the authors say what *you* want to say.

Let's take a new example so you can see how similar academic writing is to a talk show.

Malthus (1798) argued that population growth will eventually outstrip the rate at which food can be produced, thus leading to famines and wars over scarce resources. This pessimistic view was criticised by many economists for not anticipating the possibility of technological advances (Friedman, 2002) and birth control (Thomas, 1998). However, in recent times, Malthus's predictions are becoming true, especially in developing nations. Technological progress is causing more problems – such as climate change and pollution – that are harming the quality of resources required for food production. In fact, Dixon (2008) predicts that very

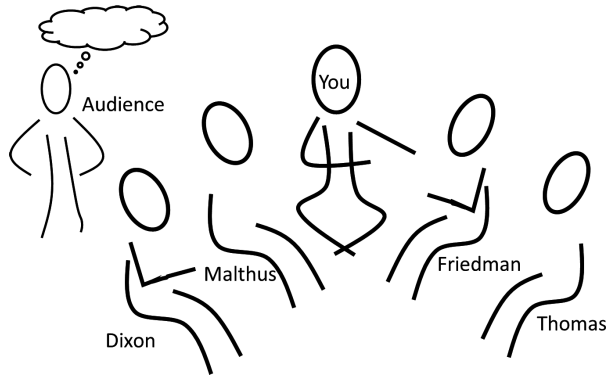


Figure 4.12 The hidden talk show behind the extract

soon major wars will be fought over increasingly scarce resources such as water.

Notice how, through my own words, I am putting the authors in conversation and leading them to a conclusion. It was easy to get the conversation going because I identified a ‘talking point’ – an aspect where one group of authors (Friedman and Thomas) can engage another group (Malthus and Dixon). You can imagine them in a room: Malthus says one thing; to which Thomas and Friedman provide a critique; followed by Dixon supporting Malthus by addressing Thomas and Friedman’s account. Perhaps you can already see that the author – which is ‘you’ in Figure 4.12 – is facilitating a conversation between the four authors, with the aim of convincing the readers (audience) that we are heading for an environmental crisis.

If you can’t yet see it, then let me translate the text for you into a talk show:

YOU: Let’s talk about whether our planet’s resources can continue to support us. What is your view Reverend Malthus?

MALTHUS: I have been warning against this for centuries. As early as 1798, I explained that population growth will eventually outstrip the rate at which food can be produced, thus leading to famines and wars over scarce resources, because...

THOMAS: I really think that is a bit pessimistic, Malthus. Come on now, there hasn’t been any ecological crisis since you published your book. In fact, technological advances helped to significantly and consistently raise food production...

FRIEDMAN: And don’t forget birth control, which reduced population growth – something you did not anticipate at all, Malthus!

YOU: But wait, just because Malthus’ prediction hasn’t come true yet, does not mean it won’t happen at all. Are we not seeing this very same

technological progress causing more problems now – for example pollution – which is reducing the quality of our natural resources?

DIXON: If I may chip in here, I would like to support you in this. Good quality water is becoming increasingly scarce due to global warming and pollution. I predict that very soon major wars will be fought over increasingly scarce resources, such as water, and probably other natural resources too.

Notice from this dialogue what the talk show host (you) does. The host raises talking points, connects different arguments and switches from guest to guest. In the end, the conversation is guided towards what the host wants to say, but without the host having to say so directly.

For more examples of the translation of an academic text into a talk show (if you want some practice) see Appendix B. For practice, I suggest you take the first text and translate it into a talk show, and then take the second talk show and translate it into an academic text. I provide my versions, but there are countless ways to do these ‘translations’.

4.2 *Writing the talk show in academic research*

Metaphors are incredibly useful when learning a new skill. If well-chosen, they capture the complexity of a skill while being easy to keep in mind. Keeping a metaphor in mind allows it to release information in manageable chunks as you need it. I discovered this when I started learning how to write academically decades ago. All I had to do was keep my personal metaphor in mind while writing and my writing would improve dramatically. Now the talk show metaphor is even more powerful as a way to understand argumentative academic writing than my personal metaphor back then, so let’s explore it more deeply.

Using metaphors is fine, and useful, as long as you are aware that a metaphor always breaks down at some point. One big difference between a talk show and written argument is that you use different registers, that is, your manner of communication differs. In academic writing, you have to avoid the informal conversational language normally used in actual talk shows.

One book I highly recommend to help you with the language of argument is *They Say/I Say* by Graff and Birkenstein (2006). In their book, they provide templates you can use to express the three main moves we make in an academic text:

- **I say:** These are all the moves you make where you convey your *own* ideas and thoughts to your readers, mainly when you warrant your claims, make sense of counterarguments, connect arguments or draw conclusions.
- **They say:** These are the moves where you convey the ideas of *other* authors. You do this mainly when you need authors to respond to each other or want to communicate their ideas to the readers.

- **You see:** In real-life conversations, it is possible to see what is happening in the conversation. In writing, however, the conversation is more ethereal and not as concrete. So, you have to constantly give signals to the reader about what is happening in the conversation. These signals are things like references, headings, summaries, transition words, hooks, linking sentences and more. These are the moves in which you draw a person's attention to something they might not see without your help, almost like in a real conversation when you *direct attention* by saying 'now you see that...' or when you ask 'you see?'

Figure 4.13 uses an adapted version of the rhetorical triangle to explain where these moves come from.

In the figure, *logos* represents the existing body of knowledge, and that knowledge speaks to the audience of readers (*pathos*) via previously published authors. Since these authors are distinct from you, when this occurs 'they say' things. As the author of your own research (*ethos*) you can also speak to the audience through 'you see' moves, or you can speak to the previously published authors through 'I say' moves. In contrast to real-life talk shows, which rely mostly on 'they say' and 'I say' moves, written talk shows – that is, academic arguments – rely much more on 'you see' moves.

Graff and Birkenstein (2006) break each of these three moves up into several templates. I would love to summarise their book here, but, for copyright reasons, I am only going to make you aware of the basic ideas and templates based on the three main argumentative moves. In this chapter, I will provide only a few of the most basic templates you need to know, but if you find this helpful, then buy a copy of their excellent book. Initially, you may follow these templates closely, but as you start to understand the thinking behind them, you will eventually develop your own style.

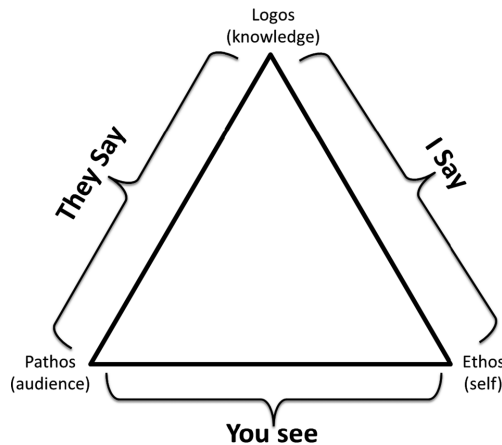


Figure 4.13 The basic moves and participants in an academic argument

4.3 *The fundamental rule behind the moves: everything is a response*

Bear this rule in mind as you write using the templates and your writing will improve significantly: everyone responds to someone. In any conversation it would be strange if someone said something that had nothing to do with what was said moments before. When the host talks ('I say'), she responds either to another guest or to something she, herself, just said. Similarly, when a guest talks ('they say'), he or she either responds to another guest or to the host. And then there is the host who responds to the audience all the time ('you see') to make sure they keep track of what is happening. This is especially clear when we translate academic papers as we did in the first talk show example with Malthus, Friedman, Thomas and Dixon, and with the examples of talk show 'translations' in Appendix B.

So a useful question to ask yourself as you write is: who am I responding to or who is this author responding to? If a statement does not directly respond to something that just happened, then one of the following is true:

- The statement is irrelevant so delete it.
- The statement belongs earlier or later, so move it.
- The statement is a new issue, so connect it so that it logically flows from what was just said. For this, you can use transition words, hooks or linking sentences.

Let's take the first talk show from before and analyse it in Table 4.4. From it you will see that all three moves featured, and that every move is a response.

Everything is a response in research writing, but unlike an actual talk show, in academic conversations there is really just one person speaking: you, the author. You are the one writing so regardless of which move is occurring, you are making it happen. All responses happen *through the author* as the next section explains.

4.4 *How the responses happen: the author as puppet master*

No matter who speaks, they speak through you: like a puppet master in Figure 4.14 (see page 75) who makes his puppets speak. The puppets are the other authors who appear to speak, but in reality, it is the master who is making them say things. Through the puppet master, the puppets speak to each other or to the puppet master. Of course, unlike Figure 4.14, in research writing you will usually have more than two 'puppets' at a time in a paragraph and many more in a section. Also, the puppets will change all the time because every paragraph and section will need new 'puppets' for new arguments.

Sometimes, the puppet master may speak to the audience directly. In some disciplines, you do so directly using the first-person; though, in other

Table 4.4 Breaking a talk show into moves

<i>Extract</i>	<i>Moves and responses</i>
Malthus (1798) argued that	You see (speaking to the audience by signalling who is talking)
population growth will eventually outstrip the rate at which food can be produced,	They say (Malthus' view which would have been a response to the previous paragraph)
thus leading to	You see (speaking to audience by signalling the implication of Malthus' view)
famines and wars over scarce resources.	They say (Malthus' responding to himself)
This pessimistic view was criticised	You see (signalling to the audience that a counterargument is coming)
by many economists for not anticipating the possibility of technological advances (Friedman, 2002) and birth control (Thomas, 1998).	They say (Friedman and Thomas responding to Malthus) and you see (signalling to the audience who is talking)
However, in recent times,	You see (signalling a counterargument)
Malthus' predictions are becoming true, especially in developing nations. Technological progress is causing more problems – such as climate change and pollution – that are harming the quality of resources required for food production.	I say (taking a position by responding to Malthus in agreement and to Friedman and Thomas in disagreement)
In fact,	You see (signalling to the audience that you will support your position)
Dixon (2008) predicts that	You see (signalling who the person is who will now support you with evidence)
very soon major wars will be fought over increasingly scarce resources such as water.	They say (Dixon responding to your position in support)

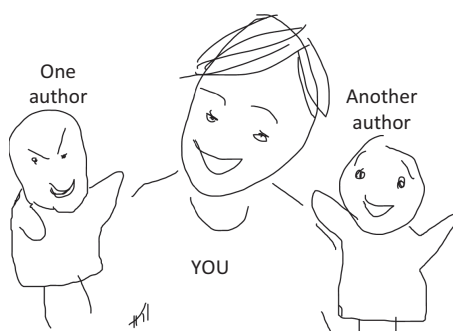


Figure 4.14 Puppet and puppet master

disciplines, this is avoided because it is analogous to what is known as ‘breaking the fourth wall’ in theatre. If you do address the audience in the first person, you could do it like this:

- My own view, however, is that ____.
- After doing an error analysis, I found ____.
- I agree with Jones (2006), but he may not have taken into account that ____.

See if this is acceptable by looking at the articles in your discipline. If not, you can signal your own voice simply by not referencing the statement, or by signalling it clearly as responding to another author, for example:

- However, it is not true that ____ as is evident from ____.
- It is more likely that ____.
- It seems Welch (2016) is right when he observes that ____.
- Yet, a simple error analysis reveals that ____.
- Gardner (1987) is both right and wrong that ____.

It is possible to convey a wide range of specific messages directly to the audience without using the first person, for example:

- To signal your commitment to your claim, use phrases like: ‘It is possible that ____’, ‘Perhaps ____’ or ‘____ might show ____’ to show that you are not certain; and, if you are more confident, use phrases like: ‘Clearly, ____’, ‘It is evident that ____’, or ‘Jones (2016) demonstrates that ____’.
- To bring your own values into the text, use phrases like: ‘Surprisingly, ...’, ‘Unfortunately, ...’ or ‘Smith (2012) makes the important point that ...’
- You can also signal your own voice when you criticise another argument, for example by stating: ‘Rogers (2009) overlooks ____’, ‘Recent research has shown that Venter’s (2014) conclusions are outdated, since ____’, or ‘One can challenge the view that ____, because ____’.
- You can signal that a view is different from yours by labelling it as such, so that your view becomes clear by implication, for example by writing: ‘Proponents of ____ suggest that ____’, ‘Socialists claim that ____’ or ‘A more orthodox perspective on the problem would be that ____’.

When you are not saying something, you need to explicitly signal that others are talking. This is one of the main reasons we use references. If there are no references, the audience will not know who is talking or will assume it is the author talking, so you will often find the following:

- Smith (2006) states that ____.
- According to Smith (2006) and Singh (2005) ____.

- Government officials should, as Sithole (2014) argues, focus on ____.
- Most theoretical economists (Friedman, 1997; Samuelson, 2013) believe that ____.

To reinforce the point about the author as puppet master, consider the first bullet point above. Even though Smith is saying something, Smith is not physically present, so he is not actually speaking – it is the author who is ‘putting words’ in Smith’s mouth. It is for this reason that you have to be extremely cautious. Just because you, as the puppet master, can make the ‘puppets’ say anything, it does not mean you should – think carefully about what you make other authors say because, if you don’t, you will probably misrepresent them. Whatever you make another author say has to be something that that author would have said in that situation, so you cannot deviate from the meaning or intention of an author’s words when you write.

Now that we have addressed the danger of making other authors ‘say’ something, there are times when they might not say something explicitly, but there is something important that is *implied* or assumed. In such a case, you need to make it clear that you are identifying an assumption or implication derived from their words, and that it is not precisely what the author said. See below ways in which to do this:

- What we can derive from Stiglitz’s (1991) study of ____ is that ____.
- Krugman (1979) does not admit it, but he seems to assume that ____.
- Economists often do not find it necessary to prove that ____ as is evident from Arrow (1988) and Hahn (1964).

Use these kinds of templates with care. If you identify an implication or underlying assumption, you need to argue why you think it to be the case, otherwise you lay yourself open to the charge that you are misinterpreting others.

Of course, the safest option is to use the exact words of other authors by quoting them. However, you will notice, from the papers in your discipline, that this is done very rarely because it is difficult to read text that consists of many quotations. Also, a person using too many quotations opens themselves up to the accusation that there is too little of their own thinking in their writing.

So when is quoting acceptable? Quoting is acceptable on condition that: (1) you need the exact words of the author in support of your argument; (2) you put those words in quotation marks, together with a reference and the exact page number on which it appears in the source; and (3) you frame the quotation with your own words and explanations (as explained next).

It should be noted that, if one really does quote properly, it is not easier than using your own words, as many may think. On the contrary, it takes a lot more work. This is because when quoting, you are ‘ripping’ a quotation out of another article, from a different context with different arguments, and

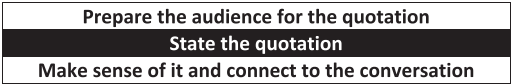


Figure 4.15 The quotation sandwich

placing it into your text. Now, you have to make the ‘ripped out’ quotation ‘fit’ in yours, without it being incongruous, by integrating it with your argument. To do this you have to do a ‘quote transplant’ or what Graff and Birkenstein (2006:42) call a “quotation sandwich”.

To make a quotation sandwich, you first prepare the reader for the quote and ensure that the conversation introduces and leads on to the quotation. Then, you quote. Thereafter, you explain what the quotation contributed to the conversation. It resembles a sandwich because, as Figure 4.15 shows, something comes above and below the quotation.

Let’s look at some templates for making quotation sandwiches. Firstly, you should introduce a quotation, in ways like this:

- In contrast, Krugman (1979:33) insists that, “_____”.
- Popper (1939:912) explains this as follows: “_____”.
- It seems that Ade (2012:215) agrees when he concedes, “_____”.

You then provide the quote, and follow up with a sentence or two that explains how the quotation contributes to the conversation. Here are some templates for explaining quotations after you introduced the quotation:

- What Jones (1978) believes is _____.
- In other words, Augustine (2014) thinks that, _____.
- It appears that Mbeki (2010) is saying that, _____.

As an example, notice how Gilland (2002:48) treats a quotation:

Food production has outpaced population, chiefly as a result of ... A half-century ago few, if any, would have considered this possible. Roberts (1951) concluded that: “within a century or so, the world’s population will almost certainly be stabilized at something like three thousand million, which is the utmost that the earth is likely to be able to feed”. Roberts did not believe that mineral fertilizers could significantly increase crop production for more than a few years.

This is broken down into the parts of the quotation sandwich in Table 4.5. Making quotation sandwiches forces you think carefully how every quotation supports your argument. If you cannot make a convincing quotation sandwich, it means the quotation does not belong in your text.

Table 4.5 Example of a quotation sandwich

Prepare	Food production has outpaced population, chiefly as a result of.... A half-century ago few, if any, would have considered this possible. Roberts (1951) concluded that:
Quote	“within a century or so, the world’s population will almost certainly be stabilized at something like three thousand million, which is the utmost that the earth is likely to be able to feed”.
Make sense	Roberts did not believe that mineral fertilizers could significantly increase crop production for more than a few years.

If you understand that you are the ‘puppet master’ in an ‘academic talk show’, you may realise that this idea of making sandwiches is not limited to quotations. It applies to anything said by others or any idea you obtained from another source. Whatever comes from other sources (they say) first belonged to ‘other talk shows’, and therefore need to be transplanted and integrated seamlessly into your ‘show’ by going via you, as the author.

For this, we follow the same procedure: you first prepare the ground for an idea; then you state or summarise it; and, finally, make sense of it and connect it to the next idea or statement. This was illustrated in Figure 4.15; except that, now, instead of just quotations, we ‘sandwich’ other authors’ *ideas* as well.

Take the following extract from Wentzel (2017:118) below. Can you see how the author makes sandwiches with other authors’ ideas?

...This political aspect of policy problems makes beliefs resistant to empirical tests so that the inconsistencies remain.

More data does not eliminate inconsistencies in beliefs (Barberis and Thaler, 2003). As recorded in the research of various experimental and behavioural economists (Cartwright, 2011), systematic biases appear when humans form beliefs and such biases often remain constant even if contradictory data emerge. This world with real people – who have limited capabilities and conflicting beliefs, engage in power struggles and exhibit diverging interests – not only guarantees that the collective understanding of any economic issue will be rife with inconsistencies, but also that these inconsistencies will be impervious to data.

This is of course reminiscent of Hattiangadi’s (1978) definition of intellectual problems as logical inconsistencies in beliefs that cannot be eliminated simply by gathering more data...

Let’s break Wentzel’s text down into layers in Table 4.6. This is not an exact analysis, because there are overlaps between the different parts of the sandwich.

If the writer did not ‘sandwich’ everything, it would be impossible for the audience of readers to see the connections and keep track of the meaning of

Table 4.6 Analysing the extract

Connect to previous idea	This political aspect
Prepare for Barberis and Thaler	of policy problems makes beliefs resistant to empirical tests so that the inconsistencies remain.
State their idea	More data does not eliminate inconsistencies in beliefs (Barberis and Thaler, 2003).
Link to and prepare for Cartwright	As recorded in the research of various experimental and behavioural economists
State her idea	(Cartwright, 2011), systematic biases appear when humans form beliefs and such biases often remain constant even if contradictory data emerge.
Make sense of what Barberis and Thaler, and Cartwright said	This world with real people – who have limited capabilities and conflicting beliefs, engage in power struggles and exhibit diverging interests – not only guarantees that the collective understanding of any economic issue will be rife with inconsistencies, but also that these inconsistencies will be impervious to data.
Link to and prepare for Hattiangadi	This is, of course, reminiscent of
State his idea	Hattiangadi's (1978) definition of intellectual problems as logical inconsistencies in beliefs that cannot be eliminated simply by gathering more data.

the conversation. As you can see from the extract above, this happens most of the time through hooks (e.g. “this political aspect”), linking sentences (e.g. “as recorded in the research of”) and when warranting evidence (e.g. explaining the evidence offered by Barberis & Thaler and Cartwright). Though not shown in the extract, it also happens when you make sense of counterarguments.

To summarise so far, everything in research writing is a *response*, and all responses are communicated through the *medium* of the writer. Let's now look at conversation as a whole and illustrate further templates for responses at different stages.

4.5 Stages of a conversation: open, explore, close and connect

No conversation starts without a reason. There must be something to which you and others can respond by bringing in different ideas, views and perspectives, which opens the conversation. As the writer or ‘host’ of your talk show, you provide this reason by stating a view or introducing some debate and this will invite the expression and exploration of these differences in the next stage. The host guides the guests through various talking points, playing

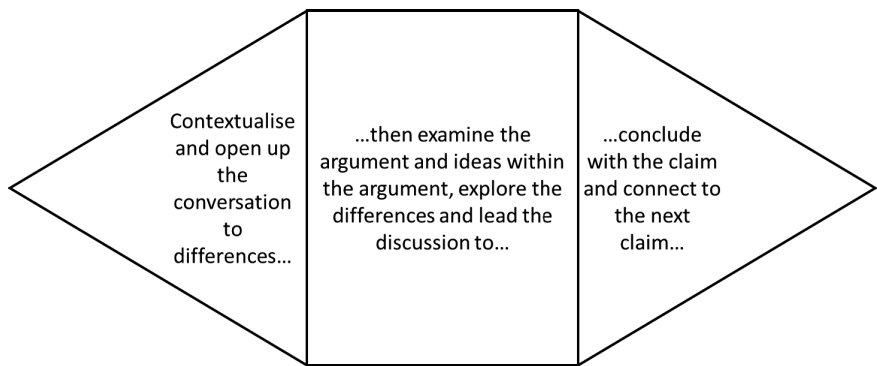


Figure 4.16 Open, explore, then close and connect

with the differences; until, finally, the host brings it to a close by concluding and connecting the conversation to the next section or chapter. Figure 4.16 summarises the process.

In a paper or thesis, this applies not only to the overall structure of introduction–body–conclusion, it also applies to chapters and sections. In fact, it is something you do with every talking point in the talk show that you have identified using the imaginary conversation technique.

Table 4.7 summarises the different moves in the different stages of a conversation. While there are some grey areas, the table shows that we find most ‘they say’ moves at the beginning of the conversation, and thereafter, when exploring the issues. ‘I say’ moves, however, usually occur when opening and closing a conversation. With ‘I say’ moves, you position yourself as part of the literature and you take a position as a fellow author among previous authors who have engaged in these arguments. While, with ‘I say’ moves, you are positioned amongst other authors, with ‘you see’ moves, you stand with the readers and guide them through the conversation.

Let’s consider these stages in more detail with possible templates.

Table 4.7 Different moves in different stages

	<i>Open up</i>	<i>Explore</i>	<i>Close down</i>
I say	State your own position		Disclaim Proclaim
They say	Set up other positions Set up a debate	Summarise positions and responses	
You see	Signals of preview, standard views and debate	Connect	Signals that conclude, direct attention and connect

a) *Open*

Three common ways to open a conversation are: (1) state your claim up front; (2) state a common or standard view in the discipline; or (3) introduce a debate. Let's consider them and possible templates for each one.

In some disciplines, it is common to state your argument in the introduction of a thesis or chapter, and often at the start of a section. This happens not only to orient the reader, but also to provoke other authors to respond. We often do this in real life when we make statements in the hope that our friends will ask us about it and so get a conversation going. While I recommend that you look at how authors in your discipline do it in the introductions of their papers, here are some of the common templates you can use:

- This article argues ____.
- As this section will suggest ____.
- In what follows, it will be shown that ____.
- Often it may simply be a statement of the claim, initially without a reference, at the start of a paper, chapter or section

Your view is not the only view that will provoke others. Sometimes authors state a common or standard view in the discipline in order to raise questions about it or to suggest that it is not as certain as people expect. Here are some templates:

- Policymakers routinely argue that ____ (Morales, 2014).
- Most poor and low-skilled South Africans today believe that immigrants ____.
- Conventional wisdom amongst anthropologists is ____ (Sen, 2009).
- Many researchers assume that ____ (Mzwakhe, 2016; Zwicky, 2008).

Be careful, unless it is a genuinely common view, you will need to provide a reference. Too many novice researchers use phrases like 'most researchers believe...' without any reference because they are too lazy to actually read the literature. In some cases, however, that kind of generic statement does not necessitate a reference. Though there are exceptions, a simple rule is that you don't need to provide a reference if it is a common view amongst non-scientist groups. Even so, sometimes it may not be so obvious that a view is actually generally adopted among non-scientist groups, in which case, you would need to provide a reference.

Of course, the easiest way to get a conversation going at the start of a paper or section is to simply introduce a debate. This forces readers and other authors to take sides and respond. One good approach is to start with

a summary of a debate. When you summarise the opposing positions, do so briefly using interpretive summarising as discussed in chapter 1. Here are some templates:

- One debate in creativity studies has been _____. On the one hand, those inspired by stage models (like Wallas, 1913; Smith, 2016) argue _____. On the other hand, experimentalists contend _____ (see Weisberg, 2014).
- When it comes to the topic of _____, most linguists agree that _____ (Romer, 1999; Kabundi, 2015). The main point of contention is on the issue of whether _____. One school of thought (Gorman, 2009) is that _____, while others claim that _____ (Fudau, 2017).
- With regard to _____ there are those who take the extreme position that _____ (Nyembezi, 2010), but an increasing number of researchers are now taking the more nuanced view that _____ (Dibakwane, 2017).

Use ‘you see’ moves to signal what kind of conversation opener you have chosen. For different openers, use words and phrases like the following:

- Preview: future tense signals (‘will argue’, ‘will be suggested’) and signals of location (‘in this paper’, ‘in this section’).
- Standard view: signals that a standard view exists (‘common’, ‘conventional’, ‘popular wisdom’) and signals of degrees of acceptance (‘most’, ‘many’, ‘majority’).
- Debate: signals that opposing views exist (‘debate’, ‘issue’, ‘contention’, ‘disagreement’) and signals that show where one view starts (‘while’, ‘whereas’, ‘in contrast’).

Opening a conversation at the start of a paper, thesis or section has to be something that invites responses from those who have other perspectives to offer. When this happens, we enter the exploration stage.

b) Explore

In the exploration stage, you will respond and make authors respond to each other or to you. Exploration involves expanding on the arguments raised by, or implied in, the opening. If you used the imaginary conversation technique, you already know the main talking points that develop the conversation. At every talking point, you will summarise the arguments of others (‘they say’) and then create interactions between them; mainly by using ‘you see’ moves of connection to signal the kind of interactions. By summarising a view, you not only expand on it, you also make it clear what motivates the responses to it.

Summarise others’ views fairly, even if you disagree with them. To summarise another view fairly, you have to suspend your own beliefs

temporarily so that your reader is not misled by your own prejudices. You also need to give the reader sufficient supporting details so that they can assess the merits of the other views on their own. Your summary is fair if the reader is not able to detect whether you agree or disagree from the summary alone. Your position should only become evident in the responses to the summarised view.

The summary should be designed to move your argument further. You will not summarise everything the other author said: only that aspect that contributes to your own argument. In other words, writers trying to make different claims will focus on different parts of an article when summarising. This was explained in chapter 1 and was called ‘interpretive summarising’. This also means that summaries will be short: as long as a sentence and rarely more than a paragraph.

Once you have presented the summary, you then respond. The danger is that novices tend to spend too much time summarising and forgetting that the real conversation happens not in the summary, but in the responses to it and the interaction between the views. When novices make this mistake, their writing reads more like a telephone directory of summaries that resemble something as shown in Figure 4.17.

Now contrast that with the actual extract with Malthus and the others from earlier. The kind of writing you see in Figure 4.17 is boring and static in that it does not move anywhere. This is because, at the end of these paragraphs, it would still not be clear what the purpose of listing four summaries was. The actual extract from earlier also moves faster, because, instead of summarising everything that every author said, it takes from every author *only* what they can contribute to the conversation *at that particular moment*.

There has to be a response to any summary, so ensure that it clearly follows from the summary and shows how the response fits with your argument. This should happen if you remember to use the sandwich technique. The response then puts a ‘spin’ on the summary, showing either how the source supports your argument or how responding to it helps to move the conversation forward.

Malthus said ... (discuss Malthus’s view for a paragraph)

Dixon found ... (discuss Dixon’s view for a paragraph)

According to Friedman (discuss Friedman’s view for a paragraph)

Thomas stated that ... (discuss Thomas’s view for a paragraph)

Figure 4.17 A list of summaries is not a conversation

Here is an example drawn from an author whose overall argument is that global and local institutions are not designed to ensure that globalisation serves the interests of the poor:

The most widely cited case for _____ has been set out by _____. Briefly summarised, their case rests on _____. (*first he briefly summarises the strongest argument of other authors that disagree with him*). Some of the problems with this approach stem from _____ (*now he criticises the methods they used to generate supporting data*). The more serious problem concerns _____ (*then he criticises the underlying assumptions which by implication strengthens his own argument*). (Taken from Watkins, 2002.)

Graff and Birkenstein (2006) explain that responses that move a conversation forward are not simply ‘agree(ing)’ or ‘disagree(ing)’. If it were like that, conversations would be boring – as you will have noticed in conversations where a person only answers your questions with ‘yes’ or ‘no’. Rather, when making arguments, every response has to be a response plus something. Such responses lie on a continuum (see Figure 4.18), depending on what you add to the response.

When you want to disagree with some authors, don’t simply disagree: rather, offer an argument with evidence and reasons to support why you disagree, or draw other authors into the conversation who will help you to do it. This ensures that you move the conversation forward and contribute something to it.

Here are some ways to disagree with a difference:

- Rogoff (2005) is wrong because he ignores the evidence offered by Makhubela (1995) that _____.
- Dorfman (2004) makes the questionable assumption that _____.
- There is sufficient reason to disagree with O’Leary (1998) because, as recent research has shown, _____ (provide references to this research).
- On the one hand, Smith (2013) argues _____. But, on the other hand, he argues _____. This argument is inconsistent since _____.
- Sabat (2015) suggests that because of _____, Mwamba (2013) draws incorrect conclusions about _____.

Argument is not simply about disagreeing and criticising, it is also about agreeing and building on the work of others. But even when you agree and

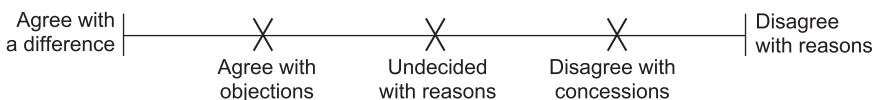


Figure 4.18 Continuum of responses

build on the work of someone, you have to make your own contribution to the conversation or introduce other authors who have something to offer and use them to elaborate further. The additional contribution may be small, but it has to be different and move the discussion forward. It could be as simple as additional supporting evidence or simply a way to ‘translate’ the argument of the author in a way that is easier to understand. The important thing is not simply to ‘parrot’ what others say. You need to show you do something more with others’ ideas in a way that is valuable to the conversation.

Here are some examples of agreeing with a difference:

- The experience of Malaysia with _____ confirms Ndlovu’s argument (2012) that _____.
- Dube (2005) appears to be correct, because as subsequent studies showed _____ (provide references to these studies).
- However, if Keynes (1936) is right, we are asking the wrong question because _____.
- Richards (2017) not only confirms the hypothesis of Davids (2001), but shows its wider application in _____.

Whenever you agree with one view, you are most likely disagreeing with another view. Be aware of this, and let this awareness show in your writing. There is very little that is one-sided in research. Sometimes, you will agree up to a point or even disagree, but still see some merit in the opposing argument. It needs to be clear from your writing where you lie on the spectrum. If you lie closer to disagreement, stress the disagreement; the closer you are to agreeing, the more you should stress your agreement. Look at the sentences in Table 4.8 – notice how each column has a different emphasis:

Table 4.8 Agreeing or disagreeing

<i>Agree with objections</i>	<i>Disagree with concessions</i>
While Mantso (2017) is theoretically correct that _____, in practice the opposite is true.	Mantso (2017) may be wrong about _____, but his views are theoretically sound.
Whereas Mbele (2004) offers evidence that _____, Vilakazi’s (2005) refutations are more convincing since _____.	Though it should be conceded that _____, the evidence still suggests that _____ [provide references].
While Jones (2005:99) is right when she claims _____, she is on dubious ground when she argues that _____.	Botha (2010:23) is probably wrong in supposing that _____, he is closer to the truth in reasoning that _____.
Smith (2015) arrives at conclusions that are true in developing economies, but not necessarily in richer countries.	Even if Smith’s (2015) conclusions do not apply to developed nations, they still apply to poorer economies.

There may be some cases where you are not convinced by either side, or believe that both sides have equally strong arguments. You may then express ambivalence. But do not see this as an easy way out from having to think hard about the issue. If you express ambivalence, you need to indicate what will be needed to clear up the ambivalence, or under what conditions the argument will sway you one way or another.

Here are some examples of being undecided with reasons:

- The claim that ____ is not convincing even though it has not been disproven. On the one hand, _____. Yet, on the other hand, _____. To make a final judgement, one would need to know whether _____.
- There is no clear evidence for either view. Econometric studies (provide references) support _____, but the contradictory claims derived from simulations (provide references) are equally convincing. It seems that under conditions of _____, the theory that _____ is applicable, while under conditions of _____, it is not.
- The question of whether _____ depends on _____. If it is the case that _____, then _____. However, in the case of _____, it is almost certain that _____.

Let us look at how to use ‘you see’ moves to help us respond through agreement and disagreement (and even ambivalence). In section 3 of this chapter, we have already seen most of the ‘you see’ moves that are used to respond to arguments, connect them and move between them, as one would expect to find during the exploration stage of a conversation. These included transition words, hooks and linking sentences. The typology in Figure 4.19 summarises these.

	Continue	Break away
Same argument	Add a new idea that elaborates on the same argument (transition words and hooks)	Add a contrasting or contradicting idea within the same argument (transition words combined with hooks)
New argument	Add a whole new argument that supports or elaborates on the previous argument (linking sentences)	Add a new argument that disagrees with the previous argument (linking sentences combined with transition words)

Figure 4.19 ‘You see’ moves when responding

Looking at agreement, one can continue with the same argument, that is, the ‘same argument’–‘continue’ cell in Figure 4.19. This mostly happens between sentences when you add something to an argument or elaborate on an argument. It also happens between paragraphs within the same section when you elaborate on, or show the consequence of, an idea. Between sentences, you will use the transition words (see the first column of Table 4.1), which serve as signals of continuation; and, between paragraphs, you will use both hooks and transition words in combination. On the continuum of responses in Figure 4.18, this would fall on the left hand side.

One can break away while staying within the same argument – the ‘same argument’–‘break away’ cell in Figure 4.19. This can happen within a paragraph when someone contradicts what was just said or adds a contrasting idea. The same also happens between paragraphs within the same section. Within a paragraph, you rely mainly on the transition words in the second column of Table 4.1 that serve as signals of breaking away; and, between paragraphs you will combine hooks and transition words. On the continuum of responses in Figure 4.18 this would fall on the right hand side.

When moving between sections and chapters, you start completely new arguments – like new episodes of a talk show on new issues with new guests. You still have to ensure that the new talk show logically relates to the previous one by showing that it either builds on it or contradicts it. If it builds on it (the ‘new argument’–‘continue’ cell), then you need to at least use linking sentences, as explained before, to show the relationship. You will also use linking sentences when the new talk show stands in contrast (the ‘new argument’–‘break away’ cell), but here it is useful to combine it with transition words that signal breaking away (as seen in Table 4.1) to make this clear.

Transition words, hooks and linking sentences are extremely important ‘you see’ moves. Without them, the audience will be lost, because in a written talk show, it is not possible for the audience to see facial expressions, notice body language, hear the tone of voice and the other ways that help us realise what is happening in a real-life conversation.

Furthermore, it is through these ‘you see’ moves, that you create knowledge. New ideas are nothing more than old ideas in new combinations. As the writer of an original paper or thesis, you put ideas and previous authors together in combinations that do not exist in other research. Simply throwing them together without connecting them is as useless as a talk show where everyone talks about anything all at the same time. Sure, this will be new, but it won’t make any sense. Through these ‘you see’ moves, you guide the interactions between ideas and authors and show how they work together to make new arguments.

c) Close and connect

At the end of the conversation, you silence the different voices, except for one, into a single view or conclusion. It usually occurs at the end of a

section, paper or chapter; but if you have several arguments in a section, you might even close your argument in the middle of that section. One thing is certain though: every conversation that is opened has to be closed and connected to the next conversation. The closing should follow from the opening of the conversation and create a sense of closure. It should be clear that the two are related, that the closing takes a position based on the provocation of differences you achieved in at the opening.

If your closing comes at the end of a paper or chapter, it is usually a good idea to remind the reader of how you opened the conversation, and give them a short version of the logical flow up to this point. If you used the imaginary conversation technique to plan the argument, then you already have this short version. However, if the opening and closing are very close together – maybe within the same paragraph or short section – this is not necessary.

Here you will often find words and phrases that signal conclusion, such as:

- In conclusion, ____.
- From ____ it is possible to conclude that ____.
- The point of this discussion was ____.
- As was evident from ____, one may argue that ____.
- What ____ imply is ____.
- ____ suggests that ____.
- Considering ____, it seems reasonable to assume that ____.

In your brief review of the conversation up to here, you cannot go through everything. You will direct the attention of the audience to the most critical parts of the conversation using words and phrases such as:

- Importantly/significantly/notably ____.
- A critical point to remember is that ____.
- ____ appeared to offer the most persuasive argument for ____.
- There was no convincing answer to Medina's (2016) critique of Cliff's (2012) methods.

Since you are now silencing other voices and differences, this is not the place to bring in new information. If you feel the need to bring in new evidence or references at this stage, your argument is incomplete. In such a case, return to the conversation that led up to this point and change the conversation in a way that it integrates the new information.

One useful resource for thinking about the 'you see' moves in closing a conversation is appraisal theory (Martin & White, 2005). Martin and White explain that closing (or contracting) a conversation does not mean we ignore other voices and differences. Instead, we indicate that they are not welcome anymore, because we want to get to a point.

According to appraisal theory, we can close a conversation in a negative way (when we ‘disclaim’ other voices) in two ways, by:

- Denying other voices by saying something like: ‘Here is the view with which we started, but, based on the arguments presented, I believe we should reject it.’ If you do this, you will tend to use words that downright negate, and exclude, other views, such as: ‘no’, ‘never’, ‘not the case’ etc.
- Countering other views by saying something like: ‘Here is the view we started with, but based on the discussion we had I believe there is something better.’ If you do this you will use words that still imply the existence of other inferior views such as: yet, although, but, however.

However, you don’t have to close a conversation in a negative way. You can do so in a positive way when you ‘proclaim’ the view you want to end with in two ways:

- Concur with one view in a way that assumes that the audience already agrees with one particular view – almost like saying, ‘I know you agree’. If you do this, you will tend to use words that suggest there is really just one view to agree with, such as: ‘naturally’, ‘of course’, ‘obviously’ etc.
- Based on the exploration stage of the conversation, endorse one view in a way that says: ‘Those who agree with this view are correct’. This involves phrases such as: ‘Smith [and other authors] showed that ...’ or ‘It has been demonstrated ...’.

After the brief review of the conversation, you will need to move to the next conversation by helping the reader understand why this argument was important to the conversation and connecting it to the next conversation using the ‘you see’ moves already discussed.

5 Conclusion

Your writing will improve if you bear one metaphor in mind – which is the extended metaphor I’ve used throughout this chapter – that research writing is like running your own issued-based talk show, with you as the puppet master and other authors as puppets speaking through you. In this particular talk show, there is only one rule: *everything must be a response*.

Like all conversations, such a talk show goes through three phases: opening, exploring and closing. In each phase, you will find three moves: ‘I say’, where you position yourself among previous authors; ‘they say’, where previous authors respond to each other and to you; and, ‘you see’, which is when you speak to your readers.

Appendix A: Article ‘be arrogant’

If people knew what the word ‘arrogant’ really meant, they wouldn’t use it to try to insult others. Whenever someone describes me as arrogant, I regard it as a compliment. In fact, I actively work to attain the label as quickly as possible from as many people as possible.

What is arrogance?

Like many words, its true meaning has been lost through years of abuse. If one considers its etymology, arrogance means to presume on one’s rank or to take upon oneself without authority. This kind of language harks back to the age of class systems, when society was divided into peasants, merchants, nobility and royalty. For example, a peasant who aspired to become a merchant or had dreams of becoming a nobleman would have been regarded as arrogant.

However, if it weren’t for ‘arrogant’ peasants, society would never have progressed, because progress is only possible if people are unhappy with the status quo. Peasants who weren’t arrogant accepted the class system and their lowly rank and therefore didn’t do anything to bring about change. All change depended on those who thought and acted ‘above their class’: that is, the arrogant ones who presumed on their rank. One could thus speculate that progress in any society is directly proportional to the degree of arrogance present among its members.

To see whether this is still relevant today, consider what the opposite of arrogance is. It is to accept one’s position in society and not to ask for anything from people who place themselves in higher ranks than us. In short, not to be arrogant is to display an ignorant contentment that reinforces the view that some people are better than others. Not to be arrogant is to be stagnant.

A society of stagnant people is motionless: a cesspool of classes, ranks and inequality. Arrogance challenges perceived rules and accepted authority, and, by doing so, brings about greater equality. A truly democratic and fair society has to be arrogant.

Given that the word ‘arrogance’ has its origin in the outdated class system, it is surprising that people still use the word in universities today. Lecturers will often call their peers or students arrogant, but one will rarely hear them describing their superiors similarly. I have always liked arrogant students, but not all of them. This is because I don’t think all arrogance is good. There is constructive and destructive arrogance.

Types of arrogance

Destructive arrogance is not much better than the stagnation it fights against. It is destructive because it fights. It fights against the perceived class system

(e.g. status, authority, intellect), and therefore acknowledges its existence, or may even accept it.

A person with destructive arrogance tries to break through to the higher ranks, or, at the very least, tries to show the people in the higher ranks that he does not accept their authority. In the process, such an arrogant person will destroy many relationships even if he still contributes to progress. Once the destructively arrogant person reaches the higher ranks, he will treat the people he perceives to be below him in the same way he was treated when he was a member of the lower ranks. This is because this person implicitly accepted the existence of a class system by contesting it. A person with such destructive arrogance is not trying to challenge and transcend the actual class system, but to fight the people in the class system who prevent him from being as important as he thinks he ought to be. Thus, it is more about ego than ethics or progress.

When people frown on arrogance they think of destructive arrogance, without realizing that arrogance can also be constructive. Constructive arrogance may, on the surface, appear to be like destructive arrogance, because it will still infuriate people. However, at their root, they are different. Where destructive arrogance acknowledges and accepts the class system, constructive arrogance does not even acknowledge the existence of any class. The only class that exists for a person with constructive arrogance, is the individual; every person is in a class of his own.

While destructive arrogance cannot continue without perceived rankings, constructive arrogance can. A destructively arrogant person seems to direct all his attention towards the classes he perceives to be above him. A person with constructive arrogance will appear to direct his attention to people of all classes, but, in reality, such a person does much more because constructive arrogance is indifferent to hierarchies of any nature.

If someone hasn't called you arrogant lately, it means that you are perceived as accepting the system and its rules. Unfortunately for you, there are so many arrogant people around that the rules are bound to change soon. In a world of constant change, you either make things happen or you wonder what happened. You are either arrogant or stagnant.

Appendix B: More examples of talk show translations

B1 Text to talk show

Original text 1

Given the skills shortages and mismatches in South Africa, there is a strong case for, not only admitting more international students to South African universities, but also for promoting their long-term residency if they have the potential to reduce skills shortages. The structural shift in the South African economy has widened the skills mismatch, as is evidenced by the

large number of vacancies at the top-end of the labour market, for which the majority of the unemployed do not have the right skills (Vertovec, 2002). The skills shortages in South Africa are further aggravated by the so-called brain drain, as a significant number of South Africans have been emigrating to English speaking Western countries. For example, between 1989 and 2003, 32 296 South Africans emigrated to these countries. A further 720 000 left the country between 1989 and 2003 (Crush and Williams, 2005). Hall (2004) asserts that the brain drain is evident in the number of nurses, teachers and other professionals leaving South Africa for Britain, Australia or Canada. Bhorat et al. (2002) assert that emigration of highly skilled people from South Africa may be three times higher than the official data shows.

Extract adapted from: Aloyo, N. & Wentzel, A. 2011. The potential skills contribution of international students to South Africa. *Africa Education Review*, 8(2):336–354.

Talk show 1

AUTHOR: Since we have such a big skills shortage in South Africa, I think we should not only let more international students study here, but also encourage them to stay.

VERTOVEC: Yes, this makes sense because the economy has changed so much that the skills shortage got worse, especially in jobs that require more advanced skills ... the same skills that unemployed people don't have, but which graduates would have.

AUTHOR: Is the situation getting any better?

CRUSH & WILLIAMS: Unfortunately not. In fact, between 1989 and 2003, 32 296 South Africans emigrated to English speaking Western countries, and a further 720 000 left between 1989 and 2003.

AUTHOR: Are the ones who leave really highly skilled people?

HALL: Yes, they are mainly nurses, teachers and other professionals who go to Britain, Australia and Canada.

BHORAT: I think it is worse than we think. Given the inaccuracies in the data, it is quite possible that three times more highly skilled people have emigrated.

B2 Talk show to text

Talk show 2

AUTHOR: Why do we have unethical behaviour in organisations?

HANNAH 2011: I think it is the effect of a few individuals, who I call 'bad apples'.

BROWN 2005: Individuals will act unethically only if their leaders condone it, so I think it is better explained by their leaders.

TREVINO 1990: All of you take a too narrow view: it is due to the low levels of ethical culture in some organisations – what I call ‘bad barrels’.

AUTHOR: I think it is safe to say that explaining unethical behaviour in organisations is quite complex, so we need to pay attention to how all of these factors interact. We really need to study how ethical behaviour of individuals reflects their responses to a web of direct and indirect influences of ethical leadership and culture at all levels of an organisation.

Original text 2

Unethical organizational behaviors have been attributed to the effects of individual “bad apples” (e.g., Hannah, Avolio, & May, 2011), their leaders (e.g. Brown, Harrison, & Treviño, 2005), aspects of organizational context, such as low levels of “ethical culture” (“bad barrels” [e.g., Treviño & Youngblood, 1990]). In this study, we extend previous research and theory to examine how ethical cognitions and behaviors of organization members reflect their responses to a web of direct and indirect influences of ethical leadership and unit-level norms, standards, and sanctions pertaining to ethical behavior transmitted across multiple levels of an organizational hierarchy.

Extract from: Schaubroeck, J.M., Hannah, S.T., Avolio, B.J., Kozlowski, S.W., Lord, R.G., Treviño, L.K., Dimotakis, N. & Peng, A.C. 2012. Embedding ethical leadership within and across organization levels. *Academy of Management Journal*, 55(5):1053–1078.

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5 How to get started and write introductions

One of the biggest problems that novice researchers experience is that they don't know how to start. This chapter will not only show you a step-by-step process for getting started, it will help you to do so argumentatively. As you'll notice, some steps involve thinking and others involve writing. All of them involve making arguments, as I'll highlight at the end of the chapter.

By the end of this chapter, you will have turned an idea for research into a feasible research question and have written the first few pages of your research proposal, article or thesis. The ten-step process presented here will make it look easy; but bear in mind that, in reality, the process is not as simple. You will find that you will regularly return to previous steps as you gain more clarity as shown in Figure 5.1 (see page 96). If that happens, it is a good sign because it suggests that you are thinking about and refining your ideas.

Some steps involve mainly thinking, while others involve actually writing parts of the introduction. Let's start with the first step, which requires only thinking.

1 Thinking: find your personal 'itch'

First, you need to find at least one thing in your discipline that makes you 'itch' to do research; something that you care about and that excites you. This was explained in the first section of chapter 3, so I won't repeat it here. Sometimes, this is just an idea that is too vague to research, but the steps that follow will help you convert it into something more specific.

2 Thinking: make others care

When you do research, you do it because you hope it will make a positive difference in the real world, which means you need to write for other people. If other people don't read your writing, your research will make no difference. This means you have to get other people, besides yourself, to care about the 'itch' you identified. If you can't do this, you will be given the biggest insult a researcher can get, which is, "So what?" The biggest insult is not, "You talk nonsense", because such a response indicates that the reader

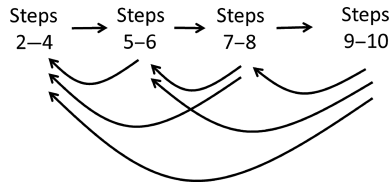


Figure 5.1 This chapter presents an iterative process of getting started

thinks what you are saying is so important that they need to respond to it. But if the reader asks, “So what?” (similar to one saying “Who cares?”) then he/she is saying that what you have written is not even worth responding to.

So here lies your first challenge: to get others to care. You don’t get other people to care by telling them about the benefits of your research, the same way a teacher does not get pupils to care about homework by telling them about the benefits of the homework. The best way to make others care is to create a problem for them. A teacher similarly makes pupils care about doing an assignment by telling students that they need to complete the assignment in order to pass the course.

The idea of a problem in most disciplines is therefore closely tied to the real world. In the real world, we need to take action when there is a gap between where we are and where we want to be, as shown in Figure 5.2. Where we are at that moment could be ‘bad’, or even ‘good enough’. In any event, it is not where we want to be, which must be being in a ‘better’ position.

Take, for example, passing a course. You have a problem if you want to pass a course, but you only just started it and it looks like it is going to be difficult to pass. So there is a gap between where you are now and where you want to be.

The bigger the gap between where you are and where you want to be, the bigger the problem. To close this gap, you have to take action, and if this action is difficult, then you have a problem.

What makes this gap a problem? Looking at the origin of the word gives us a clue. It comes from the Greek word *problema*, which means ‘obstacle to action’. What makes a gap a problem is that there are many obstacles in the real world that

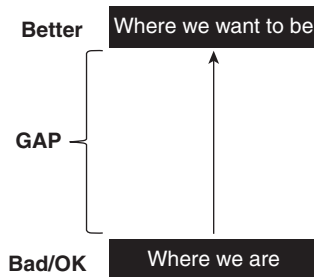


Figure 5.2 The structure of a problem

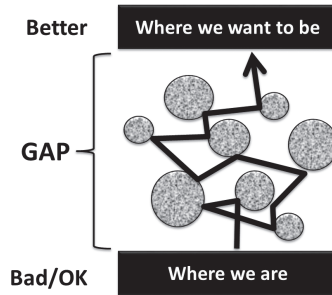


Figure 5.3 Looking inside a real-world problem

hamper action and prevent the closing of the gap, as shown in Figure 5.3. Those obstacles could be lack of funding, wrong incentives, poor time management, and many more. Because we are dealing with the real world (the world of people and things), it is important that you highlight real-world obstacles.

For example, in South Africa, we have a system where mothers can obtain a grant from the government for every child if their income is below a certain level. This causes many problems. One problem is that the grant is too small, or that the grant causes some young women to want to fall pregnant in order to obtain at least *some* income. If we frame it according to our definition of a problem, we could say that we want the grant to reduce poverty, but where we are at the moment is that the grant may be causing people to remain in poverty. So why don't we just close this gap? It is because there are many real-world obstacles in the way, such as the social and political disruption that would be caused by changing the system.

Your first important argument is that there is a real-world problem, and to argue it well, you should take the following steps.

2.1 Identify the gap

Explain the gap by showing that there is big difference between what we want and what we have. Find academic sources and credible data that you can use to support the claims that there is a gap. Without credible sources to back you up, it will be difficult for you to convince your chosen audience that the problem is serious enough to warrant their attention. Bear in mind that your audience probably has many researchers presenting problems to them, all of which compete for their attention, so you need to do all you can to convince them that they should care about your problem.

2.2 Identify the obstacles

Look inside the gap and identify the main obstacles to solving the problem and how they make it difficult to take action to solve the problem. There are

usually too many obstacles, so identify one or two to focus on. This will ensure that your writing will already be more focused.

Let's take our original example of the grant issue and the possible ineffectiveness of the system. In South Africa, we want to eliminate poverty, but we also don't want people to abuse this government assistance. If we eliminate the grant being given to poverty-stricken mothers, they cannot abuse it because it won't exist; but then poverty will worsen. But if we leave the grant as is, some people may become dependent on it, which makes it difficult to eliminate poverty. This dilemma is an obstacle that prevents the government from solving the problem and requires them to rethink the design of the grant. This is further aggravated by the lack of funding they have available to continue with this grant in its current form. The two main obstacles are, therefore, the way the grant is designed and the lack of funding. Both of these are large and complicated issues, so I would choose one to focus on.

Once you have identified an obstacle to focus on, investigate that obstacle in detail. This is because usually, even one obstacle can be much too broad. For example, if I wanted to focus on the way the grant is designed to assist the poverty-stricken mothers, I would have to find something more specific about the design that I think is a problem. It could be the criteria used to decide who is eligible for the grant or the practices used by officials when considering applicants. This helps to make the problem more specific, and assists you in identifying the audience for your research. Again, it is important to gather facts from credible sources so that you can make a convincing argument that the obstacle is real and serious.

2.3 Identify your audience

A problem is always a problem for someone and, often, a group of people or multiple groups of people. So, you need to have a person, entity or some group of people in mind when you try to discuss the problem. This also gives your writing further focus to ensure that you get to the point quickly. We always write messages, reports and e-mails with someone in mind, and research should be the same.

With any problem there are always multiple groups with much interest in the problem. Returning to the example of the grant, many groups are affected: social workers, parents, children, the government agency that designed the grant, the agency that dispenses funds, taxpayers and many more. We cannot write to all of them at the same time, because, we need to write differently for different stakeholders or affected parties. When addressing social workers, for example, we would need to write very differently to when we write for those who design the grant.

To ensure that your writing is focused and your research makes a difference, you should try to convince the group that has the power and responsibility to remove the obstacle you have identified. This is the group that falls into the high power-high interest quadrant in the next diagram. By the

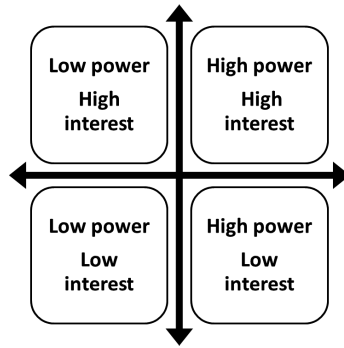


Figure 5.4 Identifying your audience

time you are finished with this step, this group should realise that your real-world problem needs their urgent attention.

Let's consider the stakeholders involved in the grant example and look at which quadrant each fits into. Taxpayers, parents and social workers may have high interest, but very little power to remove the obstacle and take action; while most children would fall into the low power-low interest quadrant. If the problem lies in the design of this grant, then it is the specific government agency responsible for the design of the grant that has the greatest power and responsibility.

Although we will be writing with this government department in mind (as our audience), this does not mean we are neglecting the other stakeholders. This is because if we can convince the government that there is a problem with the design of the grant, then they will take the action that would make the lives of all the other stakeholders better.

Once you have described the problem to your audience, you need to think how you are going to introduce the problem to them by putting it into context.

3 Thinking: identify an appropriate context

When communicating the problem, you have to be careful because not everyone will agree with you that your problem is that important, and may even refute its existence entirely. And if a reader disagrees with you early on, it is a lot more difficult to convince them at a later stage.

This is one of the main reasons we have to provide the context. Context gives us the broad background to the problem or tells the story that led up to the problem. In the context, there is usually little that readers can disagree with because you don't explain the problem yet. It creates some common ground between you and the reader, and it is from here that you can lead them to the problem. There are two main ways by which you can create context, and you should be able to do it in one or two paragraphs at most.

3.1 Narrative

Relate the chain of events from a time just before the problem emerged to the actual emergence of the problem. Brevity is of the essence here – don’t lose the reader’s interest by giving them a long history lesson.

Continuing with the example, one might start with the years just before the government introduced the grant, when poverty was at its highest historical level, especially amongst women and children. Then, in 1994, there was a change in government and the new government considered many possible solutions. One of these solutions was to design a grant that provided money to women below a certain income threshold who were raising children below a certain age. But then reports started to emerge of teenagers getting pregnant with the intention of obtaining a grant, which showed flaws in the design of the grant. Please note that while telling this narrative, I will make sure to use data and facts from credible sources that cannot be disputed. This is imperative for a researcher to do in order to substantiate the argument. What all this does is gradually lead the reader to the actual problem I want them to take seriously, as shown in Figure 5.5.

The narrative, or chain of events that led to the problem’s existence, is usually the easiest and most persuasive way of contextualising the problem. This is because it draws on historical facts that are indisputable and because most people are less critical when they are told a story. However, the narrative only works if there was a time before and after the problem. If the problem has always existed, you need to contextualise the problem using the next approach.

3.2 Inverted pyramid

Start by discussing the problem very broadly and gradually narrow down the discussion, getting more and more specific until you reach the problem. This is why I call it an ‘inverted (upside-down) pyramid’ – because you go from the very general to the very specific, as shown in Figure 5.6 (see page 101). Again, being brief is of the essence, because you will bore your readers and they will lose concentration if you spend too much time on vague and general discussions.

The problem with grants in South Africa can also be contextualised like this: you may start by giving facts that show that poverty is a worldwide problem and then narrow it down by showing that it is an even bigger



Figure 5.5 Context through narrative

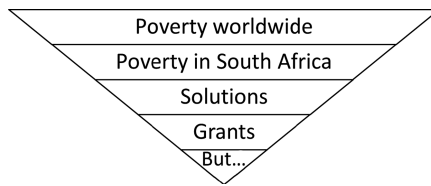


Figure 5.6 Context as an inverted pyramid

problem in South Africa. Then, show that various solutions have been considered by governments to deal with poverty, and that South Africa implemented one of these solutions: that is, the child grant that was designed in a specific way. But, after implementing the grant, it started to appear that the design was flawed.

In this approach, you start very broadly so that you build common ground with the reader. The reader will most likely agree with you because broad statements, backed up by data and facts, are difficult to dispute.

Both techniques of contextualising serve a psychological purpose: if readers agree with you at the start, they are more likely to agree with you later too. Salespeople do this with us all the time. They don't start with questions to which you can say, 'No'. For this reason, a life insurance salesman will not ask you: "Do you want to buy an education policy from me?" Instead, he will ask you a series of ever-more focused questions to which you will always answer, 'Yes', such as: "Do you love your children?" "Yes". "So you want the best for them?" "Yes". "And to give them the best it is important to ensure that they get a good university education?" "Yes"... and so on... until he asks the question: "Do you want to buy an education policy?" and now, after agreeing so much, the answer is more likely to be, "Yes, of course!"

In the same way, you are 'selling' your research to your readers so that they will care. You should therefore start with a narrative or statements that they will agree with and which will gradually lead them to the problem in the real world.

Contextualising – familiarising the audience with the place, time and setting – also helps the reader orient themselves and see where you are coming from. We do this in most stories: we don't start with what went wrong, but with 'once upon a time'. We also do this in movies: we don't start with the action, but first with a view of the place and gradually zoom in to where the action is taking place. These two examples lend themselves to that of the salesman analogy, in the protocol of going from the broad to the specific, which we do in research.

4 Writing: 'background to the research' section

After the thinking of steps 1 to 3, you are now ready to write the first section: the background to the research. Here you will put together steps 2 and 3.

First, write down the context from step 3. The context leads to the real-world problem, so then you write down the explanation of the problem from step 2. This should not take more than one page, and of this one page you should spend more time on explaining the actual problem in the real world, because it is this that convinces people that they should care.

Look at the example below from an actual piece of writing by a student, with some of the details deleted. The background starts with a general discussion of an issue in the first paragraph. It then moves to a discussion of the issue in the second paragraph, which is specific to South Africa, and how it led to a particular law. In the third paragraph, it explains the problem with that law. Facts are stated and academic references back up all the important claims.

Background to the research

The issue of ____ is one of vital importance for financial advisors given _____. The assumption here is that ____ (Brown & Reilly, 2009). But according to Campbell et al. (2001) ____, so that _____. It has thus become critical to their function of ____ (Ibbotson & Kaplan, 2000).

_____ plays a vital role in the _____ since _____ (Maginn et al., 2007). As a result, to protect _____ from being exposed to _____, the South African government implemented the _____ Act of _____.

All financial advisors in South Africa have to comply with this Act, which sets a list of guidelines with the sole purpose of _____. Unfortunately the Act appears to ignore the benefits of _____ (Bodie, 1983; Gorton & Rouwenhorst, 2006) and limits _____. So, despite these benefits, advisors cannot take advantage of them due to the limitations imposed by the Act. According to Bodie and Rosansky (1980) an _____ could have reduced _____ while _____.

In the background section, you focused on the real-world obstacles that make it difficult to solve the real-world problem. There is another kind of obstacle that needs to exist to create the need for research. This one does not lie in the real world, but in the world of knowledge. We think about that in the next step.

5 Thinking: determine why current studies are not good enough

So far, you have shown that there is a problem in the real world and you identified real-world obstacles that make it difficult to do something about it – like lack of funding, wrong policies, corruption etc. Now you have to show that research plays a role in the existence of, and solution to this real-world problem.

Real-world problems, however, cannot be solved by research, because research is not a person who can walk around and change the world. So, let us look at what role research plays in the existence of a problem. It could be that the research, up until that point, was not good enough to help people solve the real-world problem, so that further research would give people the knowledge they need to address the real-world problem. This is where you can step in.

In this step you have to move from the real world (the world of people and things in steps 2–4) to the knowledge world, the world of ideas and studies. You have to show that another obstacle in our way to address the real-world problem, is some lack of knowledge about it. In other words, you have to show that there is also a knowledge problem (also called a ‘research problem’).

If we knew everything about the real-world problem, that is, if previous research studied everything about the real-world problem completely and perfectly; we would not need further research. So it is important to show that one of the reasons we cannot solve the problem is because we don’t know enough about it.

Now, the question must be posed, how do we know if there is a lack of knowledge? We look at previous academic research studies about the real-world problem and find something that is either missing or confusing in previous studies.

As explained in Chapter 3, for a novice researcher, the easiest way to find out where the knowledge about a problem is inadequate is to search for gaps in the existing studies, so I’ll focus on that here. When you find a gap, you can do research to fill the gap, and so help to create knowledge that may help to solve the real-world problem. The most common kinds of gaps are:

- Time gaps: You find previous studies about the real-world problem, but notice that the studies are outdated and are therefore no longer relevant. New updated research is therefore needed based on the latest data.
- Place gaps: You find previous studies, but notice that the studies are only applied to some places. New research is therefore needed that applies the relevant ideas to places – cities, industries, countries etc. – neglected by researchers in the past.
- People gaps: You find previous studies, but notice that the studies are only applied to some people or groups. New research is therefore needed that applies the relevant ideas to the groups neglected by researchers in the past.
- Method gaps: You find previous studies, but notice that the studies use methods that are wrong, inappropriate or outdated. New research is therefore needed using the best available or new methods.

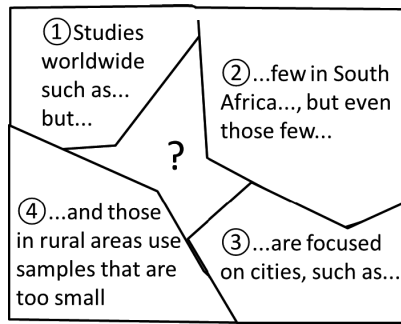


Figure 5.7 A systematic approach to showing a gap exists

As explained in Chapter 3, gap-spotting does not lead to interesting research. However, if you are still learning about research, simple gap-spotting is a good starting point. Once you gain more confidence, you can try other approaches that lead to more interesting research, as explained in chapter 3.

It is not so easy to convince people that a gap exists. The best strategy is to group the existing studies together, and systematically mention them so that it becomes obvious that there is an area where no studies exist.

Let's say, for example, you spotted a combination of a place and method gap: that there are no good studies investigating whether government grants for children of poverty-stricken mothers cause teenage pregnancies in rural areas in South Africa. You would first identify that there are studies worldwide by mentioning the main studies in other countries. From there, you would show that there are a few South African studies, but most of them were conducted in cities, and that those in rural areas cannot be trusted because their samples are too small. From that, you would conclude that it is not clear whether grants cause teenage pregnancies in rural areas in South Africa, and posit that we therefore need a study in rural areas that uses a bigger sample. Every time you mention studies, you need to provide the references for those studies, even though you won't discuss the studies in detail. This is summarised in Figure 5.7.

Another common approach that was mentioned in chapter 3 is to look for aspects of the real-world problem where previous studies disagree or have not been able to solve a puzzle. When you find such conflict or confusion in previous studies, you can make a contribution, but only if you have a way to do the research *differently* in order to settle the disagreement or solve the puzzle.

The challenge is to convince your audience that, given all the research out there about your real-world problem, you have found something inadequate. Here are some ways to do that:

- If you found that there is confusion about an issue in the form of a disagreement, then: (1) you mention the studies (by referencing them) that take one view on the real-world problem; (2) mention the studies (by referencing them) that take the opposing view; (3) explain what it is they disagree on; and (4) explain what research is still needed to settle the disagreement (and this is the research you will do).
- If you found that there is confusion in the form of an unsolved puzzle, then: (1) you explain the puzzle with reference to studies that show that there is a puzzle; (2) mention the studies (by referencing them) that have attempted, and failed, to solve the puzzle; and (3) explain what research is still needed that might help to solve the puzzle, and state that this is the research you will do.

Look at the following example based on finding a combination of a time gap, place gap and confusion about the problem:

Literature on the impact of ___ has yielded inconsistent results across the globe. For example, in his US-based study, Stiroh (2004) finds that _____. Converse to this finding, and also based on US commercial banks, De Young and Rice (2004) find that _____. Another study by Williams and Rajaguru (2013), set in Australia, not only finds that _____, but also that _____.

In addition to the inconclusive findings, many of the studies (Smith, 2015; Zalamea, 2013) investigate the impact of _____ mainly in developed countries. However, the extent to which these findings can be generalised to developing economies is limited because _____. Although South African research exists, in the form of a study by Oberholzer and Van der Westhuizen (2009), the data examined covers a time period (2001 to 2008), which means it is outdated because _____.

There are a couple of things I want you to notice from this extract:

- The author does *not discuss the real-world* problem or issues in the real world. The discussion is focused only on the studies: what has been done and what is inadequate about the research.
- There are *lots of references*. Every time the author mentions a study or makes a claim about the studies, he provided references. This means you cannot write about the knowledge problem until you have read the important literature about the real-world problem.
- The author does not discuss the studies, he simply *mentions* them so that we can see what has, and has not, been done. A more detailed discussion of the studies can be provided later in the literature review. At this point you should just give a rapid overview of the state of the research,

so you will complete your explanation of the knowledge problem in less than half-page.

- The discussion is *systematic*. Studies are grouped together and discussed in a logical order so that it leads to a conclusion of what exactly is inadequate in the research thus far. The author does not simply list the studies in some random order.

By the time you have explained the knowledge problem, you should be able to conclude in a single sentence: “So we (as researchers) therefore don’t know ____.” For example, if there are conflicting results on whether lower taxes lead to people spending more – let’s say some studies say it leads to more spending, while other studies say taxes make no difference – we can conclude: “We, therefore, do not know if lower taxes stimulate consumers to spend more.” If, for example, we found a time gap – let’s say the last study on taxes and consumer spending in Colombia was done in 2006 – we can conclude: “We, therefore, do not know if lower taxes have stimulated Colombian consumers to spend more after 2007.”

6 Thinking: identify a provisional research question

Once you can state what is inadequate about the research, it is simple to derive the provisional research question. You simply restate the statement in the form of a grammatical question, with a question mark at the end, as the next table shows.

This is probably the simplest step, and it may seem too simple, but I put it here because novice researchers often make the mistake of creating a question that does not follow from the knowledge problem. This step shows that the research question is *directly* derived from the knowledge problem: that it is, in fact, a mere restatement.

Most authors of journal articles and books today don’t state their research question, but, for novice researchers, I advise that you state your question *explicitly*. Questions are active, not passive: they pull you forward and focus

Table 5.1 A research question is simply a re-stating of the knowledge problem

Statement of knowledge problem	Research question
We don’t know what is 1+1.	What is 1+1?
We don’t know whether it will rain later.	Will it rain later?
We don’t know if the road is congested.	Is the road congested?
We don’t know if lower taxes stimulated Colombian consumers to spend more on luxury goods after 2007.	Did lower taxes stimulate Colombian consumers to spend more on luxury goods after 2007?
We don’t know how grants cause teenage pregnancies in rural areas in South Africa.	How do grants cause teenage pregnancies in rural areas in South Africa?

your mind. If you wish, you can restate the question as an aim instead, but, first, you need a question. Since the aim of research is to answer unanswered questions, the aim of any study is simply a restatement of its question, as the next table shows.

Some researchers state their objectives or sub-questions. Objectives are the result of breaking the aim into smaller tasks, which means that, if you achieved every one of the objectives, you would have achieved the aim comprehensively. If you prefer using a question instead, you would similarly break the question into smaller sub-questions. So you will either have an aim with objectives, or a question with sub-questions.

In some studies, you need to state a hypothesis, which is also derived from the question. So again, you need to have a question first. A good hypothesis should meet the following conditions at the very least: (1) it must be a grammatical statement that can either be true or false; (2) if proven to be true or false, it should answer your question; and (3) your research can then establish whether it is true or false.

For example, if my question is “What is the colour of the sky?” my hypothesis could be “The sky is green”. The latter is a statement, and my research can prove that it is false. Whether my research proves it to be true or false, I come closer to answering my question.

But, let me give you a word of caution – hypotheses are only suitable in two instances, namely: when questions have unambiguous answers; or when the possible range of answers to the questions is known in advance. Such questions usually start with: ‘What?’, ‘Who?’, ‘When?’ or ‘Where?’ It is very difficult to develop hypotheses for more open-ended questions where you cannot predict the range of answers, like ‘How?’ or ‘Why?’ questions; so in such cases, it is better to simply state the question or the aim.

If you think this step seems too simple, you are partly right. At this point, you still have only a provisional question and you will refine it in the next step. You may find that the next step shows weaknesses in your thinking, which will force you to go back to step five, or even back to step two.

Table 5.2 A research aim is simply a restatement of the research question

<i>Research question</i>	<i>The aim of this study is to determine...</i>
What is 1+1?	...what is 1+1.
Will it rain later?	...whether it will rain later.
Is the road congested?	...whether the road is congested.
Did lower taxes stimulate Colombian consumers to spend more on luxury goods after 2007?	...if lower taxes stimulated Colombian consumers to spend more on luxury goods after 2007.
How do grants cause teenage pregnancies in rural areas in South Africa?	...how grants cause teenage pregnancies in rural areas in South Africa.

7 Thinking: refine the research question

I will distort a well-known English saying by stating that, “The road to a good question is paved with stupid questions.” Your provisional question is probably a stupid question, but that is alright. If you are afraid to begin with a stupid question, you will not end up with a meaningful question.

So, how do you know if a question is stupid or not? A stupid question is meaningless – it does not make sense or is too vague to be answered by research. A meaningful question can be answered.

A research question is meaningful if it gives very specific guidance on finding the answer. This means a question has to meet four conditions in order to be meaningful:

- A meaningful question makes plausible assumptions;
- A meaningful question tells you what information you need to answer the question;
- A meaningful question tells you where and when you may get the information; and
- A meaningful question tells you by what methods you will get this information.

The first condition ensures that an answer is *possible*, while the last three conditions ensure that the question is *specific* enough to be answered. If your question meets the last three conditions, you will already be half-way to the answer, because you can already tell a lot about the answer from a meaningful question.

A meaningful question already gives you the ‘silhouette’ of the answer: it enables you to see what the answer may look like and what you need to do to find it, even if you can’t see the details yet. Like the silhouette in the next figure, you can tell a lot from some things even if you can’t see the details. You can see it is a man and a woman: the woman has long hair, the man has short hair, and they are probably standing. It appears that the woman is angry and scolding the man, but he does not seem intimidated. A meaningful question similarly helps you to derive a lot of information about the answer and how you will find it.

How do you make your question meaningful? You simply take every term in the question and define it until it is unambiguous. Let’s take an example with a meaningless question such as: “What is the secret to success?” There is no way this question can be researched, because it is far too vague. It is not clear at all what is meant by ‘success’. If we don’t know that, then we won’t know what information is needed, and if we don’t know what information we need, we won’t know where, when or by what method we can get this information. So we have to go through a process of clarifying the question so that it becomes researchable: some people call it ‘operationalising’ the question.

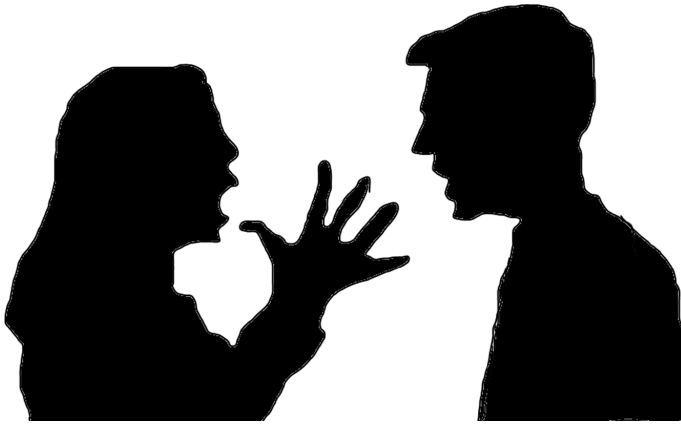


Figure 5.8 A meaningful question is a silhouette of the answer

Here is an example of how a question is operationalised, shown as a possible dialogue between a professor and a student:

PROFESSOR: What is your research question?

STUDENT: It is “What is the secret to success?”

PROFESSOR: What do you mean by ‘success’?

STUDENT: I mean financial success. So my question is, “What is the secret to financial success?”

PROFESSOR: What counts as ‘financial success’?

STUDENT: That is when you have a high net worth. So my question is “What is the secret to achieving a high net worth?”

PROFESSOR: How much is a ‘high’ net worth?

STUDENT: High net worth is at least US\$ 10 million. So my question is “What is the secret to achieving a net worth of US\$ 10 million?”

PROFESSOR: Does this apply to any person?

STUDENT: No, I want to investigate it for Colombians. So my question is, “What is the secret to achieving a net worth of US\$ 10 million in Colombia?”

PROFESSOR: Is this for all time or by a certain time?

STUDENT: No, the person must achieve it before the age of 50 to be financially successful. So, my question becomes: “What is the secret to achieving a net worth of US\$ 10 million in Colombia by age 50?”

PROFESSOR: What do you mean by ‘secret’?

STUDENT: That would be the method by which to achieve it. So my question is, “By what method can a Colombian achieve a net worth of US\$ 10 million by age 50?”

This conversation can go on a bit longer as they explore what is meant by the terms: ‘achieve’ (for example, whether it should be limited to legal

means); ‘net worth’ (perhaps what kinds of assets and liabilities should be considered in its calculation); or ‘US\$ 10 million’ (whether the wealth must be held in US\$ or if the equivalent in pesos is acceptable). With every term the researcher becomes aware of the wide range of possibilities for exploration and is forced to choose a specific direction. In fact, the above conversation could already have gone in many different directions at the start, depending on the kind of ‘success’ the student had in mind. There is no correct direction as long as the question meets all four conditions and fits in the discipline.

The point is that they came a long way from a stupid question like “What is the secret to success?” to “By what method can a Colombian achieve a net worth of US\$ 10 million by age 50?” which is a more meaningful question that will be much easier to research. To confirm this, let’s test the question against the four conditions, returning to the dialogue as an example:

PROFESSOR: So your question is, “By what method can a Colombian achieve a net worth of US\$ 10 million by age 50?” Let’s test it. What information would you need to answer the question?

STUDENT: I would need information on Colombians who have achieved this net worth before 50.

PROFESSOR: What else?

STUDENT: Well the question suggests that I should also find information on their net worth and on their life stories to see how they achieved it. I can already see that I need to clarify how exactly I will measure net worth, so this means I will have to expand my question a little.

PROFESSOR: Where will you get all this information? To get the list of people with a high net worth might be tough.

STUDENT: Yes, I didn’t realise that, so if I can’t get this I would need to change my question. But, for now, I would need to find out if there is a list of the richest Colombians. I will get their life stories from magazines, newspapers, biographies and talking to acquaintances. That is, if I can get access to these people and if they would be willing to speak to me.

PROFESSOR: And, if not, you would need to adapt your question somewhat. Anyway, when, or for what period, will you get this information?

STUDENT: I can see that the question needs clarification here. Because money loses its value over time, I should perhaps only focus on people who achieved this net worth in the last ten years.

PROFESSOR: By what method, or how exactly, will you obtain all the information you need?

STUDENT: It seems to me that it would involve lots of secondary data analysis of financial records, if I can find them, and some kind of textual analysis of the life stories of the wealthy people. Also, if they are willing to speak to me, I would conduct semi-structured interviews with acquaintances, friends or even the successful people themselves.

PROFESSOR: What does your question assume?

STUDENT: One big assumption of the question is that there are methods that one can follow to achieve wealth.

PROFESSOR: Is that a plausible assumption, and is there evidence to support it?

STUDENT: I am not sure; I will have to read some more of the literature to see. I can see your point: if there is no *method* for achieving wealth, then the question cannot be answered, because it assumes that there is a method.

PROFESSOR: So, it seems to me that you made a good start, but it is not certain yet if it will be possible to research the question. You still have lots to find out and you will probably have to change your question somewhat.

The best way to test your question is not have yourself check it against the four conditions, because we often assume our questions are clearer than they really are. Rather, show your question to someone who is familiar with your discipline, and ask them: “If this were your question, would you know what information you would need? Would you know where, when and by what method to get it?” If they struggle, or give you answers that you did not expect, it is a sign that you need to refine your question further.

As you refine your question, it gets longer and longer, as you saw in the above dialogue. Perhaps the question ends up being: “What are the ten most common methods by which Colombians have legally achieved a tangible net worth of US\$ 10 million (or its equivalent) by age 50 in the last ten years?” It may happen that the question gets too long and clumsy, and the solution is to add *delimitations*. Note that ‘delimitations’ are very different from ‘limitations’: delimitations are *boundaries* you place around the question for clarity.

What you can do is state the short version of your question, and then add a paragraph underneath it in which you explain your question (the delimitations). For example:

The research question of this study is: “What are the most common methods by which Colombians have achieved financial success before the age of 50?”

Financial success is defined as having a tangible net worth of more than US\$ 10 million, or the equivalent in another currency. Tangible net worth is the value of total assets minus total liabilities and excluding the value of intangible assets. The study will focus only on Colombians who have legally achieved this net worth in the last ten years for which data is available, that is the period 2007–2016. This study will identify the ten most common methods, categorised by kind of industry, by which they achieved their wealth.

Many students underestimate the importance of refining their questions at this very early stage, and as a result, they are never able to complete their research because they tried to research a question that was too vague to research. As you may have noticed from the dialogues above, checking your question against the four conditions for a meaningful question allows you to change and clarify your question before it is too late.

Now, and I mean **NOW**, is the time to think about what it will take to answer your question, and to determine if you will be able to answer it. From the above dialogue, the student realised that he needed access to information that might not be possible to obtain, or he might have realised that he doesn't have the skill to use the required methods. He might then either have to change his question to make it more feasible, or obtain the necessary skills or contacts before proceeding with his question. In my experience, too many students get stuck in their research later, or have to start all over again, because they don't spend enough time on this step.

8 Writing: the 'research problem' section

Having done much thinking in steps 5–7, you are again ready to write. By now, you should realise that all research is about meaningful questions that are derived from two problems: a real-world problem, followed by a knowledge problem. You need both a real-world problem and a knowledge problem in order to conduct valuable research. To see why, see what happens when one of them is missing:

- Consider a famous and rather strange question from the Middle Ages: "How many angels can dance on the head of a pin?" It is difficult to see why anyone should care about such a question. Sure, we don't know how many angels can dance on the head of a pin, so there is a knowledge problem, but so what? Even if we knew the answer it would not help us to address any problem in the real world. There is a knowledge problem, but no real-world problem.
- Or consider this question: "What will happen to sales of fuel when the oil price goes down?" While this question would help to solve a real-world problem, research would have no value because we already know the answer from previous research. In this case we have a real-world problem, but there is no knowledge problem, so more research will not contribute anything new to what we already know.

To write the 'research problem' section, you need to start where the previous section ended: with the real-world problem. By referring to previous scientific studies, show where previous research has been inadequate in terms of illuminating the real-world problem. Don't explain anything about the real world now, but explain why our *knowledge* about the real-world problem is missing something with reference to the important studies (step

5). Then draw from this to establish your research question explicitly (from steps 6 and 7). Based on your unpacking the terms found in the question, you could add a paragraph or two of delimitations (step 7).

Here is an extract from an actual proposal, with the details deleted.

Research problem and question

There are many international studies which discuss the _____. The studies which do analyse _____ have different aims to _____. For example, a study by Sapenzia (2002) claimed that _____. This study does assess the impact ____; however, it is focused on _____. Similarly, another study conducted by Petersen and Raghuram (1995) compared _____, but _____. While a further study conducted by Drucker (2005) analysed the _____, it only focused narrowly on _____ rather than on _____.

The South African studies on _____ focus on the factors that _____ the South African economy. These studies have _____ (Fedderke & Romm, 2004; Moolman et al., 2006). Unfortunately, these studies look only at factors influencing _____ and not specifically _____. Other South African studies assess _____ (Affleck-Graves et al., 1988; Bhana, 1987; Van Den Honert, 1988). Although these results can possibly be applicable to _____, the above-mentioned South African studies analyse the impact of _____ on _____, and again, not specifically on _____.

In summary, international literature on _____ does exist; however, it is limited when it comes to _____. Furthermore, although there are studies in a South African context that _____, there is no research on _____. Since _____ is unknown, the research question is: _____?

The term _____ refers to a _____. The _____ refers to the impact on _____. The desired state would be that _____. _____ will be measured by _____.

Observe the systematic discussion going from grouping together the studies, internationally and then locally, and highlighting the inadequacies. The focus is on the inadequacies in the studies, and not on the real world. The question is derived from the knowledge problem and delimitations provided at the end.

9 Thinking: anticipate how you will find the answer

Once you have a researchable question, you are ready to think about what you will do to find the answer. At a later stage, you will expand on this in the research design and methodology section or chapter. Here you explain, in as much detail as possible, exactly how you will conduct your research in order to find the answer.

Many students think that it is impossible to explain this so early in the research, but that is not true. You already know what it will take to obtain your results because you thought about it in step 7 when you refined your

Table 5.3 A meaningful question tells you a lot about your research design

<i>A meaningful question tells you...</i>	<i>So you should know and be able to explain...</i>
...what information you will need	Which variables to investigate What data is needed on each one Whether this data is available
...where and when to get this information	The sources of each individual variable For what period and place the data will be gathered
...by what method you will obtain the information	How exactly you will obtain the data How you will analyse the data

question. If at this stage, you cannot write at least two or three paragraphs about how you will generate results, it is likely that your question is not meaningful. In such a case, you should return to step 7.

What can you tell us already about your research design? Table 5.3 shows this, based on the last three conditions of a meaningful question.

Even though you won't write the research design and methodology section yet, you are already thinking about how to argue about what needs to be done to answer the question. The things you need to explain here are exactly the things you had to think about when refining your question: variables, data sources, time period, data gathering methods and techniques of data analysis. 'Data' refers to any credible quantitative or qualitative information that you can use to help you answer the question.

If you are vague at this point, you will probably be vague when you actually have to write the research design and methodology section or chapter (as explained in chapter 7). This will harm your credibility, because vagueness creates the impression that you don't know what you are doing. It is also a sign that your question is not researchable, and that you are trying to hide this.

10 Thinking and writing: explain why your research is significant

In this final step, you close the circle and write the 'significance of the research' section. You go back to the real world, and you ask yourself: who will benefit from this research? Now you need not limit yourself to those with the power to solve the problem; you can include everyone with an interest in the problem. Tell us how they would benefit from the research once it is completed. How would your research make a positive difference to their lives in the real world?

11 Evaluating the quality of your arguments

To execute this ten-step process successfully, you have to make several arguments, which means you will use the techniques from chapter 4. When

you are done, you can also evaluate your arguments using Toulmin's model from chapter 1. In case you didn't see the arguments, let's review them, using the Toulmin model, and see what needs to go into each argument

- Background:
 - Claim: There is a problem in the real world.
 - Data: Facts about the real world from credible sources.
 - Warrant: For some group, show how the facts lead to a problem for them.
 - Counterargument (optional): Refute those who might say that this problem does not exist or is not that urgent.
- Research problem:
 - Claim: There is something about the real-world problem that we do not know as researchers.
 - Data: References to previous studies about the real-world problem.
 - Warrant: A systematic review of the studies that show how the studies are inadequate in some way.
 - Counterargument (optional): Respond to those who think that there is nothing wrong with the current research.
- Research design:
 - Claim: You have a good idea what needs to be done to answer the question in the most appropriate way.
 - Data: Possible designs that were used in other studies.
 - Warrant: Show that the design is appropriate given the question and available data.
 - Counterargument (optional): Respond to those who might argue that your chosen design is not appropriate.

An important part of argumentative writing is to make connections between your arguments. Since your arguments will be in separate sections, you should use linking sentences, such as those shown below:

- 1 Background section argument: **There is a problem in the real world for some group of people...**
Link: ...and this real-world problem is unsolved because...
- 2 Research problem section argument: **...the current research is inadequate...**
Link: ...and to address this inadequacy I will...
- 3 Research design section argument: **...do the following to generate new knowledge...**

Notice how the linking sentences make the one argument flow into the next. If you read the bold sentences all in one go, you will notice that it

reads like one very long, but flowing sentence that makes sense: “There is a problem in the real world for some group of people, and this real-world problem is unsolved, because the current research is inadequate, and to address this inadequacy, I will do the following to generate new knowledge.”

12 Overview of the process

Let’s summarise the process. Research starts with something personal: an issue or problem that you care about and want to find out more about (step 1). You have to make other people care about it, specifically those with the power to take action, by showing that there is a problem for them in the real world about which something needs to be done (steps 2–4). Then you look at the research about this real-world problem, and show that previous studies are inadequate, and as a result, we have a knowledge problem (step 5).

The real-world problem (written about in the ‘background’ section) and the knowledge problem (written about in the ‘research problem’ section) are often confused by novice researchers. In the background section, you write about the facts that relate to the experience of people trying to solve a real-world problem. In the ‘research problem’ section, you don’t write about the real world, as such, but rather about the studies done in the past and where they are missing something. This is summarised in another way in Table 5.4.

If you can summarise the background section in the following three sentences, you are ready to write it up:

The problem in the real world for _____ (audience) is _____ (obstacle). The main facts that suggest that there is a problem are _____. This group should pay attention to it because _____.

If you can summarise the knowledge problem in the following three sentences, you are ready to write it up:

Table 5.4 Contrasting the real-world problem and the knowledge problem

	<i>Real-world problem</i>	<i>Knowledge problem</i>
Where is the problem?	In the <i>everyday</i> reality of people and things	In the <i>studies</i> on the real-world problem
Who experiences it?	Ordinary people, practitioners and those with the power to address the problem	Academics with an interest in the research
How do we prove it?	Provide facts in support of the seriousness of the problem	Show that the studies are inadequate

The studies about _____ (the real-world problem) are inadequate because _____. We therefore do not know _____. So, the question is: _____?

Once you rephrase the knowledge problem as a question (step 6), you refine your question by testing it against the conditions for a meaningful question (step 7), and you write up the 'research problem' section (step 8). If you did a good job at refining your question, you should have a lot of clarity on how you will conduct the research (step 9). Finally, you close the circle and return to the real world in step 10.

Below is a possible template based on the steps discussed. If you are writing a proposal, this will take about three to four pages; and, if it is an article, it will likely be one to two pages. To be certain, look at the research in your discipline.

1. Background to the research

Argue that there is a problem in the *real world* for a particular group. Use facts from credible sources to support your case that there is a problem that this group should take seriously, and why.

2. Research problem

Now go to the *literature* and explain what aspect of this real-world problem has not been researched adequately. Clearly identify the gap in the research by a brief, but systematic, overview of the literature. Convert it into a question.

3. Research design

What will it take to answer this question? What needs to be done and is it feasible? What data do you need to find or create, and is this feasible? Did you check? Where will you get this data from? Is it accessible? Did you check? Provide some details of what will go into your model, survey, or whatever design you will use. Did previous studies also use this approach or do you need to adapt it? How? Did you check this is feasible? What are these studies?

4. Significance of the research

Who will benefit from this research and how?

5. List of references (10–20 academic references are fine at this stage)

Sections 1, 2 and 3 should contain lots of references. Provide their details here in the correct format.

The conventional proposal usually also contains a literature review, but we'll get into that in the next chapter. Anyway, you cannot write sections 1 and 2 without reading a lot of literature, so, in a way, you would already have done a preliminary literature review after this.

Even with this process, your research is a journey that will only make sense when you reach the end. There will be many tough times, you will find yourself repeatedly backing out of blind alleys and letting go of leads that have disappointed once again. Feeling overwhelmed, dejected and lost is an important part of this journey, because this is no ordinary journey. You are not simply going to a well-known destination; you are not merely discovering things that others know; you are discovering things that nobody knows. It is a journey of creation, and you don't simply reach the endpoint, you create it. And once you are at the endpoint, you will realise that it is not the end after all, simply the beginning of new journeys, for yourself or others.

6 Writing an original and argumentative literature review

The literature review is often the worst written part of most dissertations and articles. It often reads as if most authors just listed lots of studies, or dumped studies and theories on top of each other, like a pile of bricks, without any clear logic to the writing (as shown Figure 6.1). As a result, when reading many articles and theses, one can skip the literature review, because it adds little to the overall argument. Another reason novice researchers struggle with the literature review is because the literature can be overwhelming. It is difficult to know what to write about, how to organise the writing and how to give it that elusive ‘argumentative’ (or ‘critical’) quality.

In this chapter, I will show you how to overcome all these challenges and write a literature review that: is argumentative and flows logically; will deepen your understanding; and will actually contribute to the originality of your research. A word of warning though: the approach I follow here is quite unconventional and creative. It will not work for you if you wish to write a conventional literature review.

1 What is a literature review?

One reason for unfocused and badly written literature review sections and chapters is its label. Calling it a ‘literature *review*’ is misleading. This is because the literature review is more than just a ‘review’; and not contained in any one chapter or section. It is located everywhere in a paper or thesis where you try to make an argument. If you read well-written articles in good peer-reviewed journals, you will notice that very often there is no separate section called a ‘literature review’. This is partly because literature is consulted throughout, and thus used in all sections; each time serving a different purpose.

In the background section of a dissertation (as explained in chapter 5), literature is used to provide the facts that support the argument that the issue under investigation is a real-world problem to some group. When the research problem is discussed, references to the literature make the case that existing research is inadequate in some way. In the research design section,

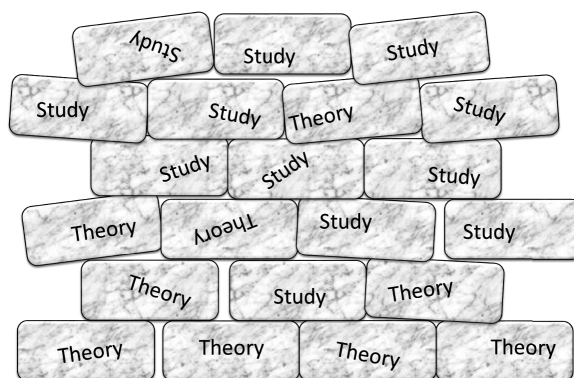


Figure 6.1 A poor literature review

literature is used to justify a particular research approach, and it appears again in the discussion of the results to help you make sense of the results.

And yet, despite the fact that literature is reviewed in all parts, there is still a chapter labelled ‘literature review’. Due to its inappropriate title, its purpose is commonly misunderstood, and its work confused with the work of literature in other parts. So to write this chapter better, let’s first consider its function and why it is much more than a review.

1.1 What a literature review does

In a dissertation, the literature review chapter usually comes after the introductory chapter in which you argue for a specific research problem and question, and precedes the research design chapter in which you explain how the question will be answered. In the introductory chapter, the question is proposed; while in the research design chapter, the question is taken for granted. This is quite a leap, and something makes it possible.

Positioning the so-called literature review chapter in between these two chapters tells us a lot about what the purpose of this chapter is – and indicates that there is much more to it than simply reviewing the literature. The literature review chapter is quite simply where you make the case that the proposed question (from the previous chapter) should be accepted, so that research can be started in the following chapter. Figure 6.2 summarises this.

To state it differently, the literature review is not merely a review: it is a **series of connected arguments in support of the research question**. Simply ‘reviewing’ the literature therefore misses the purpose of the literature review. A mere review leads to list-like writing. In addition, writing one in this way is simply a laborious chore and leaves you with a poor understanding of your question, which will then become glaringly evident later in your research design and in your interpretation of results.

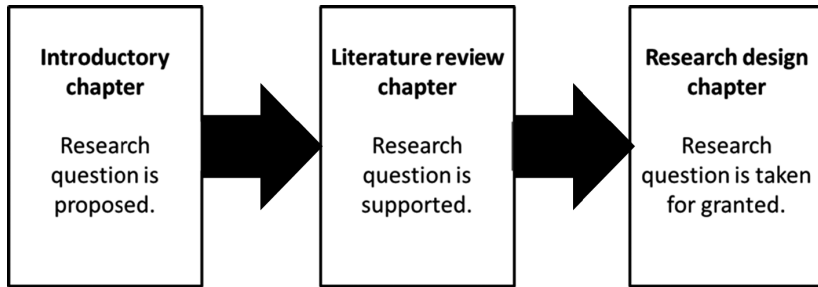


Figure 6.2 The placement of the literature review is significant

1.2 Why the research question needs a series of supporting arguments

You may remember from chapter 5 that a meaningful question meets four conditions. The first condition was that a meaningful question should make plausible assumptions. All questions are built on a foundation of assumptions, that is, they are based on statements that we take for granted. We didn't pay much attention to this in chapter 5, but it is central to the writing of a literature review. For example, the question, 'Why were you late?' *assumes* that you were, in fact, late. Even the simplest of questions make several assumptions as illustrated in Figure 6.3.

Here is the potential problem with any question: if any of the assumptions that underlie a question are implausible, the question becomes meaningless. For instance, if I ask 'When did you stop smoking?', but you never smoked; my question is problematic because it cannot be answered. In research, you will find it impossible to write a good literature review if your question is meaningless.

While the simplest questions, such as 'What is one plus one?', are based on a handful of obvious assumptions, research questions are more complex and can easily make ten times more assumptions, many of them not so obvious. Not being aware of the doubtful assumptions of a question means that you may be trying to answer a question that has no answer.

Assumptions can easily be wrong, because when we take something for granted, we have not proven it is true beyond any doubt. Even though this

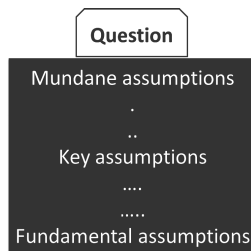


Figure 6.3 Questions are based on assumptions

may cause problems, we cannot pursue knowledge without making assumptions. The pursuit of knowledge needs to start somewhere; so, at some point, we simply have to take some things for granted. To prevent assumptions from causing problems, we need to know how to find them and recognise that they are always open to doubt.

If assumptions are open to doubt, so is anything that is based on assumptions, such as questions. A question based on implausible assumptions will be dismissed by the discipline. Fortunately, you can reduce the chances that this will happen by identifying the assumptions in your question and, *with the aid of the literature*, argue that these assumptions are currently accepted, or at the very least, plausible.

A question is based on three kinds of assumptions. Of these, only the key assumptions need to be argued: those that your readers might be willing to question. You may remember from chapters 1 and 4 that I explained that an argument is any statement that allows for differences to emerge. There are two types of assumptions, however, that don't allow for these differences, and therefore cannot be arguments, namely: mundane and fundamental assumptions. Mundane assumptions are regarded as too obvious to argue, and fundamental assumptions are regarded by the discipline as too important to argue. Only the key assumptions are open to different perspectives.

Assumptions like 'plus means adding together' or 'people exist' are mundane since they are so obvious to the readers that they have become facts. Other assumptions are so fundamental to the discipline that most researchers will not dare question them. For example, most research in theology could not have existed without the fundamental assumption that 'God exists', hence very few researchers in theology would dare question it.

What you need to find are the key assumptions that your discipline regards as open to question or potentially arguable. A statement such as: 'The free market works better than the government', is open to argument in many disciplines. However, a discipline like Libertarian Economics would regard that statement as either a fact or simply too fundamental to question. Hence, whether an assumption can be argued depends on the discipline, which means you have to know the literature well enough to be able to distinguish the fundamental and key assumptions in your question.

Assumptions are connected to others, so research questions make several inter-related key assumptions. These arguments need to be argued in a logical sequence so that, by the time the last argument is reached, the audience is convinced of the validity of your question. As a result, a good literature review will produce a series of logically connected arguments that support the research question.

1.3 Why an argumentative literature review is easier to write well

As soon as you realise that you need to create arguments, you won't just summarise and review theories and studies. You will think and write

differently. Your writing will naturally become more coherent and more critical as you start by looking for connections between ideas and making ideas and authors engage with each other.

When you realise that you have to argue the question and its assumptions, your literature review will become like the talk show I explained in chapter 4. You, as the author, will act as the talk show host, directing the conversation between guests (other authors) around ‘talking points’ (the assumptions). The talking points should be arranged sequentially until they lead up to the point where the question is accepted as plausible. Given that assumptions are open to doubt, the author (or talk show host) will not merely produce a list-like literature review, but will draw in other authors (guests of the show) as needed to argue with each other until a point is settled.

In the next section, based on this understanding, I explain a technique for writing better literature reviews. It involves: firstly, generating the content of the literature review; then connecting and structuring this content; and finally, drawing up a writing plan.

2 Generating the content

Finding the assumptions that underlie the question generates the raw content of a literature review because they have to be argued to show that the question is meaningful. A person practised in finding assumptions would easily find many assumptions in a question they understand. However, this can be difficult, even for experts. What is required here is the open-mindedness I highlighted in chapter 3 – *shoshin* (‘beginner’s mind’) in Zen Buddhism. As explained, in such a state, there is an awareness of preconceptions and an ability to be detached from them.

By employing this way of thinking when writing a literature review, you are encouraged to become aware of the ‘taken-for-granted’ ideas that can imprison your thinking, and through this awareness, go beyond them and open them up to argument. It is therefore not unusual to learn a great deal when approaching a literature review in the way explained here.

Unfortunately, we are not trained in the art of finding assumptions, and therefore initially one is forced to guide the process with a rather mechanical approach I call ‘relational analysis’. Relational analysis involves analysing the relations between the terms in a meaningful research question.

So, before this technique can be used, the question needs to be made meaningful as explained in chapter 5. This means the question should be defined so well that it meets these conditions: (1) the question alone suggests what information is required to answer the question; (2) the question itself suggests where and when to get that information; and (3) the question alone suggests by what method to obtain or generate the information.

The easiest way to do this is simply to take each noun, verb, adjective and adverb in the question and define what is being said until the question meets the three aforementioned conditions. After such a process, a question like,

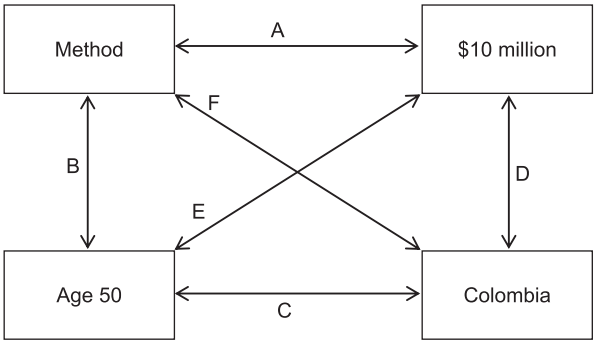


Figure 6.4 Relational analysis map of the question: ‘By what method can a Colombian today accumulate \$10 million in a bank account by age 50?’

‘What is the secret to success?’ may become, ‘By what method can a Colombian today accumulate \$10 million in a bank account by age 50?’

The next step is to break the question up into its key terms. The question may be separated into four terms: (1) method; (2) Colombia; (3) \$10 million; and (4) age 50. There is some discretion to be used here, but the researcher familiar with the literature should know which terms are most relevant. Since all the terms appear in the same question, the questioner assumes that there is a relation between them. These terms are then mapped in relation to each other, as shown in Figure 6.4.

The more key terms, the more relations, so it is usually good to keep the key terms between three and five. Each of the pair-wise relations (labelled A to F) is then questioned to determine what is assumed about the relationship between each pair of terms in the question. In the case of relation A (in Figure 6.4), one may ask, ‘What is the relation between method and \$10 million that is assumed in the question?’ and realise that it assumes that a method exists that one can follow to achieve financial success.

Once a list of assumptions has been generated, one identifies the obvious assumptions (and fundamental assumptions, if there are any). These assumptions will either: not be argued or be treated only as background when arguing the other assumptions, so they can be removed from the list. Any assumptions that are very similar should also be joined at this point. After this, the remaining assumptions, which I label ‘key assumptions’, are assumptions about which there are different perspectives. The question usually favours one perspective over others, and hence each such assumption will call for an argument to be made.

After having combined duplicate assumptions and removing ones that are too obvious or fundamental, one might emerge with the assumptions below for each relation in Figure 6.4:

- A: There is a method that guarantees financial success.
- B: Any method takes time to deliver results.
- C: Most Colombians will live much longer than 50.

D: People evaluate success in terms of money.

E: Success is only meaningful if you are still young.

F: Colombia is predictable enough for a method to work.

The list above shows only one assumption per relation, but it is quite possible for more than one assumption to be generated for each relation, or that some relations may not have any non-obvious assumptions.

The other condition for a question to be meaningful is that the assumptions are plausible. Since you now have a list of key assumptions, this would also be the best time to determine if any of them are implausible. If so, the research question needs to be changed, because a single implausible assumption will make the question problematic, meaning that it will not be able to be answered. Then repeat the process until all the assumptions are plausible given the literature of the discipline.

Now, relational analysis is not the only way to find assumptions. If you prefer to organise your literature review around a problem, you can use the conflict diagram explained in chapter 3 to find them, and follow the rest of the process as explained in the rest of this chapter.

3 Planning the structure and flow

At this point, you have found the key assumptions, which are the claims that should be argued. This is the basic content, but now you need a writing plan that tells you how to connect it logically and elaborate on it. In this section, I explain how to use the content generated to design a structure and logical flow for the literature review.

3.1 Technique for connecting the content of the literature review

We now look for a logical flow between assumptions, using the technique of linking sentences, as explained in chapter 4. To find the logical sequence of your arguments: first, write each assumption as a full sentence that describes the claim being made; then, insert a linking sentence or phrases (shown in *italics* below) between the assumptions, so that all of them together reads like a long sentence that makes sense. Doing this with the example might lead to something like this:

Success is best evaluated in terms of money (D), *and such success is only meaningful if you can achieve it before the age of 50 (E); but fortunately this is possible, because there are methods that guarantee financial success (A) that can work, because Colombia is predictable enough (F) so that people following such methods will live long enough to apply this method (B) and enjoy its results (C).*

If you try this out for yourself, you will notice that there are often many different ways to connect the claims. There is much discretion to be used here, and the only real test is the quality of the connections. The connections between claims are rarely simple words like ‘and’ or ‘furthermore’; connections usually have to be phrases that make the one claim flow into the next one without it appearing contrived. If you can’t find a connection, or the connection feels forced, then diagnose the problem using Figure 4.10.

3.2 Technique for structuring the content of the literature review

Even when there are strong connections between arguments, it can still be difficult to follow if the literature review gets too long. In such a case, the next step I propose is breaking it up into sections.

Now, finding sections is a matter of judgement: one looks at the logical flow, and tries to identify if any of the assumptions form a natural grouping or if there are obvious breakpoints between them. If such groupings are found, each should be given a descriptive label. These groups will become the sections of the literature review.

It may well be that a single assumption contains so much that it could be a section by itself. There is a lot of discretion to be used again here; and the more knowledgeable you are about the literature, the easier it will be to identify appropriate groupings. Continuing with the example, a possible result is shown in Table 6.1.

In the case of more complex literature reviews, with a large number of assumptions; it is more useful to group assumptions into sections first, and then to work out the logical flow between sections; and then the flow between the assumptions within these sections. This may give rise to sub-sections. Such an alternative approach is explained in the next section.

Table 6.1 Using sections

Section heading	Content and logical flow inside the section
1. The nature of success	Success is best measured in terms of money (D) and such success is only meaningful if you can achieve it before the age of 50 (E) but fortunately this is possible because...
2. Methods to achieve success	...there are methods that guarantee financial success (A) which can work because...
3. Achieving success in Colombia	...Colombia is predictable enough (F) so that people following such methods will live long enough to apply this method (B) and enjoy its results (C).

3.3 An alternative approach when you have a large number of assumptions

To illustrate the alternative approach for when you have more than six assumptions, let's take another example from an actual research project with the question: 'How does the psychosocial vulnerability of South African social work students affect their functioning?' and see how it may work.

The first few steps are the same. First, we identify the main elements of this question, and then map them in a relational diagram as shown in Figure 6.5.

We use this diagram to elicit the assumptions of the question. The following assumptions arose from it:

- In South Africa, there are different demands on students' functioning;
- Both personal and academic functioning are important to social work students;
- Personal functioning is critical to social work students;
- South African social work students face unique vulnerabilities, which are not found in many other countries;
- Social workers in South Africa face different challenges;
- Vulnerability affects functioning;
- Social work students more exposed to vulnerabilities than those studying towards other professions.

Here is where the two approaches diverge. Instead of linking the assumptions, you first group them. Put the assumptions that seem to belong together in separate groups and give every group a descriptive label. Each group will then become a section. The label is ideally a full sentence that captures the overall claim of that section. The more descriptive it is, the easier the next step will be. These groups will become the sections of the literature review. Once again, there is a lot of discretion to be used here and the more familiar you are with the literature; the easier it will be to identify appropriate groupings. The result may be something like that seen in Table 6.2 (see page 128).

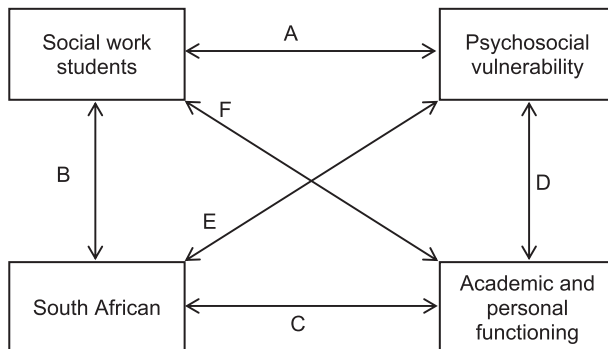


Figure 6.5 Relational analysis map of the question: 'How does the psychosocial vulnerability of SA social work students affect their functioning?'

Table 6.2 Grouping assumptions

South African social work students' functioning is affected by unique vulnerabilities
– In South Africa, there are different demands on students' functioning; – South African social work students face unique vulnerabilities not found in many other countries; – Social workers in South Africa face different challenges.
Social work students are psychosocially vulnerable
– Vulnerability affects functioning; – Social work students more exposed to vulnerabilities than those studying towards other professions.
Social work students need both academic and personal functioning
– Both personal and academic functioning are important to social work students; – Personal functioning is critical to social work students.

At the moment, the three groups are in no particular order. The next task is to create the logical flow between the sections. Take the descriptive labels of every group and place them in a sequence that makes the one logically lead to the other. Since each label is a full sentence that describes the claim being made, there is a simple method to test if the logical flow works. If one can insert a linking phrase between the descriptive section labels (with some minor changes) that makes all the labels read like a long sentence that makes sense, then the order is probably logical. For example:

Social work students need both academic and personal functioning ... and this functioning is hampered because ... **social work students are psychosocially vulnerable** ... and because of South Africa's unique socio-economic challenges ... **South African social work students' functioning is affected by unique vulnerabilities.**

One then repeats the same process within *each* section to ensure that there is logical flow there as well. This is achieved by using connecting sentences (in italics) to create a logically flowing sentence from all the assumptions. The next table (Table 6.3) was the result.

You will notice that, once again, I made changes to some of the ideas. This will invariably happen, because when you think about the arguments and their connections, you will discover things that are not clear or find missing ideas and/or connections. As a result, you will make many changes, and this takes time. But it is time well spent, because a good outline, where the arguments and the connections are clear, will make both your reading and writing much easier later.

Table 6.3 Finding flow within the sections

Sections	Sub-sections or main ideas within sections
Social work students need both academic and personal functioning...	Both personal and academic functioning are important to social work students, <i>but of these two kinds of functioning, personal functioning is critical (more important) to social work students</i>
...but this functioning is hampered because...	
...social work students are psycho-socially vulnerable...	Psychosocial vulnerability affects the functioning of students <i>and because social work is more exposed to harsh realities</i> , social work students more exposed to vulnerabilities than those studying towards other professions
...and because of South Africa's unique socio-economic challenges...	
...South African social work students' functioning is affected by unique vulnerabilities	Social workers in South Africa face different challenges <i>so those studying towards a social work qualification will find that there are different demands on their functioning which in turn leaves them open to facing unique vulnerabilities not found in many other countries</i>

4 Organising the writing

At this point we have the content and structure that ensures a logical flow and supports the question. Now we need to expand these claims and design a writing plan.

4.1 Technique for expanding on the content of the literature review

Every assumption raised in the literature review is a claim that needs to be argued. The best way to expand on each claim is to see it as a talking point between authors in a talk show. To get a conversation going on each point, you conduct an imaginary conversation with a 'friend' who responds to every claim you make with an open-ended question designed to get you to respond with more information. Similar to what was explained in chapter 4, ensure that your 'friend' avoids closed-ended questions, because these tend to have answers that close the conversation down.

The process is simple: state your claim and then imagine an open-ended question that it might evoke. For example, if I state, 'Risk can be managed', then a very likely responding question (and follow-up question) would be, 'Can it really? How?' Some questions will have such follow-up questions, and this is fine, as long as these questions follow each other logically.

Table 6.4 Questions to trigger conversation

Claim	Response (trigger question)
Success is best measured in terms of money <i>and such success is only</i>	Why money and not something else?
meaningful if you can achieve it before age 50 <i>but fortunately this is possible because</i>	Why is youth so important?
there are methods that guarantee financial success <i>which can work because</i>	Like what? And how do you know they work?
Colombia is predictable enough <i>so that people following such methods</i>	How do you know this? And why is predictability necessary?
will live long enough to apply this method <i>and</i> enjoy their results	Why do you say that?

In Table 6.4 I show the unfolding of this process from the example based on the question: ‘How can a Colombian today accumulate \$10 million in a bank account by age 50?’ Each time a claim is made, there is an open-ended response to it.

Each question–response will trigger your writing about that claim, tell you what to write about and so keep you focused on it. This kind of imaginary conversation directs the talk show within each section. The writing plan is then simply, as shown in Table 6.4, the logical sequence of claims – with their connecting sentences in between them – and trigger questions.

The connecting sentences between claims need not always be argued, but they need to appear with some elaboration between the claims being connected. These connections are very important as it is often in the connections we make between ideas that we create meaning and have new insights.

Table 6.4 gives us a writing plan. When writing according to this plan, each trigger question is used to open up the conversation. Imagining a person asking logical follow-up questions will guide your exploration of each claim. Then, at the end of the ‘talk show’, the claim is closed down with a conclusion and a connecting sentence that leads to the next claim. This open-explore-close structure and its templates were explained in chapter 4.

4.2 Technique for organising the writing of the literature review

Once the writing plan has been created, it needs to be written up with the aid of reputable sources. To help organise the large number of sources, you can create a matrix that lists all the relevant sources vertically and all the sections (main claims) horizontally (as shown in the next table). Using this method, you will

Table 6.5 A matrix to organise your sources

	<i>The nature of success</i>	<i>Methods to achieve success</i>	<i>Achieving success in Colombia</i>
Source ₁	X		
Source ₂		X	
Source ₃	X		
Source ₄		X	X
...			
Source _x			X

indicate to which section each source contributes something in the same way a talk show host might decide which guests to draw in at which talking points.

From our earlier example based on the question ‘How can a Colombian today accumulate \$10 million in a bank account by age 50?’, we identified three possible sections, so a possible matrix would look like the one in Table 6.5. In the place of an ‘x’ you could also put an identifier that refers you to your interpretive summary of that source.

When writing the first section for example, one would: only take out source₁ and source₃; put the others away; and look only for those parts that can be used to answer the relevant trigger questions of that section. Once all the trigger questions have been answered in this section, source₁ and source₃ are put away and source₂ and source₄ are taken out to talk about the trigger questions of the second section ... and so on.

While writing, it is important to trust the writing plan and follow it from start to end without returning to previous parts. I repeat, *do not continuously go back and read the previous parts*. The writing plan will keep you focused and the connecting sentences will ensure coherence, so give yourself permission to just write and continue writing. Not only will you write much faster this way; but, by the time you are done, you will be able to look at your writing more objectively. It is exactly when we read our own writing repeatedly that we lose objectivity, and following the plan without looking back will help you see problems more clearly once you are done. Then put the writing away for at least two days. If you do this, you will be surprised how much more clearly you will see your mistakes.

Now revise your writing, but revise it in layers. Revision takes quite long if you do it correctly, but it leads to a much better product. Explaining how to revise your writing is a topic by itself, but a short version of the revision process is set out in the appendix.

4.3 What if I need to use a conventional structure?

By now, it should be obvious that this process does not lead to the conventional literature review structure. There is no section labelled ‘conceptual

framework', because the concepts are analysed at the points at which they most appropriately move the argument forward. The same applies to other common sections such as 'theoretical framework', since theories and empirical studies are integrated with the argumentative flow where they best contribute to the understanding of the question.

While this breaks with convention, it does prevent the usual list-like and incoherent literature reviews that one finds in theses and even some good journals. Because the content, flow and structure are all derived from the question, it also addresses the biggest challenge people have with the literature review: how to make it fit with the question and make it all 'hang together'.

But what if your supervisor insists that you use a more conventional approach? Let's use the social work example again, and show how you can transform the product produced by my approach into something with a more conventional structure. In the example about social work students we had three sections, as represented in Figure 6.6.

In order to address this problem of transforming the unconventional structure into something more conformist, let's suppose you used theory in all three sections, but also referred to empirical work in all three sections. Let's say, however, that your supervisor insists that you should have a separate section called 'theoretical framework' and discuss the empirical studies separately from that. Putting aside the issue of whether this makes sense, you can transform the outline by simply keeping the theories and empirical studies in the same order, but moving them to separate theoretical and empirical sections as shown in the Figure 6.7.

The arrows represent the connecting sentences, and you will probably have to rethink the connections, but the result will still be a literature review that is coherent and argumentative.

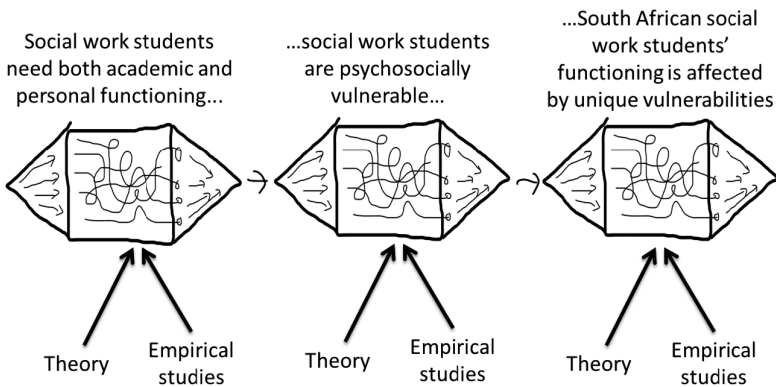


Figure 6.6 Mixing theoretical and empirical arguments

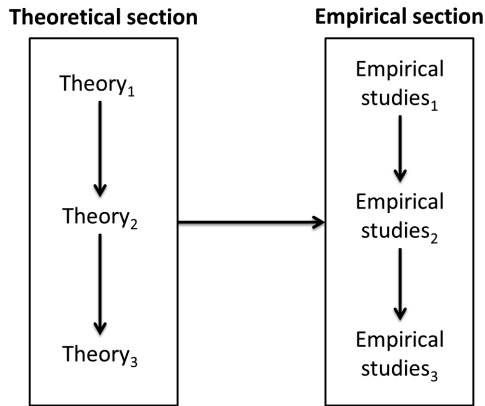


Figure 6.7 Separating theoretical and empirical arguments

5 Conclusion

In sum, what should be clear from this discussion is that a good literature review can be a creative process that deepens the understanding of both the writer and the reader. Literature reviews written in this manner delve into the research question, and unlike most literature reviews in journal articles or dissertations, have the capacity to be original contributions in and of themselves. Such reviews will also avoid the list-like quality of many literature reviews and be a pleasure to read and actively engage with.

Appendix: Revision and editing

It is best to revise in layers, paying attention to one level at a time. First, do a global revision, and ignore everything else, then move on to the next level.

When most people revise their writing, they tend to spend most of their time on levels 4 and 5, when, in fact, levels 1–3 are the most important. This is because it is useless to write a document with impeccable spelling and grammar, if it has no clear argument.

Level 1: Global:

- You first revise by checking if the overall argument (thesis) comes through clearly and that the basic structure is logical.
- Check that the introduction clearly provides context and pre-reveals the thesis.
- Check that the conclusion summarises the thesis and that it is the same as the one pre-revealed in the introduction.

Level 2: Chapters (one chapter at a time, if you have chapters)

- Ensure that all the chapters' arguments are derived from the thesis.
- Does each chapter's argument come through clearly?
- Does the structure used in the chapter organise the ideas in a logically flowing way?
- Check that the introduction of each chapter provides context and pre-reveals the chapter's argument. Check that the conclusion summarises the chapter's argument and that it is the same as the argument pre-revealed in the introduction.
- Are all chapters clearly connected?

Level 3: Sections

- Ensure that, in every chapter, all the sections' arguments are derived from the chapter's argument.
- Does each section's argument come through clearly?
- Does the structure used for each section flow logically?
- Check that the introduction of each section provides context and pre-reveals the section's argument. Check that the conclusion summarises the section's argument and that it is the same as the argument pre-revealed in the introduction.
- Are all sections in the chapter clearly connected?

Level 4: Paragraphs

- Ensure that in every section all the paragraphs' arguments are derived from the section argument.
- Does every paragraph convey only one idea?
- Is every paragraph clearly linked to the previous paragraph?

Level 5: Sentences

- Does every sentence clearly flow from the previous sentence?
- Check grammar.
- Check spelling.

7 Principles of writing argumentatively about your research design and methodology

Needless to say, the most important contribution of your research is your answer to your question, as captured by your results and your interpretation of these. But before you can get to the answer, you need to think how you will approach the process of finding it. This is the task of any section or chapter on research design and methodology.

Your research design is the overall *strategy* you will use to generate the answer to your research question, while your research methodology provides the *details* about how exactly you will put the research design in practice. For example, if your research design is a survey, then your methodology would involve thinking about the specifics of questionnaire design, sampling, pilot testing, collection of data, and validity and reliability issues.

As you can see from Figure 7.1 (see page 136), the research design continues from where the current scientific literature stops. If you have a meaningful research question, it means the existing literature brings you close to the answer, but won't take you all the way there. Hence, your research is needed to provide the last piece of the puzzle. In your research design, you explain how your overall strategy to generate the new knowledge, together with the existing literature, will answer your research question. When discussing your research design, you tell us what the bridge between the literature and your answer looks like, and when discussing your research methodology, you tell us how exactly you will build that bridge.

In a book like this – which is a *general* guide to argumentative research writing – it is tricky to talk about research design and methodology because so much of it is discipline-specific. So, instead of going into specifics, this chapter will focus on argumentation – the one thing that is often missing from research design chapters, but is expected in all disciplines.

Too often, chapters on research design and methodology read like a to-do-list rather than a continuation of the overall argument of the thesis or paper. To help you avoid list-like writing, in this chapter I discuss the arguments that you should make in any research design chapter, regardless of discipline. To obtain the details of specific designs you can consult any one of the many relevant books in your discipline.

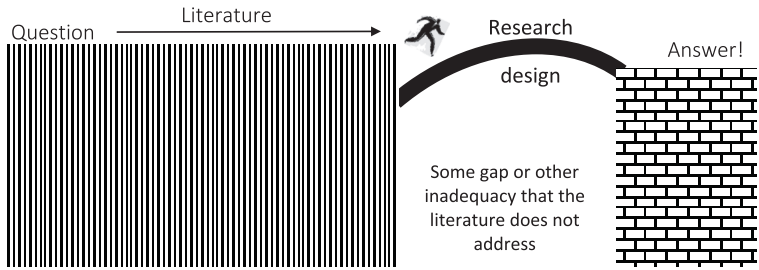


Figure 7.1 The link between the literature and the answer

1 A tool for generating arguments

It is much easier to write argumentatively if you know what you want to argue. The first step is therefore to generate your claims. While this book has introduced you to some generative tools (like the imaginary conversation and relational analysis), they are not so easy to apply to writing about research design and methodology. For this reason, I want to introduce you to a tool that can be used to generate arguments in almost any situation, which I will then apply to writing about research design. This tool is an adapted version of the rhetorical triangle that you first saw in chapter 4 (see Figure 7.2).

This triangle reveals that there are three elements in an argument: *logos* (which I interpret as the knowledge of the discipline created by previous authors); *ethos* (what you as the author of your own paper bring to the

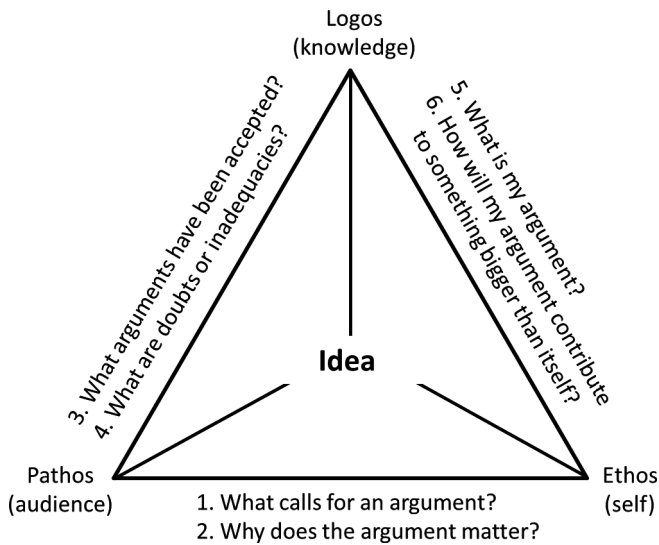


Figure 7.2 Using the rhetorical triangle to generate arguments

argument); and *pathos* (which I interpret as those argumentative actions that try to connect to the audience of readers). The sides of the triangle represent the interactions between these elements.

Looking at the triangle in this way shows that, when you write, there is an interaction between: you as the author (self) and previous authors (knowledge); you and your readers (audience); and previous authors and your readers. All this interaction, or conversation, is organised around an idea. This idea can be something as large as the overall thesis, or as small as a single argument in a paragraph. In this chapter, we take this organising idea as the arguments of the research design and methodology chapter.

In each of the three interactions, there are two general questions which open up the conversation between two parts of the triangle:

- *The self–audience interaction* focuses on the reason why readers should pay attention to your argument. The readers will naturally ask you what it is that calls for an argument and why that argument matters.
- *The audience–knowledge interaction* focuses on what the readers will want to know about previous arguments. The readers will ask what is certain and agreed upon and what is doubtful or inadequate in previous arguments.
- *The knowledge–self interaction* focuses on what your own argument contributes to previous arguments. You should be able to explain what your argument is and how it contributes to the larger conversation happening around the idea.

When faced with the need to generate arguments around an idea, these questions are effective in opening up the conversation so that the differences can be discovered. Responding to these questions, with the aid of other authors, gets the ‘talk show’ started and leads to an exploration of the differences (as explained in chapter 4). In this exploration phase, you engage with the evidence and warrants, counterarguments and different perspectives. Finally, the conversation comes to a close as you take a position. This is summarised in Table 7.1.

When writing about your research design and methodology, you will be making argument after argument – not simply telling the reader what you will do. In the next sections, I will show you how to adapt the rhetorical triangle to find the arguments to which you can apply the techniques of argumentative writing.

2 Using the tool when writing about research design

The research design is your overall strategy to find the answer, so it does not discuss details about how the research will be done. Your overall argument here is that you know what the most appropriate research design is to investigate your research question.

Table 7.1 Opening, exploring and closing the conversation

Opening	Exploring	Closing
1. What calls for an argument?		The reason why the idea needs to be argued.
2. Why does the argument matter?		The reason why the audience should care about it.
3. What arguments are accepted?		Which arguments are taken for granted or agreed on.
4. What are the doubts or inadequacies?		Where previous arguments failed, are incomplete or in doubt.
5. What is my argument?		A statement of your claim and argument that supports it.
6. How does my argument contribute to the larger one?		How your argument moves the discussion forward.

2.1 Selecting an appropriate research design

The easiest way to find the most appropriate design is just to look at what was used in previous research that is related to yours. If this does not help, it is best to let your research question guide you. The research question, if meaningful, will help you to decide on the most appropriate research design (as explained in chapter 5).

One condition for a meaningful question is that it tells you *what* information is needed to answer the question. This does not refer to articles that you still need to read, but to the data or evidence that you need to generate or gather in order to answer the research question. So, if your question is meaningful, you should be able to write down in detail what information you still need to obtain. This missing information can either be empirical (about real-world entities) or non-empirical (about abstract ideas, concepts, theories or methods). If your research requires you to find out about people or things in the real world, your design is empirical. If it requires you to find out about things that have no corresponding referent in the real world, your design is non-empirical. Knowing this, helps you decide on the most appropriate design to answer your question.

Another condition a meaningful question must meet is that it should tell you where and when to get the information to answer the question. The ‘where’ is very useful to find an appropriate design. To think about *where* you will get the information, you need to consider two things:

- Does the information you need already exist somewhere in some form (in other words can it be gathered from secondary sources)? Or will you have to generate the information you need by yourself (in other words, will you gather it from primary sources)?

- What is the nature of this information you will need? Is it numerical (numbers or reducible to numbers)? Is it textual (in the form of written text)? Is it human (obtained from listening to humans and studying their artefacts)? Or is it a combination of some/all of them?

As Table 7.2 shows, the data you need determines the most appropriate design to use when you do an empirical study, which is the majority of studies in the social sciences. There is no exact recipe for determining the correct design, but the table does help to narrow it down. You will notice that the table is not exhaustive and some designs can fall into different cells since they can be adapted to generate different kinds of data. It should be mentioned, though, in case you are considering a non-empirical study, some of the likely designs are a systematic review, philosophical analysis, conceptual modelling or theory construction. We shall focus more on empirical studies however, given their prevalence.

For example, let's suppose my study primarily needed information on the poverty rates of African countries. This is numerical information that already exists which I'll get from the World Bank. I would enter it as shown in Table 7.3, and look up the options for this cell as given in Table 7.2. In the end, I would most likely choose some kind of secondary data analysis. Or, suppose I needed to obtain information about levels of racism in a town

Table 7.2 Different kinds of data require different designs

	<i>Primary sources</i>	<i>Secondary sources</i>
Numerical data	Survey, experiment, comparative study	Secondary data analysis, statistical model
Textual information	Comparative study, evaluation research	Textual/content analysis, historical study, comparative study, evaluation research, case study
Obtained from humans	Survey, life history, action research, ethnographic study	–

Table 7.3 Different kinds of data

	<i>Primary sources</i>	<i>Secondary sources</i>
Numerical data		Poverty rates of African countries, sourced from the World Bank.
Textual information		
Obtained from humans	Views about racism in Benoni, sourced from my interviews with residents.	

called Benoni. If that information did not exist, the options in Table 7.2 suggest that I would need to conduct a survey or an ethnographic study.

If you are lucky, most of the information you look for will fall into one cell, and this means you will probably just need to use one design. But if you require lots of information falling in two different cells, it probably means that you need to use more than one design in combination.

2.2 Using research designs in combination

Your research question may call for a combination of designs if the required information needs to be generated from more than one design. In such cases, you need to think how the designs will interact and argue why this combination will work. There are two approaches to combining designs (as summarised in Figure 7.3):

- *Sequential approach*: This is when you use one design first to generate the information and then feed that into the next design. For example, you might do content analysis of policy documents to determine what the purpose of a policy intervention was and then use that information to design interview questions to explore why the policy was effective or not. The content analysis in this case provides the information on which the surveys or interviews will be based.
- *Concurrent approach*: This is when you execute different designs at the same time and then integrate the information. It happens when different designs generate different information that needs to be combined to answer the question, but here, the information generated from one design does not need the information from another design.

Much has been written about quantitative research (using numerical data) versus qualitative research (using textual and human information) and mixed designs. Practically speaking, it is best for you to first decide on your research design/s before getting into whether your research will be qualitative, quantitative or mixed. Your writing will be more coherent if you first

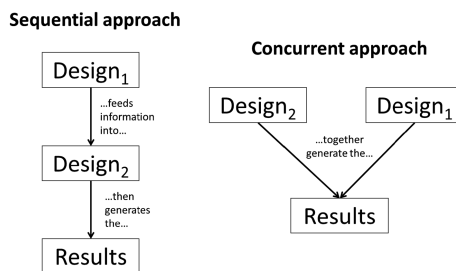


Figure 7.3 Combining designs

derive the appropriate research design from your question, and then justify in hindsight, why a quantitative, qualitative or mixed approach is best.

2.3 Generating the arguments about research design

To generate arguments, we adapt the rhetorical triangle from Figure 7.2, but focus the questions not on some general idea, but on the research design, specifically. The word ‘design’ therefore replaces ‘idea’ in the middle of the triangle. The questions are then adapted to reflect this focus, as shown in Figure 7.4.

Table 7.4 (see page 142) shows how the trigger questions that open the conversation will lead to an exploration, and ultimately, to a closing as you take a position. If the argument is short, not all questions need to be answered because they may have been answered earlier in the same study (most likely questions 1 and 2 in the table). In the case of a longer thesis, it is usually a good idea to review the answers to refresh the readers’ memories and to connect to previous arguments. Also, readers who generally go straight to the research design would appreciate this.

While the trigger questions are numbered 1–6, it is not always necessary to argue for their answers in that order. In this case, it does seem to be an order that makes sense, but as always, the test is whether the claims (in the closing column) follow each other logically and lead into the next argument you want to make. In the extract to follow (see page 142), I use linking phrases and sentences (in *italics*) to test the logical flow.

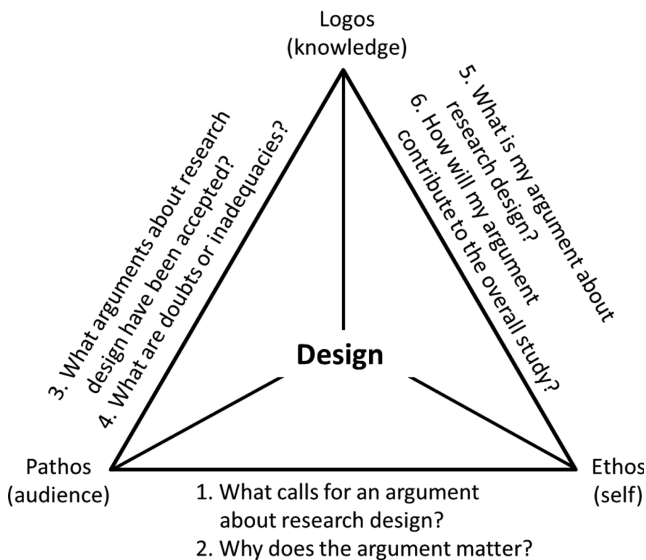


Figure 7.4 Generating arguments about research design

Table 7.4 Opening, exploring and closing the conversation about research design

Opening	Exploring...	Closing
1. What calls for an argument about research design?	... that there is a question that has not been answered by previous research and their research designs.	We need to design new research for this unanswered question.
2. Why does the argument matter?	... why the wrong research design could lead to the wrong answer (and thus stop us from solving the problem).	The correct research design will generate the results that can assist in solving the problem.
3. What arguments about research design are accepted?	... the research designs, or certain versions of them, that have been preferred by previous researchers.	Some research designs, and ways to approach them, are acceptable, and can help to choose the best design for this study.
4. What are the doubts or inadequacies?	... the disagreements between previous researchers on the appropriateness and merit of using certain designs.	There are issues with the designs in previous studies that need to be taken into account when designing the research for this study.
5. What is my argument about research design?	... by weighing up the previous designs, issues with those designs and choosing the appropriate design given the uniqueness of the study.	Description of the most appropriate design that has been selected for this study.
6. How does my argument contribute to the study?	... why it is expected that following the chosen design will answer the question.	If executed correctly, the results generated, using the chosen design, will answer the question.

We need to design new research for this unanswered question. *If we do this correctly*, it will generate the results that can assist in solving the problem. *Previous research related to this problem* has agreed on certain research designs and ways to approach them and this can help to choose the best design for this study. *However, we cannot just follow previous researchers exactly because* there are issues with the designs in previous studies that need to be taken into account when designing the research for this study. *After taking all this into consideration*, I have selected ___ as the most appropriate design for this study because ___. *If executed correctly*, the results generated with this design will answer the question.

The last sentence leads us logically into the next section on research methodology, which explains how exactly the research design will be executed. This is also what we consider in the next section.

3 Using the tool when writing about research methodology

Research methodology is not only specific to each research design – different disciplines may also execute the same design in different ways. Discussing the details would require a small library of books, so here you will need to consult the textbooks commonly used in your discipline. That said, the discussion of the research methodology is still a series of arguments in which you try to convince your readers that your approach will lead to an answer to your question. We can therefore expect to see very similar kinds of arguments across disciplines and methodologies.

To find those arguments, we apply the rhetorical triangle again in the Table 7.5. As this shows you, the triangle can be applied again and again at lower levels – from the overall thesis, to chapters, sections and subsections. You may, for example, apply the triangle again in question 5 where you argue how exactly you will conduct the research.

When discussing your research methodology, you will likely discuss the questions in a different order because some supervisors expect you to follow a specific structure. For example, some supervisors might want you to discuss the details of the validity of the study (question 4) and reliability and generalisability of the results (questions 5 and 6) under separate sections or in a different sequence. Even so, you will still make arguments, and they will be arguments similar to the ones in Table 7.5.

Research methodology textbooks make sampling, validity, reliability and generalisability quite complex since there are different types of each and even different terminology depending on whether the study is quantitative or qualitative. If you are not yet familiar with these concepts, refer to the appendix. I over-simplify there, but after reading it, you should get the ‘bigger picture’, which will help you understand the concepts of sampling, validity, reliability and generalisability; and thus grasp the textbooks’ details faster. Bear in mind that each of these concepts is an argument in itself.

4 Beyond research methodology

Once you successfully executed your research, you will have produced your results. You then present the results in the format preferred by your discipline and interpret them to show how they answer your question. These results should be interpreted in such a way that it makes your overall argument (thesis) clear in your conclusion.

The presentation, and especially the interpretation of your results, involves much argumentation. Because this is so discipline-specific, it is beyond the scope of this book to discuss the details. However, by now,

Table 7.5 Opening, exploring and closing the conversation about research methodology

<i>Opening</i>	<i>Exploring</i>	<i>Closing</i>
1. What calls for an argument about research methodology?	...why chosen design has to be executed in a specific way.	The design has to be executed in a specific way to generate results.
2. Why does the argument matter?	... why executing the research design incorrectly could produce results that won't answer the question.	The correct execution will produce the required results.
3. What arguments about the research methodology are accepted?	... what previous researchers have agreed on with regard to conducting this kind of research – theoretically and methodologically.	Conducting the research in this way will ensure that the results are valid.
4. What are the doubts or inadequacies?	... where previous researchers have disagreed or where the methodology is problematic or might not be suitable given the context of the study.	The methodology needs to take certain issues into account, has limitations to be overcome and should be adapted in a certain way to the context.
5. What is my argument about the research methodology?	... how exactly you will conduct the research to ensure that the results are reliable.	Detailed explanation of the methodology, especially sampling and procedures of data collection and data analysis.
6. How does my argument contribute to the research design?	... how you will know that your results answer the question (e.g. tests) and what you will look for in the results to answer the question.	The results that will be generated can be trusted to answer the question, and the degree to which they can be applied to other contexts.

you have all the tools necessary to ensure that this too has an argumentative quality.

Appendix: Simplifying sampling, validity, reliability and generalisability

Once you selected the appropriate research design, you need to think about how you will execute this design exactly. The main issues to consider are sampling, validity, reliability and generalisability; and the relationships between these are captured in Figure 7.5.

Without going into detail, I will discuss each arrow in Figure 7.5.

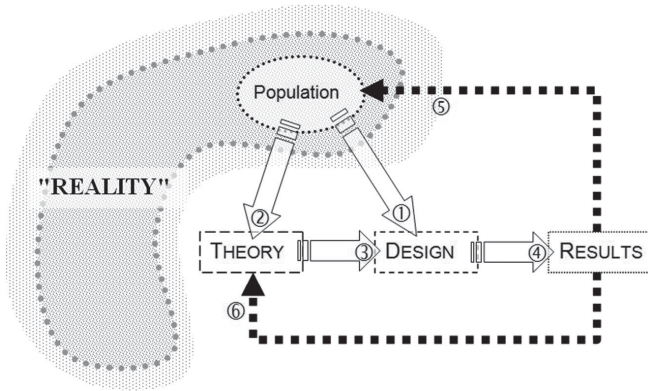


Figure 7.5 Sampling, validity, reliability and generalisability

Sampling

All empirical research investigates the real-world reality, but obviously not all of it. Each discipline explores a smaller aspect of this reality: for example, economists study ‘economic reality’ and this is shown as ‘reality’ in Figure 7.5.

Within this narrowed ‘reality’, you will study an even smaller part, which I labelled ‘population’. It should be interpreted quite broadly in that it refers to the entire collection of objects, events or people that you will investigate as a result of your question. This could be: a particular time period; a specific country or region; a particular set of variables; certain defined phenomena; or a group of people etc.

Even a population is too broad to research, so we usually study only a small section of this population. For example, we may study only certain variables like HIV prevalence, focusing: only on a certain period (for example, the period from 2014–2016); only on a particular city in a country (such as New York); on the phenomenon only under certain conditions (such as in poverty); on interviewing only a small number of people in a larger group (e.g. young adults). Whenever we do this, we are ‘sampling’.

Due to the size of the real world, it is inevitable that we will sample regardless of the research design used. This is shown as arrow 1 in Figure 7.5, and you can see that sampling from a population informs the research design.

When you discuss sampling issues, you should address at least the following:

- Identify the population.
- Explain how, or based on what, you will sample. Justify your approach to sampling.
- How will you know whether the sample is sufficient or representative? Are there any tests you could run to check it?

Validity

When the results you generate from your research design are valid, it means that the results truly reflect the concepts and relationships you intended to investigate. This sounds strange, but if you don't exactly know what you want to research, you are likely to generate results that answer a different question than you intended.

The simplest way to ensure that your results will, in fact, answer your question is to clearly define the concepts and relationships that are relevant to your research question before you start gathering or generating the results. The correct definitions and relationships are found in the theory and other studies related to your problem. So, theory needs to inform your research design (arrow 3) and this theory would have been derived by previous authors from an investigation of the population (arrow 2).

To show how important it is to clearly define your terms and relations, let's take a study about how education affects youth unemployment as an example. In this case, you would need to go to the literature to find definitions for: (1) education; (2) youth; and (3) unemployment.

- If, for example, you don't know what you mean by 'education', you won't know if you should gather information about primary, secondary or tertiary education, or about academic performance, completion rates or some other indicator. So, you may end up mixing them up and producing very confusing results. The literature would help you here in identifying the most appropriate way to define 'education' and you need to explain your definition to the reader with reference to those studies.
- You need to know, in advance, who is regarded as part of the 'youth' or else you may end up interviewing people who are not youth, and generate results that are not valid. It would be useful to look at what similar studies have done and justify your definition with reference to those studies.
- 'Unemployment' would be another tricky concept in this study. Is any person without a job unemployed, or only if that person is jobless while actually looking for work? If you choose a definition that is inconsistent with the literature, your results will be regarded as invalid. So ensure that you know how your discipline defines the concept and explain it with reference to this literature.

If you used a survey in such a study, knowing these definitions will help you to identify the participants correctly and to ask the right questions. For example, if you did not know the correct definition of unemployment, you would only ask a participant if they have a job, and forget to also ask them if they have been looking for a job. If you used some statistical model, knowing these definitions will ensure that you include the correct variables in your model and that you measure these variables correctly.

Continuing with the example, you would also need to consult the literature to establish what it states about the relationship between education and youth unemployment, the mechanisms by which education affects youth unemployment and what obstructs this relationship. Without this, you would not know what to look for in your investigation, and you may end up studying something different from what you intended.

If you used a survey in such a study, knowing about the relationship between the variables will help you to determine the correct questions to ask. For example, if you established, from the literature, that the effect of education on youth unemployment depends on the subjects that students study, you would ask questions about their subject choices. If you did not know this, you would not ask these questions at all and your results would therefore be invalid. If you used some statistical model, knowing these relationships will help you to know what the expected signs of the coefficients are and how to identify missing variables (like control variables or interaction variables). If you do not know this, you may miss some important variables in your model specification, or struggle to evaluate your results later.

Besides reading the literature, it is also important to check with experts whether they find your study valid. For example, check with them if they agree with your definitions and the relationships between them, or if they agree with your variables (if you construct a model) or agree with your questions (if you conduct a survey).

To ensure readers are convinced about the validity of your results, you need to, at least, explain the following as part of your research methodology section:

- Define the concepts in your question and sub-questions very clearly with reference to the accepted theory and similar studies, and explain how these definitions will inform your study.
- From the literature, identify the critical variables, and relationships between them, that have bearing on your research question and explain how your research will explore these variables and relationships.
- If you are using a quantitative research design, there may be tests you can run to test validity of the results. Which tests will you use?

Reliability

Not only should your results be valid, but they should be consistently valid. You need to ensure that regardless of who does the study, the results will still be more or less the same. If others repeated your study at a different time or place, and obtained the same results, your results are reliable. That is why reliability is sometimes called ‘repeatability’.

To convince the readers that your results can be repeated, you should:

- Detail the step-by-step process you will follow to generate your results. This is the most important way to ensure reliability, and, if you are

using a qualitative design it may be the only way to ensure reliability. You need to explain, *in detail*, what exactly you will do – almost like a recipe. It needs to be in so much detail that another researcher can visualise the research process and have sufficient detail to execute the same research from your description of the process alone. I cannot emphasise this enough because many novice researchers go wrong here by providing insufficient detail. If others do not have enough detail, your study cannot be repeated or will be repeated incorrectly, and this will make it unreliable since other researchers may generate results that are different from yours.

- How will you record the information and how will you ensure accurate recording? Also explain what possible errors might occur and how you will check for and minimise any such errors. If you use a quantitative design, there may be tests you can run to check for errors. If so, explain the tests you will run.
- Discuss to what extent and under what conditions your results could be confirmed and repeated by other investigators.
- If you are using a quantitative research design there may be tests you can run to test reliability of the results. Which tests will you use?

Arrow 4 in the diagram shows how the chosen research design leads to results. This represents reliability, because if other researchers followed your research design, they should get very similar results under the conditions you specified.

Generalisability

All studies sample from a larger population as discussed previously. For the study to be useful, the results generated from the sample should be applicable to the population, to some degree at least. This is shown as arrow 5 in the diagram – the results should be able to tell us something about the population or at least a larger sample than the one used in the study. Explain to what extent you think the results of your study could be generalised to a different place, a larger group or a different time.

Other issues

Arrow 6 is unlikely to need any discussion at this point as it only becomes relevant once you have your results. Once you have them, you would use the theory to make sense of the results or explain how your results confirm or falsify the theory.

Depending on what design you use, there are more specific kinds of sampling, validity and reliability and different tests for each. Also, there are differences in their meaning, depending on whether the study is qualitative or quantitative, so it is important to consult a methodology textbook. Also, there are specific tests – especially if your design involves a survey or is quantitative, and you need to ensure that you discuss this in the research design and methodology chapter.

8 In conclusion

The purpose of this book was to focus on what makes research writing difficult across all social science disciplines and how to overcome these challenges with an argumentative mindset. And this is what it did. It did not, however, cover all the skills and theories related to research and writing a thesis or a paper. Thus, this textbook's generalised focus is important in that it provides an approach to argumentative writing across all subject matter. However, its lack of specificity means that, ideally, this book should be used in conjunction with more conventional research methodology textbooks that are specific to your discipline. For example, chapter 1 did not discuss specific databases or the conventions of each. Likewise, chapter 7 did not discuss specific research designs. The emphasis was always on the general challenges and the argumentative work that is expected, regardless of discipline. While I tried to be as general as possible in chapters 5–7, there are, nevertheless, differences between disciplines and journals in each of these areas. As I mentioned throughout the book, even if you are equipped with the techniques of this book, there is no substitute for knowledge of your particular discipline. The two working in conjunction should, however, forge the prospect of a formidable argument.

From now on, don't just write – argue!

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